# Facing Unequal Opportunities: Does Experience Shape Redistributive Preferences?

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We study whether firsthand experience of unequal opportunities changes how people judge merit and redistribute income. In our experiment, two workers complete a real-effort task, with one assigned to an easy version (advantaged) and the other to a hard version (disadvantaged). A bonus is initially awarded to the higher-performing worker. Third-party spectators are randomly assigned to experience either the easy or the hard version before deciding how to redistribute the bonus. By holding the workers' circumstances fixed while varying spectators' own experience, the design isolates the causal effect of experiencing advantage versus disadvantage. Spectators who complete the hard version attribute performance more to unequal opportunities, view the initial allocation as less fair, and redistribute more toward the disadvantaged worker, even when doing so is personally costly. An extension compares one-sided experience with experiencing both versions and with receiving information through qualitative reports or quantitative statistics. Experiencing both sides moves beliefs and allocations the most, while information alone, even when it conveys the size of the gap accurately, moves them less. Redistributive preferences thus depend not only on what people know about unequal opportunities, but also on how they come to know it.

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# 1 Introduction

Fairness views shape whether people regard inequalities as acceptable and how they want such inequalities to be addressed (Cappelen et al., 2007, 2013; Konow, 2000; Stantcheva, 2021). In modern societies, the dominant view is meritocratic. Inequalities are more accepted when they result from individual effort and choices, but considered less fair when they arise from circumstances beyond individual control, such as random luck (Almås et al., 2020; Almås et al., 2025). Yet meritocratic judgments become difficult when individuals face unequal opportunities. In such settings, observed performance becomes an imperfect signal of effort and ability, as it also reflects the opportunities individuals face. The same effort can produce very different outcomes depending on the conditions individuals face and effort itself is often shaped by opportunities (Altmejd et al., 2021; Bursztyn et al., 2017; Falk et al., 2026; Glover et al., 2017; Parsons et al., 2011). Poor access to training, discrimination, or a weak labor market may both make success harder to attain and discourage hard work, whereas advantageous opportunities can make success easier and effort more rewarding. This creates a fundamental dilemma for societies that aspire to be meritocratic. How should outcomes be rewarded when performance reflects not only effort, but also the unequal opportunities through which effort is translated into success?

A growing experimental literature shows that people account for unequal opportunities only partially. Spectators who know that workers faced unequal opportunities still largely reward observed performance (Andre, 2024; Cappelen et al., 2024; Dong et al., 2024; Pogliano, 2024; Pruess et al., 2025; Bhattacharya and Mollerstrom, 2025; Brun, 2026). This may reflect differences in fairness views, but also innacurate beliefs about how opportunities shape performance. People often underestimate the role of circumstances and attribute outcomes to effort and ability instead (Jones and Harris, 1967; Cappelen et al., 2024; Pruess et al., 2025). This raises the question of where such beliefs come from. In other domains, personal experience is an important source of economic beliefs, from inflation and unemployment to stock-market returns (Malmendier and Nagel, 2016; Kuchler and Zafar, 2019; Malmendier and Wachter, 2024), and experiences of disadvantage have been linked to political preferences (e.g., Cruces et al., 2013; Moriconi et al., 2025). If beliefs about opportunities are shaped by experience, differences in exposure may help explain differences in what people consider fair. Whether firsthand experience of unequal opportunities changes how people judge merit and redistribute, however, remains an open question.

In this paper, we address these questions by implementing a two-phase spectator game experiment (Almås et al., 2020; Cappelen et al., 2007, 2013). In the first phase, participants we call "agents" complete a visual counting task under unequal opportunities. Agents are randomly assigned to one of two versions of the task, in which they count a target letter in a grid of similar-looking characters. In the hard version, the target letter is more difficult to distinguish from the surrounding characters. We deliberately design this difference to be difficult to infer from the instructions alone, while allowing its consequences for performance to accumulate over the course of the task. Although effort and ability still matter within each version, task assignment strongly influences performance: agents assigned to the easy version complete 12.5 tables on average, compared with 3.5 among those assigned to the hard version. We then form pairs

consisting of one agent from each version, selecting pairs in which the agent assigned to the easy version performed better. A $6 bonus is initially awarded to this higher-performing agent.

In the second phase, a separate sample of participants we call "spectators" learn about the two versions of the task and decide how to redistribute the bonus between a pair of agents. Before making their decisions, spectators are randomly assigned to experience either the easy version (`EASY`, the experience of advantage) or the hard version (`HARD`, the experience of disadvantage) for four minutes. Their performance during this experience does not affect their earnings, which limits incentives to justify their own success or failure and leaves little room for motivated belief formation. We elicit beliefs about the task before the experience and again later in the experiment, to examine whether experience changes how spectators interpret the determinants of performance. Initially, most spectators underestimate the importance of task assignment: 86% consider the two versions similar, and only 11% believe that the easy version provides a performance advantage. After the experience, the share recognizing the advantage of the easy version rises to 35% in `EASY` and 49% in `HARD`. Experiencing the disadvantaged version thus leads to stronger belief revision about the role of opportunities in determining performance.

We then examine whether experiencing unequal opportunities changes redistributive preferences. Each spectator makes two consequential redistribution decisions: one that entails no personal cost ("costless") and one that carries a fixed cost ("costly"). We start with the costless decision, which comes first in the experiment. Spectators in `EASY` transfer on average $1.53 of the $6 bonus to the disadvantaged agent, while spectators in `HARD` transfer $1.80, an 18% increase that remains unchanged after controlling for socio-demographic characteristics. The difference arises mainly at the intensive margin: the share of spectators who redistribute a positive amount is similar across treatments (70.5% in `EASY` and 75.6% in `HARD`), while the share who split the bonus equally increases from 21.7% to 31.7%. Our design limits the scope for self-serving explanations, as spectators' own performance does not affect their earnings. Instead, exploratory analyses point to changes in how spectators interpret the sources of performance differences. Accounting for beliefs about the relevance of luck and unequal opportunities substantially attenuates the treatment effect, while stronger emotional responses, particularly anger and irritation, may also play a role.

We next examine whether spectators are willing to incur a personal cost to implement the allocations they consider fair. Among those who redistribute when doing so is costless, 45.1% in `EASY` and 57.6% in `HARD` continue to redistribute when a fixed cost of $0.20 is introduced. Even after controlling for the exact amount transferred in the costless decision and socio-demographic characteristics, spectators in `HARD` are 9 percentage points more likely to continue redistributing. We use both decisions to estimate a structural model of the weight spectators place on implementing their fairness view relative to their own income. Spectators in `EASY` place essentially no weight on fairness once acting on it is costly. In contrast, the estimated fairness weight is positive and statistically significant in `HARD`, indicating a greater willingness to bear a personal cost to implement the allocation spectators consider fair.

Our main analysis shows that spectators who experience disadvantage redistribute more, and are more

willing to bear a cost to do so, than those who experience advantage. We then investigate which aspects of experience matter for redistributive judgments, and whether learning about unequal opportunities through information generates similar responses. To do so, we run an extension with three additional treatment arms. In `MIXED`, spectators complete both versions of the task in randomized order, allowing them to directly experience and compare advantaged and disadvantaged conditions. Spectators in `MIXED` transfer on average \$2.13 of the \$6 bonus to the disadvantaged agent, 39% more than in `EASY` and 18% more than in `HARD`. We use this treatment as a benchmark for two forms of information provision that convey the opportunity gap without firsthand experience. In `REPORTS`, spectators receive qualitative accounts of agents' experiences, while in `STATISTICS`, they receive quantitative information about performance differences between the two versions. Redistribution in both treatments falls short of `MIXED`. Nevertheless, `REPORTS` increases redistribution by 24% relative to `EASY`, whereas transfers in `STATISTICS` are similar to those in `EASY`, despite participants receiving detailed information about the performance gap.

**Related Literature.** This paper builds on and contributes to several strands of the literature. First, we add to the expanding literature on fairness and inequality acceptance under unequal opportunities (Andre, 2024; Bhattacharya and Mollerstrom, 2025; Brun, 2026; Cappelen et al., 2024; Dong et al., 2024; Pogliano, 2024; Pruess et al., 2025). Across this literature, a common design feature is that studies vary the opportunity structure workers face, or the information spectators receive about it, while holding spectators' own exposure to those opportunities fixed. The resulting evidence shows that spectators only partially account for unequal opportunities when evaluating merit and correcting inequalities. Dong et al. (2024) show that redistribution responds to the nature and magnitude of opportunity differences, although this response remains incomplete. Cappelen et al. (2024) trace this incompleteness to two distinct sources: a failure to recognize how opportunities shape performance and a continued willingness to reward performance even when this relationship is recognized. Andre (2024) shows that spectators continue to reward choices shaped by unequal circumstances partly because they remain uncertain about the relevant equal opportunity counterfactual. Bhattacharya and Mollerstrom (2025) similarly show that people reward work even when luck determines who receives the opportunity to work in the first place. Brun (2026) shows that accounting for the relevant counterfactual is cognitively demanding, with more cognitively able individuals better equipped to make the inferences needed to account for unequal opportunities. Pogliano (2024) shows that these judgments are also shaped by beliefs about the origins of ability, with beliefs about the importance of genetic endowments affecting redistributive decisions. Pruess et al. (2025) show that spectators underestimate how small relative advantages compound over the course of a task, leading them to redistribute less when inequality stems from unequal opportunities than when luck acts directly on outcomes. We complement this literature by varying spectators' own exposure to the opportunity structure while holding the workers and distributive problem fixed. We show that inaccurate beliefs help explain this unresponsiveness to opportunity gaps, as direct experience changes how spectators interpret performance differences and increases compensation of the disadvantaged worker. Moreover, experiencing both sides of the opportunity gap produces stronger redistributive responses than receiving qualitative or quantitative information about it. This sheds light on how beliefs enter

redistributive decisions, and on how task-specific fairness judgments are formed.

We also contribute to the emerging literature on costly fairness and the willingness to implement redistribution (Dimant et al., 2024; Dwenger et al., 2024; Piraino and Ryan, 2024). Our magnitudes are comparable to Piraino and Ryan (2024): for a similarly modest personal cost, roughly half to two-thirds of those who redistribute when it is free continue to do so once redistribution becomes costly. Our contribution is to show that this willingness to act on fairness views is itself malleable. This speaks directly to the recent call to distinguish the fairness ideal individuals hold from the behavioral weight they place on that ideal when acting on it entails personal costs (Fehr and Wu, 2026). To our knowledge, we are the first to empirically examine how this willingness changes due to interventions and to explore its magnitude in a setting featuring unequal opportunities.

Second, we contribute to the literature focusing on how experiences shape meritocratic beliefs (Amasino et al., 2024; Cassar and Klein, 2019; Deffains et al., 2016; Espinosa et al., 2020; Fehr and Vollmann, 2025; Jia, 2022). Previous findings have focused on self-serving biases and motivated reasoning that arise from personal experience of success or failure. Our study departs from this research by isolating the effect of imperfect beliefs. We show that even with no self-interested motives in play, experiencing disadvantageous opportunities can change beliefs and drive people towards more redistributive decisions.

Finally, we connect to the literature on experience effects and belief formation (Kuchler and Zafar, 2019; Malmendier and Nagel, 2016; Malmendier, 2021a; Malmendier and Wachter, 2024). We extend this literature to redistributive judgments, showing that firsthand experience with unequal opportunities affects how individuals interpret performance differences and the allocations they consider fair. Moreover, our comparison between experiences and information provision relates to Graeber et al. (2024), who document that qualitative information is more impactful than similar quantitative information. We replicate that finding and extend it by showing that direct personal experience is even stronger than both, consistent with evidence on the neural mechanisms that underlie learning (Chen et al., 2026).

The paper is structured as follows. Section 2 presents the experimental design and procedure. Sections 3, 4, and 5 present the results of our study, analyzing treatment differences in beliefs and behaviors. Section 6 discusses the main findings, relates them to the literature, and Section 7 concludes.

## 2 Experimental design

We adopt the spectator design by Cappelen et al. (2007), which comprises an Agent and a Spectator Phase. This design takes place in two phases. In the first, two participants, called 'agents', perform a certain task and are assigned initial rewards based on their performance. In the second, a third subject, called 'spectator', is informed about the agents' task and initial reward allocation, then given an opportunity to reallocate the earnings between the two agents. The primary variable of interest is the spectator's redistribution decision. We now discuss both phases in detail.[1]

[1] We implement the experiment using the oTree platform (Chen et al., 2016).

## 2.1 Agents

Agents complete an effort task with different levels of difficulty (as in Deffains et al., 2016; Fehr and Vollmann, 2025). The effort task consists of a counting task (as introduced in Abeler et al., 2011). Agents count how many times a character appears in a table like the one below (see Figure 1). After receiving general instructions, we assign each agent to perform either an easy version of the task or a hard version of the task. In the instructions shown to participants, we refer to the two task versions only as Version A and Version B, rather than as the easy and hard versions. For the easy version, we assign agents to count letters As. For the hard version, we assign agents to count letters Bs. We make counting Bs relatively harder by both including decoys that can be confused with Bs and slightly increasing the size of As.[2] This feature of our task is not immediately apparent, mimicking the hidden nature of structural barriers and privileges in society.

**Figure 1: Counting Task Example**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| B | B | β | α | α | A | β | I | 8 | I |
| I | I | I | 8 | α | A | A | α | 8 | I |
| B | I | ß | β | A | ß | β | β | A | α |
| I | β | 8 | α | 8 | I | β | ß | I | α |
| 8 | β | α | α | β | β | 8 | B | A | ß |
| β | 8 | β | 8 | β | 8 | β | I | β | α |
| 8 | B | β | ß | 8 | A | I | β | A | 8 |
| ß | B | A | A | A | β | ß | 8 | A | I |
| A | β | 8 | A | I | A | B | A | A | 8 |
| β | α | B | I | ß | B | I | A | A | 8 |

*Notes:* This figure depicts an example of the table used in the counting task. The task is to count how often a specific character appears in the table. The goal is to solve as many tables as possible: the final score is the total number of tables solved correctly. Each table is filled with characters. The participants must count how many times a target character appears and submit the count. When the answer is correct, a new table appears with new characters in shuffled positions. There are two versions of the task. The type of tables are the same in both versions, only the target character differs. In Version A, the participant counts how many times uppercase "A" appears in the table (e.g., 17 in the example below). In Version B, the participant count how many times uppercase "B" appears in the table (e.g., 14 in the example below).

We inform agents that better performance can increase their chances of getting an additional payment. Agents have 4 minutes to complete as many tables as possible. Agents receive a fixed participation fee of

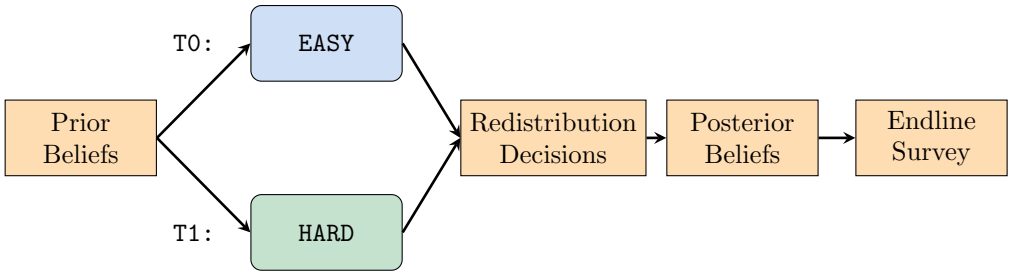
[2] We also impose a penalty for incorrect responses within a given table. After the first incorrect attempt, agents must wait 20 seconds before submitting again. After the second incorrect attempt, agents receive a new table. The purpose of these penalties is to discourage guessing. We expect the penalties to fall largely on agents assigned to the `HARD` version. This is indeed the case. In Version A, participants complete correctly 86% of the tasks they attempted, in Version B 49%.

$1.3 and can receive a bonus payment determined by the spectators from Section 2.2 of max $6.

## 2.2 Spectators

The spectator phase consists of three main parts. In the first (experience) part, spectators work on effort tasks. In the second part, participants act as third party spectators: we elicit their beliefs about agents, and decide how to redistribute the agents' income. In the third phase, participants answer socio-demographics questions. Figure 2 shows the complete experiment flow. See Table A2 for the full set of questions included.

**Figure 2: Experiment Flow**




*Notes:* This figure depicts the experiment flow for the Spectator phase.

First, we explain spectators the same counting task as the one completed by the agents (see Figure 1). The versions of the tasks are always referred as version A (i.e., the easy version) and version B (i.e., the hard version). The explanation does not disclose which version of the task each spectator will practice with. We require all spectators to respond correctly to a set of comprehension questions before advancing.

Next, we ask spectators three questions to measure their prior beliefs. We focus on perceived differences between the versions of the task and on determinants of success in a competition where individuals are assigned two different versions of the task, asking spectators to rank determinants (effort, ability, version of the task) from most to least important.[3]

**Experience.** We inform spectators they will have a chance to get familiar with the task. We refer to this part as the experience. We neither mention nor implement performance incentives. Nevertheless, we invite spectators to try to solve as many tables as possible within 4 minutes (i.e. the same time given to the agents). Because spectators' performance during the experience does not affect their own earnings, the design limits incentives to justify a favorable personal outcome and the scope for self-serving beliefs. We randomly assign spectators either to practice with the easy version or with the hard version. This is the main intervention of our study and creates two treatment groups: `EASY` and `HARD`.

We include an attention check in the middle of the experience. We record both the answer submitted and the time taken to answer to it. We also ask spectators to report the version of the task they practiced with at the end of the experience. After the experience is over, we inform spectators that they get a $1.00

[3] The exact wording of these questions can be found in Table A2.

bonus payment to reward their effort so far.

**Redistribution Decisions.** After the experience, we introduce spectators to agents from Section 2.1. We explain that agents were randomly assigned to either the easy or hard version of the task. To ensure correct understanding, we require correct response of three comprehension questions before proceeding. We elicit spectators' beliefs about agents at group-level (i.e., not about a specific pair of agents).[4] We then present a pair of agents. The pair is composed of one agent assigned to the easy version of the task and one assigned to the hard version. A \$6.00 bonus payment is initially assigned to the agent within the pair that completed more correct tasks. We create pairs in such a way that the agent that completed the easy version is always assigned the initial bonus. We ask spectators to decide whether and how much of the initial bonus to redistribute. We include three comprehension questions and an attention check before the first redistribution decision. We show spectators the conditions to which each agent was randomly assigned, which agent performed better (without providing the detailed information about the performances) and the initial bonus distribution.[5]

The decision about the redistribution for the pair of agents is our main outcome. It is followed by an open text form, in which we ask spectators to explain their reasoning. We then introduce a second redistribution decision for a new pair of agents. The structure is identical to the first decision: each agent completed a different version of the task, and the bonus is initially assigned to the best-performing agent. The only difference is that changing the initial allocation now entails a private cost for the spectator. Specifically, spectators must pay a fixed cost of \$0.20 (deducted from the \$1.00 bonus received earlier) if they choose any positive redistribution.[6] If spectators maintain the initial allocation unchanged, they incur no cost. The screen automatically displays the spectator's resulting bonus while they make the decision. This second decision allows us to distinguish what spectators consider fair from whether they are willing to act on that fairness view when doing so is personally costly.

**Posterior Beliefs.** We elicit posterior beliefs about the task. We use questions that mirror the prior beliefs elicited before the experience. Specifically, we reiterate questions on perceived differences between the versions of the task and on determinants of success in a competition where individuals are assigned two different versions of the task. The first question is identical to the one used for eliciting prior beliefs, while the second uses a different elicitation mode to mitigate experiment demand effects and consistency

[4] We ask: (1) the average number of tasks completed by agents assigned to the easy version; (2) the average number of tasks completed by agents assigned to the hard version; and (3) the likelihood that an agent assigned to the easy version outperforms an agent assigned to the hard version. For each belief, we also ask the perceived confidence. We inform spectators that one of the three estimates will be compared to the results of our agent's study, and that they will get a bonus if their estimate is exactly correct.

[5] These decisions are probabilistically consequential, as we inform the spectators that we will randomly select 1 out of every 10 participants, and, for that person, we will randomly implement one of their two decisions for a pair of real agents.

[6] This follows a similar cost structure to that implemented in Cappelen et al. (2023), where spectators need to pay to compensate a third party. The costless decision is always elicited first because it replicates the standard spectator game framework and provides a clean measure of the spectator's fairness view before private costs are introduced. This fixed ordering could induce consistency or status-quo effects, potentially increasing the absolute rate of continued redistribution. Since the order is identical across treatment arms, however, such effects should not mechanically explain treatment differences in costly redistribution.

biases, as suggested by (Haaland et al., 2023).

We also ask beliefs about agents at group- and individual-level (i.e., about the specific pair of agents for which the spectators previously decided for)[7] , including counterfactual beliefs about who would perform better if unequal task assignment were removed. These questions capture whether spectators attribute the observed performance difference to merit or to unequal opportunities. Finally, we ask spectators how fair they consider the initial allocation of the full bonus to the higher-performing agent.

**Endline Survey.** The endline survey is composed of the elicitation of experienced emotions, additional and non-consequential redistribution decisions, and socio-demographic questions.

First, we focus on the ability to imagine the agents' experience and on spectators' emotions (anger, compassion, sadness, indifference, irritation, elicited on a 4-point Likert scale) when learning about the initial situation of the two agents. Second, we ask spectators to make four hypothetical redistribution decisions. First, for a pair of agents in which both completed the same version of the task and the bonus is initially assigned to the best performer. Second, for a pair of agents in which both participated in a coin toss and the bonus is initially assigned to the winner of the coin toss. These two settings involve pure merit and pure luck, respectively, and are often used in the literature (see e.g., Almås et al., 2025; Harrs and Sterba, 2023). Third, for a pair of agents that completed different versions of the task (i.e., one completed version A and the other completed version B) in which the one completing version B performed better and was initially assigned the bonus. Fourth, for a pair of agents that completed a different task under unequal opportunities: answering questions after reading learning materials of either low or high relevance (as in Dong et al., 2024). After this, we ask spectators to self-report the fairness views they follow out of a close list: libertarian, egalitarian, non-compensating meritocrats, compensating meritocrats, and others. Finally, we collect socio-demographic information: sex, age, race, state of residence, education, dyslexia, employment status, personal and household gross annual income, political affiliation, political leaning in social and economic issues (see Annex G for more details).

## 2.3 Implementation

We collected data in November and December 2025. A total of 811 individuals were recruited via *Prolific*, with 160 agents (i.e., 80 pairs) and 651 spectators (336 in `EASY` and 315 in `HARD`).[8]

[7] We try to differentiate between beliefs about the group-level and the individual-level. It should be noted that our study is mostly suited to capture the former, as we do not provide much information about the specific pair agents.

[8] Most participants completed the entire survey (91.1% in `EASY` and 85.9% in `HARD`). The lower completion share in `HARD` than in `EASY` could reflect differential attrition. Most dropouts are in the first screen (7.3% in `EASY` and 9.5% in `HARD`), when participants are still not aware of the treatment assignment. We find a slightly higher attrition in participants in `HARD` immediately after the experience (0.8% in `EASY` and 3.5% in `HARD`), and no difference towards the end of the survey (0.8% in `EASY` and 0.8% in `HARD`). The fact that the hard task is more difficult may cause frustrated participants to abandon the study. Since demographic characteristics are collected at the end of the experiment, we cannot directly assess whether the differential attrition is selective. Among participants who completed the experiment, however, the treatment arms remain balanced on almost all observed demographic characteristics. This provides no indication of strong differential selection on observed characteristics, although it cannot rule out selection on unobserved characteristics such as patience or tolerance for frustration. We believe that selective attrition could, if anything, attenuate our estimate. Participants who are most frustrated or negatively affected by the hard task may be both more likely to abandon the study and, had they remained, more likely to recognize the opportunity gap and compensate the disadvantaged agent.

Agents received \$1.30 as a flat payment, plus the share allocated to them by the spectators in the redistribution task, which averaged to a total payment of \$4.30 per worker. Spectators received a flat payment of \$3.90, plus a bonus of \$1.30 during the experiment. We removed 1 spectator from the sample due to failing the attention check, as preregistered.[9] We also take additional measures to limit bot participation and the use of AI tools. [10]

The median completion time was about 24 minutes. Tables A3 and A4 in the Annex report spectator demographics and socio-economic characteristics by treatment arm. Most pre-randomization characteristics are balanced across treatment arms, although individual income differs across `EASY` and `HARD`; all main specifications therefore control for baseline characteristics.[11]

## 2.4 Hypotheses

We adapt the model from Cappelen et al. (2013). We consider a spectator who observes a pair of agents. We focus on the transfer towards an Agent B, who completed a harder version of the task and is initially assigned no earnings. The spectator decides on a distribution $(1-y, y)$ in treatment $T$, where $y$ is the share to Agent B and $1-y$ is the share to Agent A.

The spectator derives utility from implementing what they perceive as a fair allocation, but may face a fixed cost when doing so:

$$V(y, \cdot) = -\frac{\beta(\cdot)}{2}\big(y - m(\cdot)\big)^2 - c(k)\,\mathbb{1}\{y > 0\}, \tag{1}$$

where $m(\cdot) \geq 0$ is what the spectator considers to be the fair transfer to Agent B; $\beta(\cdot) \geq 0$ is the weight attached to fairness relative to spectator's own income; and $c(k)$ is the cost on spectator of the redistribution.

We distinguish two environments $k$ that affect the cost of redistribution ($c(k)$). In the *costless* environment, redistribution has no cost on the spectator ($c = 0$). Then, the spectator optimizes by implementing what they consider to be fair ($y^* = m$). This is the traditional environment explored in the fairness literature (Cappelen et al., 2007), and allows eliciting fairness views. In the *costly* environment, redistribution has a cost on spectator ($c > 0$). We model the cost as independent of the amount transferred ($c\,\mathbb{1}\{y > 0\}$), as done by recent work (Dwenger et al., 2024; Dimant et al., 2024; Piraino and Ryan, 2024). This captures settings in which taking redistributive action requires incurring a fixed burden, such as making an appeal,

[9] The pre-analysis plan specified excluding participants who met at least three of six criteria: failing the mid-experience attention check; taking more than 40 seconds to answer that check; failing the post-experience attention check; taking more than 40 seconds to answer that check; failing to make two attempts during the effort task; and failing the pre-redistribution attention check.

[10] Specifically, the survey includes a CAPTCHA check before participants enter the study. In addition, copy-pasting is disabled in the open-text response boxes, which makes it harder for participants to directly paste AI-generated text into the survey.

[11] Covariates include sex, age, race, education, employment, individual and household income, and political party. To ensure that the results are not driven by differences in individual income, we reweight `EASY` and `HARD` to a common income distribution and find that the estimate remains unchanged. Moreover, income is balanced across treatment arms when grouped into low-, middle-, and high-income categories, suggesting that the imbalance arises from the fine-grained income classification rather than from a broad income gradient.

filing a claim, intervening in a dispute, or publicly challenging an allocation. In these settings, the main cost is not the marginal amount transferred, but the act of intervening itself. Conditional on paying the fixed cost, the optimal transfer is $y = m(\cdot)$, which minimizes the distance from the spectator's fairness benchmark. The spectator therefore decides whether to leave the initial allocation unchanged or to pay the cost and implement the fair allocation. Redistribution occurs iff the utility gain from moving from $y = 0$ to $y = m(\cdot)$ exceeds the fixed cost:

$$\frac{\beta(\cdot)}{2} m(\cdot)^2 > c(k). \tag{2}$$

The spectator who considers Agent B deserves higher incomes ($m$) or has a stronger concern for fairness ($\beta$) is more willing to redistribute. The two parameters $m$ and $\beta$ play distinct roles. $m$ governs how much the spectator redistributes when redistribution is costless. $\beta$ governs whether the spectator is willing to pay to act on that view when redistribution is costly. This distinction motivates two separate hypotheses.

**Hypothesis 1: Spectators who experience the disadvantaged task consider it fair to transfer more to Agent B than spectators who experience the advantaged task.** Our first hypothesis is that $m(\texttt{HARD}) > m(\texttt{EASY})$. Since the costless decision directly reveals $y^* = m$, this hypothesis is tested by comparing costless redistribution across treatment arms:

$$\mathbb{E}\big(y \mid T = \texttt{HARD},\, k = \text{costless}\big) > \mathbb{E}\big(y \mid T = \texttt{EASY},\, k = \text{costless}\big). \tag{3}$$

**Hypothesis 2: Spectators who experience the disadvantaged task, compared to spectators who experience the advantaged task, place greater weight on implementing what they consider fair** ($\beta$)**.** Our second hypothesis is that $\beta(\texttt{HARD}) > \beta(\texttt{EASY})$.

Testing this hypothesis requires separating $\beta(\cdot)$ from $m(\cdot)$. Since the costless decision reveals the spectator's preferred transfer, we define spectator $i$'s observed fairness view as $\hat{m}_i = y_i^0$, where $y_i^0$ is the transfer to Agent B when redistribution is costless. Let $y_i^1$ denote the transfer to Agent B when redistribution is costly. Costly redistribution occurs if

$$\beta(\cdot) > \frac{2c}{\hat{m}_i^2}. \tag{4}$$

Starting from this, we use two complementary empirical approaches. First, we estimate treatment-specific values of $\beta(\cdot)$ directly by maximum likelihood. Adding a logistic error term to the threshold rule, for each treatment $t$ we estimate

$$\Pr(y_i^1 > 0 \mid T = t) = \Lambda\left(\beta(t) - \frac{2c}{\hat{m}_i^2}\right), \tag{5}$$

where $\Lambda(\cdot)$ is the logistic cumulative distribution function. Higher values of $\beta(t)$ indicate greater willingness to sustain redistribution when costly, conditional on what the spectator considered fair when redistribution was free.

Second, the same threshold condition yields a reduced-form implication. Under the null hypothesis that $\beta(\texttt{HARD}) = \beta(\texttt{EASY})$, treatment should not affect the probability of costly redistribution once $\hat{m}_i$ is held fixed. If instead $\beta(\texttt{HARD}) > \beta(\texttt{EASY})$, then

$$\Pr\big(y_i^1 > 0 \mid \hat{m}_i,\, T = \texttt{HARD}\big) > \Pr\big(y_i^1 > 0 \mid \hat{m}_i,\, T = \texttt{EASY}\big) \quad \forall\, \hat{m}_i > 0. \tag{6}$$

We test this implication by restricting the sample to spectators with $\hat{m}_i > 0$ and estimating whether treatment predicts staying in redistribution after conditioning on fixed effects for the exact costless transfer.

# 3 Results

## 3.1 Prior and Posterior Beliefs about Unequal Opportunities

We start by reporting how spectators perceive the environment prior to the intervention (see Panel A in Table 1). We find a high degree of misperception about the role of opportunities in explaining task success. Most spectators do not perceive one version to be more advantageous than the other and state that the two versions are approximately the same (mean=.862, SD=.345). In fact, 81.9% of the participants (SD=.386) rank the version of the task as the least important factor to determine success, stating that individual effort (mean=.252, SD=.434) and, especially, ability (mean=.664, SD=.473) are the most relevant.[12] These beliefs are similar across treatment arms (difference=.016 and p-value=.563 for the version similarity, see row 1 in Table 1; difference=-.006 and p-value=.854 for the importance of the version, see row 3 in Table 1).

We then focus on actual performance in task completion (see Panel B in Table 1). Participants assigned to the easy version outperform those assigned to the hard version, both in the agent (difference=9.01 and p-value<.001) and spectator sample (difference=7.79 and p-value<.001, see row 4 in Table 1). On average, spectators assigned to the easy version complete 10.53 tables (12.53 tables in the agent sample), while those assigned to the hard version complete 2.75 tables (3.52 tables in the agent sample). Figure B1 in the Annex shows completion by version assigned. 98.2% of spectators assigned to the easy version complete more tables than the average performer among those assigned to the hard version, which highlights how determinant the version assigned is.

Finally, we report beliefs elicited after the experience (see Panel C in Table 1). Spectators update their beliefs on the relevance of the version of the task assigned. The share of participants who consider both

[12] It should be noted that these statements are collected following a detailed explanation about the task (including an example image), and after answering three comprehension questions referring to the goal and scoring of the task.

**Table 1: Beliefs and Task Performance**

| | Total | EASY | HARD | Diff. |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| **Panel A. Prior Beliefs** | | | | |
| Versions are similar | .862 | .854 | .870 | .016 |
| | | | | (.027) |
| Easy version is better | .109 | .107 | .111 | .004 |
| | | | | (.024) |
| Version ranked least important | .819 | .821 | .816 | -.006 |
| | | | | (.030) |
| **Panel B. Task Performance** | | | | |
| Tables completed | 6.77 | 10.53 | 2.75 | -7.79*** |
| | | | | (.28) |
| **Panel C. Posterior Beliefs** | | | | |
| Versions are similar | .562 | .637 | .483 | -.154*** |
| | | | | (.039) |
| Easy version is better | .415 | .345 | .489 | .144*** |
| | | | | (.038) |
| Version weighs less than ability and effort | .604 | .690 | .511 | -.179*** |
| | | | | (.038) |

*Notes:* This table reports spectators' beliefs about the two versions of the task and their performance during the experience. Column (1) refers to the whole sample; columns (2) and (3) to spectators assigned to the easy (EASY) and hard (HARD) version. Column (4) reports the coefficient on HARD from a linear regression of each outcome on a treatment indicator, with robust standard errors in parentheses. Panel A reports beliefs elicited before the experience and Panel C beliefs elicited after the redistribution decisions. All belief rows are shares. "Versions are similar" and "Easy version is better" are based on the same question in both panels, asking on a 1–7 scale which version of the task provides an advantage for performance. "Versions are similar" equals one if the spectator chooses the midpoint (no advantage); "Easy version is better" equals one if the spectator chooses any value indicating an advantage for the easy version. The remaining category, an advantage for the hard version, is chosen by fewer than 3% of spectators and is omitted. In Panel A, "Version ranked least important" equals one if the spectator ranks the version of the task last among ability, effort, and version as determinants of success. In Panel C, "Version weighs less than ability and effort" equals one if the spectator assigns less than 50 out of 100 points to version when dividing the explanation of performance between version on the one hand and ability and effort on the other. "Tables completed" is the number of tables solved correctly during the four-minute experience. * $p < .10$, ** $p < .05$, *** $p < .01$.

versions are similar drops to 63.7% of those assigned to EASY (difference with prior beliefs=-.217 and p-value<.001) and 48.3% of those assigned to HARD (difference with prior beliefs=-.387 and p-value<.001). The difference in posterior beliefs between treatment arms is -15.4 p.p. and is statistically significant (p-value<.001, see row 5 in Table 1).[13]

After the experience, spectators state that the easy version yields an advantage (mean=.415 and SD=.493), and this is especially pronounced among those assigned to HARD (difference=-.144 and p-value<.001, see row 6 in Table 1). This change is also reflected in how spectators explain success in the task. After the experience, 69.0% of spectators in EASY and 51.1% in HARD consider luck less important than ability and effort combined (difference=$-.179$, $p < .001$, see row 7 in Table 1).

Table B1 in the Annex disentangles how the beliefs we collected only once differ between treatment arms.

[13] We show the pattern in these prior and posterior beliefs remains stable across several socio-demographic characteristics in the Annex. Table B2 shows prior beliefs on the similarity of the versions of the task are equal in both treatment arms across subgroups determined by socio-demographic characteristics. Table B3 shows how the effect of the intervention is similar across those subgroups. It should be noted that although the statistical significance varies, the treatment effect has the same sign for every subgroup. That is, in all subgroups exposure to the hard version of the task yields less perceptions that the version of the tasks are similar.

Around 40% of participants believe that the agent assigned to the easy version would win if assigned the same version than the other agent. Given that participants are aware that version assignment is random, we take this as evidence of existence of attribution bias in our sample. Almost 35% think the same for any agent assigned to the easy version. Both values are correlated ($\rho$=.319 and p-value<.001), suggesting that participants have attribution bias both in the individual and group-level. We find that spectators in HARD are 20 p.p. less likely to believe this at the individual-level (p-value<.001, see row 3 in Table B1 in the Annex) and 13 p.p. at the group-level (p-value<.001, see row 4 in Table B1 in the Annex). We also show that spectators are more accurate at predicting performance for the task they are assigned to (see row 7 and 8 in Table B1 in the Annex).[14] We take it as evidence that spectators learn specifically about the version of the task they are assigned to and that they extrapolate that knowledge to other versions of the task.

## 3.2 Perceived fair allocation: Test of Hypothesis 1

Figure 3 focuses on the redistribution decisions. Subfigure (a) shows histograms of the transfers to the disadvantaged agent by treatment arm. Overall, around half of the spectators assign either nothing to the disadvantaged agent (27.0%) or equalize earning across agents (26.6%). A further 43.9% make a positive transfer but still leave the advantaged agent with a larger share, while only 2.5% give the disadvantaged agent more than the advantaged agent. The distribution of transfers shifts to the right in HARD: spectators assigned to EASY are more likely to choose low transfers to the disadvantaged agent, while spectators assigned to HARD are more likely to choose transfers above $2.00.

Subfigure (b) compares mean transfers across treatment arms. Spectators in EASY assign $1.53 to the disadvantaged agent who completed the hard version and was outperformed by the agent who completed the easy version. Spectators in HARD assign on average $1.80 to the disadvantaged agent, which is 18% higher than to the EASY group (difference=.27 and p-value=.004).

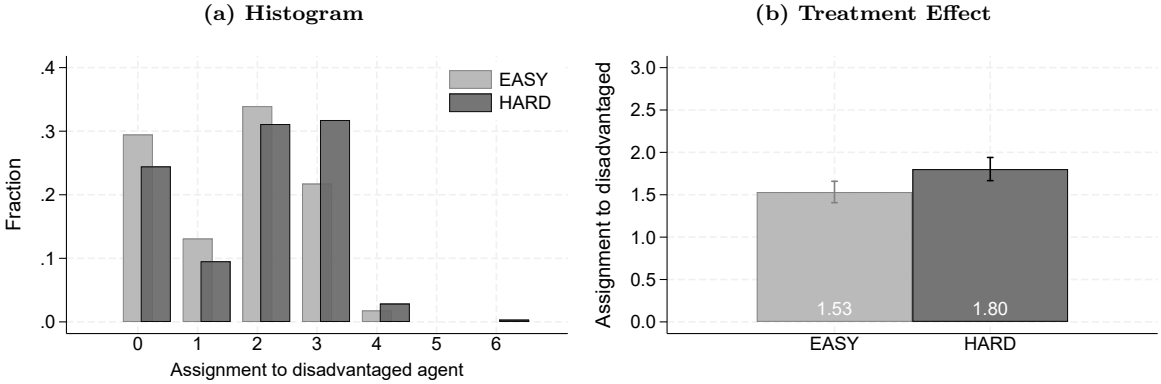
Table 2 reports the treatment effects of our intervention. Column 1 shows the estimated treatment effect in the redistribution decision. Column 2 introduces socio-demographics covariates, with negligible changes in the estimates: treated spectators assign 18% more to the disadvantaged agent (difference=.27 and p-value=.007). Panel A of Table 5 shows these results alongside a summary of the treatment effects for all interventions in our study.

### 3.2.1 Heterogeneities and Robustness Checks

We explore whether differences in redistribution decisions differ across socio-demographic characteristics. Table B4 in the Annex shows treatment effects for different subgroups by sex, age, race, educational attainment, employment status, income, and political affiliation. The treatment effect has the same sign

[14] Remember that on average agents completed 12.53 tables in EASY and 3.52 tables in HARD. For the easy version, spectators in EASY predict 9.55 tables and those in HARD predict 5.24 tables (difference=-4.30 and p-value<.001, see row 7 in Table B1 in the Annex). For the hard version, spectators in EASY predict 8.41 tables and those in HARD predict 4.41 tables (difference=4.00 and p-value<.001, see row 8 in Table B1 in the Annex).

## Figure 3: Redistribution Decisions




*Notes:* These figures show redistribution decisions in a spectator game. Subfigure (a) plots an histogram of the bonus transfer (out of \$6.00) to the disadvantaged agent. Each bar plots the fraction of spectators assigning a discrete value. Subfigure (b) plots the mean bonus transfer (out of \$6.00) to the disadvantaged agent. The light bars plot decisions among participants in `EASY`. The dark bars plot decisions among participants in `HARD`. The error spikes correspond to the 95% confidence interval as estimated via ordinary least squares regression with robust standard errors.

## Table 2: Redistribution Decisions and Perceived Unfairness

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| **Panel A. Total Sample** | | | | |
| | Redistribution | | Unfairness | |
| `HARD` | .270*** | .270*** | .077*** | .060** |
| | (.095) | (.100) | (.025) | (.025) |
| | [.004] | [.007] | [.002] | [.018] |
| Covariates | No | Yes | No | Yes |
| Effect magn. | 18% | 18% | 25% | 19% |
| Observations | 651 | 651 | 651 | 651 |
| $R^2$ | .012 | .071 | .015 | .115 |
| **Panel B. By Stated Fairness View** | | | | |
| | Redistribution | | | |
| Subsample: | Meritocrat | | Non-Meritocrat | |
| `HARD` | .307*** | .337*** | .123 | .054 |
| | (.114) | (.121) | (.155) | (.175) |
| | [.007] | [.006] | [.428] | [.758] |
| Covariates | No | Yes | No | Yes |
| Effect magn. | 23% | 26% | 6% | 3% |
| Observations | 447 | 447 | 204 | 204 |
| $R^2$ | .016 | .115 | .003 | .177 |

*Notes:* This table reports treatment effects on redistribution decisions and the stated unfairness of the initial allocation. Panel A reports treatment effects for the full sample. Columns (1) and (2) report redistribution decisions, measured as the dollar transfer to the disadvantaged agent. Columns (3) and (4) report perceived unfairness of the initial allocation, measured on a scale from 0 to 1. Panel B reports treatment effects on redistribution separately by self-reported fairness view. Columns (1) and (2) refer to meritocrats, while columns (3) and (4) refer to non-meritocrats, defined as egalitarians, libertarians, and others. The independent variable is the spectator's assignment to `HARD`. All estimates are from linear regressions. In both panels, columns (2) and (4) include socio-demographic covariates: sex, age, race indicators, education, employment status, individual and household income, and political party affiliation. Effect magnitude reports the treatment coefficient as a percentage of the mean outcome in `EASY` within the corresponding estimation sample. Robust standard errors are reported in parentheses. p-values are reported in square brackets.

for every subgroup. Exposure to the hard version of the task is associated with higher redistribution in every subgroup, though the statistical significance varies (partly due to statistical power).

To check the robustness of our results, we contrast redistribution decisions with stated fairness of the initial allocation. We ask spectators to rate the fairness of the situation in a 4-point Likert scale and we convert the values into a measure of intensity of unfairness that ranges from 0 to 1. We find that spectator in HARD perceive the initial allocation as 19% less fair than spectators in EASY (p-value=.018, see column 4 of Table 2), which is in line with their higher redistribution.

We also check whether our main findings reflect behavior due to meritocratic views. For doing so, we use self-reported fairness views that we collect in the endline survey. A majority of participants are meritocrats (68.7%). Columns 1 to 4 in panel B of Table 2 show treatment effects for the subsample of meritocrats and non-meritocrats. The treatment effect is concentrated among meritocrats, redistribution is 26.0% higher in HARD than in EASY (p-value=.006, see column 2 of Table 2). In turn, non-meritocrats show no difference in their decisions (p-value=.758, see column 4 of Table 2).

We further analyze the effects of our intervention for decisions under a different context, as a falsification test. We include a non-consequential redistribution decision that also features two agents under unequal opportunities, but differs in the type of task performed. Instead of completing the task featured in the experience intervention, the agents answer a multiple-choice questionnaire after receiving either low or high relevance learning materials.[15] We find a much subdued effect ($\beta$=.170 and p-value=.088, see column 4 of Table B5 in the Annex). These results suggest that experiencing unequal opportunities loses effectiveness when individuals decide outside the exact context of that experience, which aligns with previous findings on experience effects (e.g. see Malmendier, 2021a, for a review).

We further examine redistribution relative to spectators' own fairness judgments in a setting where inequality results entirely from luck. In the endline survey, spectators make a hypothetical redistribution decision in which the \$6 bonus is initially assigned by a coin toss.[16] On average, spectators transfer \$1.80 to the agent who lost the coin toss, and only 47% divide the bonus equally. These decisions do not differ significantly between EASY and HARD (p-value=.519). Comparing each spectator's redistribution under unequal opportunities with their allocation in the coin-toss setting, we find that average transfers fall short of this benchmark by \$0.26 in EASY and by \$0.06 in HARD (p-value=.087). The share of spectators who transfer less than their own coin-toss allocation falls from 37.2% in EASY to 29.2% in HARD (p-value=.030). Thus, while substantial inequality remains in both treatments, redistribution after experiencing disadvantage is closer to spectators' own allocation choices when outcomes are determined purely by luck.

### 3.2.2 Mechanism Exploration

We explore several channels that are consistent with the HARD-EASY difference in redistribution. Specifically, we examine whether treatment-induced differences in beliefs, fairness judgments, and emotional responses are related to the observed redistribution gap. As these measures are collected after treatment, this

[15] See Annex G for a complete transcription of the context and decision screen.

[16] This decision is hypothetical and elicited after the main redistribution decisions, so spectators may anchor on their earlier choices.

analysis is exploratory. Table 3 reports how the estimated treatment effect changes when we account separately for each of these potential mechanisms.

We start by examining the role of beliefs about the determinants of the initial allocation. We already documented that spectators in HARD perceive luck as more relevant for success in the effort task (see Table 1). When the posterior relevance of luck is added to the redistribution regression, the treatment effect falls by 71% and is no longer statistically significant at the 10% level (see column 3 in Table 3).[17] Differences in how spectators interpret the role of opportunities in performance thus account for an important part of the redistribution gap. Consistent with this, the perceived relative performance gap between the two versions, elicited before the redistribution decisions, is positively correlated with the posterior relevance of luck ($r = .38$ in EASY and $r = .46$ in HARD, both $p < .001$).[18] This suggests that posterior beliefs largely reflect perceptions spectators already held when making their redistribution decisions.

We also check whether our intervention has shifted what participant consider as fair. We start by exploiting two non-consequential redistribution decisions in contexts of equal opportunities (i.e., initial allocation due to luck and merit, respectively). As these contexts feature agents assigned exactly to the same tasks, there is no room for updated beliefs on the impact of opportunities to impact them. Differences can also reflect evolving perspectives on the merit of performance in the tasks. As mentioned before, we find no differences for decisions in contexts in which luck determines the initial allocation, but find a slight gap when merit is the reason (difference=.18 and p-value=.064). This result suggests that the intervention may lead spectators to place less weight on merit in the effort task, which can partly account for the difference in redistribution under unequal opportunities.[19] Consistent with this interpretation, controlling for the merit premium reduces the HARD--EASY redistribution coefficient by 43% (see column 4 in Table 3).

Lastly, we explore whether explanations related to self-serving biases can account for the difference in redistribution decisions we report. This is particularly relevant, as studies focusing on other types of experiences find them to be important Deffains et al. (2016); Fehr and Vollmann (2025). One of the goals of our intervention is to remove room for motivated beliefs and focus on bounded belief formation instead. First, we assess if our intervention affects spectator's proximity to the agents, as this could make non-moral reasons to affect redistribution decisions. We focus on the disadvantaged agents and proxy this using spectators' self-reported understanding of the disadvantaged agent's experience. Following the intervention, spectators in HARD report to understand the disadvantaged agent 17% more than spectators in EASY (difference=.127 and p-value<.001). Including this measure leaves the HARD--EASY redistribution gap largely unchanged (see column 5 in Table 3). This provides little evidence that perceived proximity

[17] The same holds when transfers are measured relative to each spectator's own coin-toss allocation: once the relevance of luck is accounted for, the difference between EASY and HARD in the shortfall relative to this benchmark disappears.

[18] The perceived relative performance gap is the difference between spectators' estimates of the average number of tables solved by agents assigned to the easy and the hard version, divided by the mean of the two estimates.

[19] This might also explain why, in the additional redistribution decision in which the disadvantaged agent won, we find no treatment difference (1.51 and 1.62, p-value=.414). Spectators in HARD may be less willing to treat performance differences as a sufficient basis for entitlement, even when the higher performer is the disadvantaged agent.

to the disadvantaged agent is closely associated with the treatment difference in redistribution. Second, we examine whether emotional responses to the agents' initial situation are related to our main findings. Table B6 in the Annex shows that our intervention generated negative emotional responses in spectators. In particular, spectators assigned to the hard version of the task reported more anger (98% above the mean of the `EASY` group and p-value=.001) and irritation (88% above the mean of the `EASY` group and p-value<.001). We test whether each of these responses is associated with the different redistribution decisions by treatment arm (see Table B7 in the Annex for each emotion collected). Including negative emotional responses attenuates the treatment difference by at most 40%, with irritation being most relevant in our sample (see column 6 in Table 3).[20]

The results point most strongly to changes in how spectators interpret the role of opportunities and merit in performance. Emotional responses may also accompany these changes, whereas perceived proximity to the disadvantaged agent appears less closely related to the redistribution gap.

**Table 3: Mechanism Exploration**

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| `HARD` | .270*** | .270*** | .078 | .154* | .303*** | .167* |
| | (.095) | (.100) | (.097) | (.083) | (.105) | (.099) |
| Relevance of Luck | | | 1.554*** | | | |
| | | | (.170) | | | |
| Assignment when Equal Opp. | | | | .581*** | | |
| | | | | (.040) | | |
| Understanding of Agent | | | | | -.183 | |
| | | | | | (.189) | |
| Irritation | | | | | | 1.610*** |
| | | | | | | (.227) |
| Covariates | No | Yes | Yes | Yes | Yes | Yes |
| Reduction in TE | - | 0% | -71% | -43% | 12% | -38% |
| p-value | .004 | .007 | .422 | .064 | .004 | .092 |
| Observations | 651 | 651 | 651 | 651 | 651 | 651 |
| $R^2$ | .012 | .071 | .194 | .403 | .072 | .137 |

*Notes:* This table reports treatment effects on redistribution decisions. The dependent variable is the dollar transfer to the disadvantaged agent. The independent variable is spectator's assignment to `HARD`. Each row reports estimates from a linear model. Column (1) include no covariates. Columns (2) to (6) include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

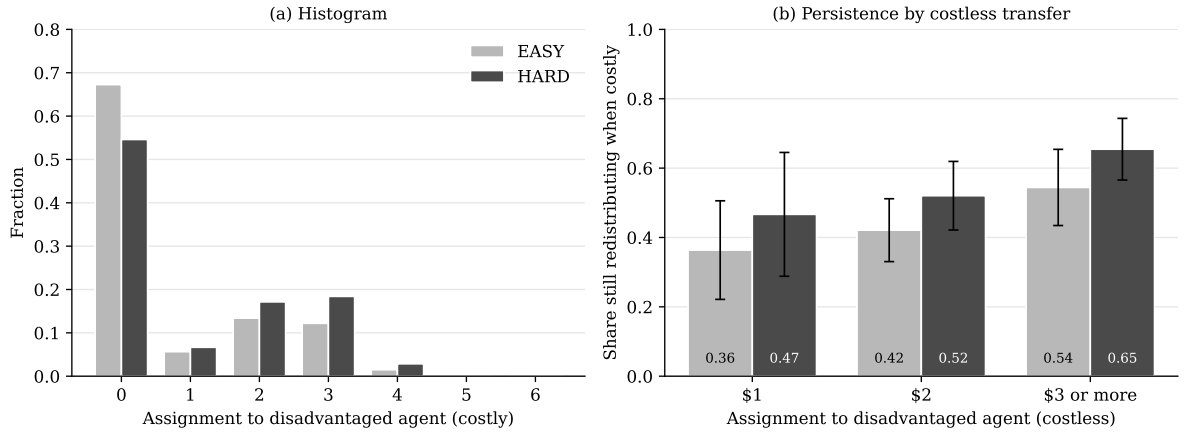
## 3.3 Willingness to implement fairness: Test of Hypothesis 2

Our main finding relates to a costless redistribution decision, in line with common practice (see Cappelen et al., 2020, for a review of the literature). We further assess a similar decision that entails a cost for the decision-maker. Panel B of Table 5 summarizes the treatment effects for this costly decision, and Figure 4 shows how the cost reshapes behavior. Mean transfers to the disadvantaged agent fall relative to

[20] Although spectators were asked to report the emotions they felt when considering the agents' initial situation, these measures may also capture residual affect generated by the task itself, particularly frustration among spectators assigned to `HARD`.

the costless decision (from \$1.53 to \$0.75 in EASY), mainly because more spectators leave the allocation unchanged: 67.3% in EASY and 54.6% in HARD transfer nothing, while the distribution of positive transfers keeps a similar shape (Panel a). Most spectators who keep redistributing transfer the same amount as when redistribution was costless (see footnote 21). As a result, spectators in HARD assign 47% more to the disadvantaged agent than those in EASY (difference=.36, $p < .001$, see column 2 in Panel B of Table 5). Persistence increases with the amount spectators consider fair, as the threshold model predicts: among those who transferred \$1 when redistribution was costless, 36.4% in EASY and 46.7% in HARD keep redistributing, compared with 54.4% and 65.5% among those who transferred \$3 or more (Panel b). Crucially, spectators in HARD are more likely to persist at every level of the costless transfer. The treatment effect on costly redistribution therefore does not arise only because spectators in HARD want to transfer more, but also because they are more willing to act on a given fairness view.

**Figure 4: Costly Redistribution Decisions**




*Notes:* Panel (a) plots the distribution of transfers (out of \$6) to the disadvantaged agent when redistribution costs the spectator \$0.20. Panel (b) restricts the sample to spectators who make a positive transfer when redistribution is costless and plots the share who still make a positive transfer when it is costly, by the amount transferred in the costless decision. The light bars plot decisions in EASY; the dark bars in HARD. Error bars are 95% confidence intervals.

Comparing decisions with and without a cost, we assess whether our intervention affects spectators' willingness to implement the distribution they consider to be fair.[21] We proceed in two complementary ways. First, we estimate the fairness weight parameter ($\beta$) directly using a threshold model. Second, we estimate a reduced-form regression that compares spectators in EASY and HARD with similar views on what is fair to transfer to the disadvantaged agent ($\hat{m}$).

Table 4 reports the results for both approaches. Panel A shows that the fairness weight is close to zero and not statistically significant in EASY (p-value=.651, see column 1), while it is positive and statistically significant in HARD (p-value=.001, see column 2 in Panel A of Table 4). The difference between the two estimates is statistically significant. Table 6 shows these results alongside a summary for all interventions in our study. Panel B shows the corresponding reduced-form result. Conditional on what spectators consider to be fair, those in HARD are 9.0 p.p. more likely to continue redistributing when a cost is

[21] Our analysis rests on the assumption that redistributive decisions in a costless environment reflect what spectators consider to be fair. The fact that 77.9% of spectators who pay implement exactly the same income distribution as implemented at no cost lends support to the assumption. Allowing for minor deviations of \$1, the share of broadly consistent decisions rises to 95.3%.

introduced (p-value=.070, see column 2 in Panel B of Table 4). Together, the two approaches show that spectators in HARD are more willing to redistribute at a personal cost than those in EASY.

**Table 4: Willingness to Redistribute**

| **Panel A. Structural Threshold Model** | | | **Panel B. Reduced-form Model** | | |
|---|---|---|---|---|---|
| | EASY | HARD | | (1) | (2) |
| $\hat{\beta}$ | -.059 | .417*** | HARD | .103** | .089* |
| | (.131) | (.131) | | (.046) | (.049) |
| $\hat{\beta}(\texttt{HARD}) - \hat{\beta}(\texttt{EASY})$ | .476** | | Dep. var. mean | .514 | .514 |
| | (.185) | | $\hat{m}_i$ | Yes | Yes |
| Observations | 237 | 238 | Covariates | No | Yes |
| Stay-in rate | .451 | .576 | Observations | 475 | 475 |
| Log likelihood | -162.084 | -161.098 | $R^2$ | .037 | .134 |

*Notes:* Panel A reports treatment-specific estimates of the fairness-weight parameter from the threshold model estimated separately by treatment group. Stars on the $\hat{\beta}(T)$ row indicate whether the treatment-specific estimate differs from zero. Stars on the difference row indicate statistical significance of the test $\hat{\beta}(\texttt{HARD}) = \hat{\beta}(\texttt{EASY})$. Panel B reports linear probability model estimates of staying in redistribution, among spectators that redistribute at no cost. The dependent variable equals one if the spectator chose a positive transfer in both the costless and costly decisions. The sample contains only spectators that redistribute under no cost. Panel B specifications control for the exact amount redistributed in the costless decision. Covariates include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

### 3.3.1 Mechanism Exploration

Similar to the costless redistribution decision, beliefs about the determinants of success may account for a substantial share of the treatment effect on willingness to keep redistributing (see Table B8 in the Annex). In particular, controlling for the perceived relevance of luck produces the largest attenuation of the HARD coefficient (-37%, column 3). Emotional responses also matter: anger is strongly associated with staying in redistribution and explains a non-negligible share of the treatment difference (-30%, see column 6 in Table B9 in the Annex). By contrast, self-reported understanding of the disadvantaged agent and the equal-opportunity merit decision account for much less of the effect. Overall, the results suggest that experiencing disadvantage might affect willingness to pay for redistribution partly by changing how spectators interpret the source of the inequality, and partly by increasing the emotional force attached to correcting it.

# 4 Extension

## 4.1 Rationale

Our main experiment shows that spectators experiencing the hard version redistribute more than those assigned to the easy version. However, this comparison also raises a natural question: is the effect driven by experiencing disadvantage alone, or by understanding the opportunity gap between advantage and disadvantage? To address this question, we introduce a MIXED treatment arm. Unlike spectators in EASY or HARD, participants in MIXED experience both versions of the task. This gives them direct exposure to the opportunity gap, as they can compare the two versions from their own experience.

Our mechanism analysis further suggests that beliefs about the role of luck in determining performance play an important role in explaining the redistribution gap. This analysis is, however, exploratory, as beliefs are shaped by the experience itself rather than experimentally varied. To investigate this mechanism further, we examine whether providing information about the opportunity gap, without direct experience, can similarly change beliefs and redistribution. This approach is also closer to many real-world settings, where people rarely experience both sides of unequal opportunities. Instead, they learn about these inequalities through statistics, personal accounts, media stories, or policy communication. Information may reveal that opportunities matter, but it may be less effective at conveying what those opportunities feel like and how they shape performance. We therefore introduce two further treatment arms with different forms of information provision. In `REPORTS`, participants receive qualitative descriptions of agents' experiences, capturing a narrative form of information through others' accounts. In `STATISTICS`, participants receive quantitative information about performance differences across the two task versions, capturing a statistical form of information about the opportunity gap.

Together, these extensions help us better understand what makes experiencing unequal opportunities consequential for redistribution. By comparing different forms of experience and information provision, we explore whether recognizing the opportunity gap is sufficient to change redistributive behavior, or whether other features of experience, such as emotional engagement and perspective-taking, also matter.

## 4.2 Experimental Design

We add three additional treatment arms (`MIXED`, `REPORTS` and `STATISTICS`), for which the experimental flow mirrors that of the main experiment. The main difference is in the experience.[22] While `MIXED` consists of a different experience, `REPORTS` and `STATISTICS` replace the experience with information provision, one providing qualitative reports (`REPORTS`) and one providing quantitative statistics (`STATISTICS`).

**Mixed.** In the `MIXED` treatment arm, we expose participants to both versions of the task. Specifically, spectators spend two minutes working on the hard version and two minutes working on the easy version (in randomized order).

**Reports.** In the `REPORTS` treatment arm, we provide qualitative information about agents' experiences instead of requiring task completion. All information is provided on a single screen. At the top of the screen, we include a summary of the most recurring themes described by agents who completed Version A, followed by two representative quotes from these agents. Below that, we include a separate summary for agents who completed Version B, again accompanied by two representative quotes. See Figure F3 in the Annex for more details. All of this information is drawn from a pilot study.[23] To ensure understanding,

[22] Another relevant change is that we remove incentives for belief elicitations in the extension. The main reason for this change is that with information provision, the incentives become unnecessary.

[23] The reason for this is that, due to budget constraints, we did not collect qualitative experiences of the agents in the main study.

we require participants to correctly answer two comprehension questions about the information before proceeding.

**Statistics.** In the `STATISTICS` treatment arm, we provide quantitative information about agents' experiences instead of requiring task completion. The entire information is provided in one screen. For each type of agent, we detail the average number of tasks completed, the maximum and minimum performance, the percentage of correct completions, and the probability that an agent of that type outperforms an agent of the other type. See Figure F4 in the Annex for more details. All of this information is drawn from the data of the agents of the main study. To ensure understanding, we require participants to correctly answer two comprehension questions about the information before proceeding.[24]

**Hypotheses.** We hypothesize that the new treatments shift both components of redistributive preferences relative to `EASY`. First, we study whether spectators in `MIXED`, `REPORTS`, and `STATISTICS` consider a larger transfer to the disadvantaged agent to be fair, as revealed by costless redistribution. Second, we investigate whether spectators in these treatments arms place greater weight on implementing what they consider fair, $\beta$, as revealed by costly redistribution conditional on the costless transfer. We further use the comparison across `MIXED`, `REPORTS`, and `STATISTICS` to explore whether direct comparative experience is associated with stronger responses than information provision.

## 4.3 Implementation

We collected data in April 2026. A total of 602 individuals were recruited via *Prolific*, with 100 agents (i.e., 50 pairs) and 502 spectators (165 in `MIXED`, 165 in `REPORTS` and 172 in `STATISTICS`).

Tables A5 and A6 in the Annex detail the spectator's demographics and show that the sample is mostly balanced across treatment arms, although sex differs significantly across arms, with a higher share of women in `STATISTICS`. When comparing the sample of our extension to our main study, we find some differences in socio-demographic characteristics. We control for all such characteristics when comparing across samples. Since the extension was conducted in a separate wave from the main experiment, comparisons between `MIXED`, `REPORTS`, or `STATISTICS` and the original `EASY` and `HARD` treatments may also reflect differences between recruitment waves. Comparisons among the three extension treatments, which were randomly assigned within the same wave, are not subject to this limitation.

## 4.4 Results

We start by reporting the prior beliefs about the task. We reproduce similar findings as the main study. 76.9% of the participants rank the version of the task as the least important factor to determine success,

[24] Notice that the treatment duration for `REPORTS` and `STATISTICS` is shorter than for the experience treatments. We believe this is a feature that reflects the real world. Visual information is typically consumed more quickly, whereas experiencing something requires more time for engagement and information processing.

stating that effort and ability are of greater importance (see row 1 in Table C2 in the Annex). Additionally, 82.9% of participants also state that the two versions are approximately the same, and they do not perceive one of the two as more advantageous (see row 3 in Table C2 in the Annex).

#### 4.4.1 Mixed Experience

Within `MIXED`, participants experience both versions of the task in randomized order. We find no statistically significant differences in redistribution between participants who completed the easy version first and those who completed the hard version first, either when redistribution is costless (p-value=.635) or costly (p-value=.362). Beliefs about the opportunity gap are also similar across sequences (see Table C21 in the Annex). We therefore pool both groups in the analysis that follows.

First, we examine how the mixed intervention affects beliefs about the tasks (see Table C3 in the Annex). Compared with both `EASY` and `HARD`, spectators in `MIXED` are significantly less likely to report that the two versions of the task are similar (see row 4 in Table C3 in the Annex), more likely to state that the `EASY` version provides an advantage (see row 5 in Table C3 in the Annex), and assign greater importance to luck in determining success (see row 6 in Table C3 in the Annex). These differences suggest that direct exposure to both versions of the task makes the unequal opportunity structure more salient than one-sided experience alone. Comparison with `HARD` shows that experiencing the hard version is not equivalent to experiencing the contrast between the two versions. Spectators in `MIXED` consistently assign more relevance to luck in explaining performance differences.

We then turn to redistribution decisions. Spectators who experience both versions of the task redistribute much more towards the disadvantaged agent. When redistribution is costless they transfer \$0.60 more (equivalent to a 39% rise compared to `EASY` and p-value<.001, see column 4 in Panel A in Table 5), and when it entails a personal cost they transfer \$0.75 more (98% more compared to `EASY`, see column 4 in Panel B in Table 5). We further compare redistribution in `MIXED` and `HARD` to examine whether experiencing both sides of the opportunity structure is associated with stronger redistributive responses than experiencing disadvantage alone. Spectators in `MIXED` allocate \$0.34 more in the costless condition and about \$0.41 more in the costly condition, that is 19% and 37% higher relative to the corresponding baseline means (p-value=.005 and p-value=.002 respectively, see columns 2 and 4 in Table C4). Our results suggest that the effect of comparative experience is relevant, particularly when redistribution is costly.

Table 6 shows a summary of the structural estimates for all interventions. Estimates point in the same direction as the previous results. The weight assigned to fairness is highest for spectators in `MIXED` ($\beta$=.798, see column 3 in Panel A in Table 6), and the differences with `EASY` and `HARD` are both statistically significant. This suggests that direct comparative experience also affects the weight individuals attach to implementing their fairness views. We also find that the greater willingness to redistribute in `MIXED` persists after accounting for our estimates on what spectators consider to be fair. Spectators in `MIXED` are 20.9 p.p. more likely to keep on redistributing than those in `EASY` (p-value< .001, see column 2 in

Table C10 in the Annex), while differences relative to HARD are smaller (10.7 p.p. and p-value=.052). Emotional response and beliefs about luck only account partially for the stronger willingness to pay in MIXED relative to EASY (see Table C11 in the Annex).

The remaining outcomes show a consistent pattern with our previous findings. Spectators in MIXED judge the initial allocation as less fair than those in EASY and HARD (see Table C5 in the Annex). They also report a substantially larger performance gap between the two versions, assign a lower probability for the disadvantaged agent to outperform the advantaged agent, are less likely to believe that the disadvantaged agent could have overcome the performance gap through hard work alone, and revise counterfactual beliefs similarly to those in HARD (see Tables C6 and C7 in the Annex). Emotional responses are also larger. Spectators assigned to both versions of the task report more anger, irritation, and compassion, and lower indifference to the situation (see Table C8 in Annex). Experiencing both versions appears to narrow the empathy gap, as spectators in MIXED report greater ease in imagining the disadvantaged agent's experience than those who experience only one version (see Table C9 in the Annex).

**Table 5: Treatment Effects on Redistribution Decisions**

**Panel A. Costless redistribution**

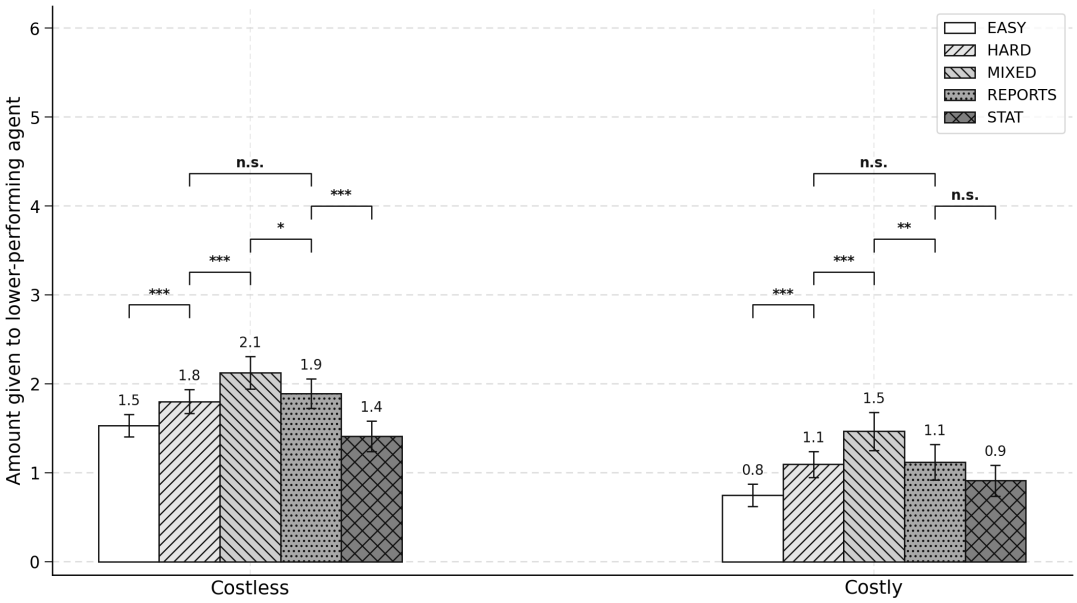
| | HARD | | MIXED | | REPORTS | | STATISTICS | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| TREATED | .270*** | .270*** | .595*** | .596*** | .358*** | .375*** | -.120 | -.093 |
| | (.095) | (.100) | (.114) | (.116) | (.106) | (.117) | (.108) | (.113) |
| Covariates | No | Yes | No | Yes | No | Yes | No | Yes |
| Effect magn. | 18% | 18% | 39% | 39% | 23% | 24% | -8% | -6% |
| p-value | .004 | .007 | <.001 | <.001 | .001 | .001 | .265 | .411 |
| Observations | 651 | 651 | 501 | 501 | 501 | 501 | 508 | 508 |
| $R^2$ | .012 | .071 | .053 | .149 | .021 | .086 | .002 | .106 |

**Panel B. Costly redistribution**

| | HARD | | MIXED | | REPORTS | | STATISTICS | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| TREATED | .345*** | .356*** | .717*** | .736*** | .371*** | .437*** | .163 | .221* |
| | (.099) | (.104) | (.126) | (.130) | (.121) | (.136) | (.110) | (.119) |
| Covariates | No | Yes | No | Yes | No | Yes | No | Yes |
| Effect magn. | 46% | 47% | 96% | 98% | 49% | 58% | 22% | 29% |
| p-value | <.001 | <.001 | <.001 | <.001 | .002 | .001 | .137 | .063 |
| Observations | 651 | 651 | 501 | 501 | 501 | 501 | 508 | 508 |
| $R^2$ | .019 | .091 | .068 | .159 | .020 | .080 | .004 | .081 |

*Notes:* This table reports treatment effects on redistribution decisions. The dependent variable is the dollar transfer to the disadvantaged agent. Panel A reports treatment effects for the costless redistribution decision. Panel B reports treatment effects for the costly redistribution decision, in which spectators must pay a fixed cost to change the initial allocation. The independent variable is spectator's assignment to a treatment arm different from EASY. Columns (1) and (2) report the treatment effect of HARD. Columns (3) and (4) report the treatment effect of MIXED. Columns (5) and (6) report the treatment effect of REPORTS. Columns (7) and (8) report the treatment effect of STATISTICS. Each row reports estimates from a linear model. Columns (2), (4), (6), and (8) include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Figure 5: Redistribution Decisions by Treatment Arm**




*Notes:* This figure shows redistribution decisions in each treatment arm. The left group of bars plots the amount assigned to the disadvantaged agent when redistribution is costless. The right group of bars plots the amount assigned to the disadvantaged agent when redistribution is costly. The bars plot decisions among participants in `EASY`, `HARD`, `MIXED`, `REPORTS`, and `STATISTICS`. Error spikes correspond to the 95% confidence interval as estimated via ordinary least squares regression with robust standard errors. Brackets report pairwise comparisons between treatment arms. $^{*f}p < .10$, $^{**f}p < .05$, $^{***f}p < .01$.

**Table 6: Estimates on Fairness Weight by Treatment Arm**

**Panel A. Treatment-specific estimates**

| | `EASY` | `HARD` | `MIXED` | `REPORTS` | `STATS` |
|---|---|---|---|---|---|
| $\hat{\beta}(T)$ | -.059 | .417*** | .798*** | .271 | .442** |
| | (.131) | (.131) | (.182) | (.171) | (.187) |
| Observations | 237 | 238 | 136 | 138 | 117 |
| Stay-in rate | .451 | .576 | .669 | .536 | .573 |

**Panel B. Pairwise comparisons**

| | Diff. | SE |
|---|---|---|
| $\hat{\beta}(\texttt{HARD}) - \hat{\beta}(\texttt{EASY})$ | .476** | (.185) |
| $\hat{\beta}(\texttt{MIXED}) - \hat{\beta}(\texttt{EASY})$ | .857*** | (.224) |
| $\hat{\beta}(\texttt{MIXED}) - \hat{\beta}(\texttt{HARD})$ | .381* | (.225) |
| $\hat{\beta}(\texttt{REPORTS}) - \hat{\beta}(\texttt{MIXED})$ | -.527** | (.250) |
| $\hat{\beta}(\texttt{STATS}) - \hat{\beta}(\texttt{MIXED})$ | -.356 | (.261) |
| $\hat{\beta}(\texttt{STATS}) - \hat{\beta}(\texttt{REPORTS})$ | .171 | (.254) |

*Notes:* This table shows estimates from the threshold model for each treatment arm. The sample contains only spectators that redistribute under no cost. The threshold term $(2c/\hat{m}_i^2)$ has a fixed $-1$ coefficient, and $c$ is \$0.20 in our experiment. Standard errors are reported in parentheses. Stars on the $\hat{\beta}(T)$ row indicate whether the treatment-specific estimate differs from zero. Stars in Panel B indicate statistical significance of tests of equality between the corresponding treatment-specific $\hat{\beta}(T)$ estimates. $^{*}$ $p < .10$, $^{**}$ $p < .05$, $^{***}$ $p < .01$.

### 4.4.2 Information Provision

We explore whether information about the opportunity gap can reproduce the effect of directly experiencing that gap. To do so, we compare `REPORTS` and `STATISTICS` with `MIXED`. We do so because both `REPORTS`

and STATISTICS describe the contrast between the easy and hard version of the task, as MIXED does. We also report some results comparing these treatments with EASY and HARD to assess how information compares to one-sided experiences.

First, we confirm that information provision affects beliefs about the task. Spectators in both REPORTS and STATISTICS perceive less advantages of the easy version of the task and assign less importance to luck in determining success than those in MIXED (see Table C13 in the Annex). Both forms of information provision affect awareness of unequal opportunities less than direct experience. This suggests that learning about the opportunity gap through either qualitative or quantitative information is not as powerful as personally experiencing the contrast between the two versions.

Participants receiving information on the opportunity gap transfer less to the disadvantaged agent than those who personally experience it (see Figure 5, and Table C12 in the Annex). Qualitative information provision leads to lower redistribution than direct comparative experience. Relative to MIXED, spectators in REPORTS transfer \$0.23 less when redistribution is costless (p-value=.075, see column 2 of Table C12) and \$0.29 less when redistribution is costly (p-value=.058, see column 4 of Table C12), equivalent to 11% and 20% reductions, respectively. At the same time, REPORTS still generates more redistribution than exposure to the easy version of the task (see column 6 of Panels A and B of Table 5). Compared with exposure to the hard version, point estimates in REPORTS are slightly higher, but the differences are not statistically significant (difference of \$0.14 for costless decisions and p-value=.209; difference of \$0.09 for costly decisions and p-value=.489, see columns 2 and 4 of Table C20).

Quantitative information is less effective. Relative to MIXED, spectators in STATISTICS transfer \$0.72 less when redistribution is costless (p-value$<$ .001, see column 2 of Table C12) and \$0.56 less when redistribution is costly (p-value$<$ .001, see column 4 of Table C12), corresponding to reductions of 34% and 38%, respectively. Compared with exposure to the easy version, redistribution in STATISTICS is not significantly different in the costless condition, although it is marginally higher when redistribution is costly (see column 8 of Panels A and B of Table 5). Compared with exposure to the hard version, STATISTICS leads to significantly lower redistribution in the costless condition (difference=-\$0.33 and p-value=.004, see column 2 of Table C20), while the costly difference is smaller and not statistically significant (difference=-\$0.17 and p-value=.179, see column 4 of Table C20). Overall, statistical information conveys the existence of the opportunity gap, but does not reproduce the redistributive response generated by direct comparative experience.

Qualitative and quantitative information affect redistribution differently. With controls, spectators who receive quantitative information transfer \$0.49 less than those receiving qualitative information when redistribution is costless (p-value$<$ .001, see column 2 of Table C12). When redistribution is costly, the difference is smaller: spectators in STATISTICS transfer \$0.27 less than those in REPORTS (p-value=.058, see column 4 of Table C12). This suggests that qualitative information is more effective than quantitative information, especially when redistribution is costless.

Evidence on whether information provision affects the willingness to implement what is considered fair is mixed (see Table 6). We estimate a lower fairness weight both in REPORTS and STATISTICS than in MIXED, but the difference is not statistically significant at conventional levels for the latter. Our results suggest that direct comparative experience induces a stronger willingness to implement fairness than receiving information, especially when the information is qualitative. Reduced-form estimates yield similar results (see Table C18 in the Annex). Spectators in REPORTS are less likely to keep redistributing once a cost is introduced than those in MIXED (difference=-10.3 p.p. and p-value=.097, see column 2 in Table C18 in the Annex). The difference between STATISTICS and MIXED is not statistically significant (difference=-9.2 p.p. and p-value=.176, see column 2 in Table C18 in the Annex). Mechanism exploration sheds little light on the reason for the differences we find (see Table C19 in the Annex). Understanding of the disadvantaged agent attenuates the gap with MIXED to some extent, especially for REPORTS, but anger, the equal-opportunity merit decision, and beliefs about luck do not consistently account for the lower persistence in the information treatments. Overall, the persistence results suggest that direct comparative experience is associated with greater willingness to pay to implement redistribution, especially relative to qualitative information.

The remaining outcomes are in line with the previous findings. Compared with MIXED, both REPORTS and STATISTICS spectators judge the initial allocation as more fair; these differences are statistically significant at the 1% level (Table C13). For numerical beliefs, MIXED spectators are more accurate than those in REPORTS and less accurate than those in STATISTICS in estimating the proportional performance gap between the two versions,[25] while there are no significant differences in the perceived probability that Agent B outperforms Agent A (Table C14). Relative to MIXED, both REPORTS and STATISTICS spectators are less likely to revise hard-work and counterfactual beliefs in the direction observed under direct exposure, with the strongest differences appearing for the hard-work and individual-level counterfactual measures (Table C15). Emotional responses also point to weaker engagement under information provision. Both information treatments are associated with lower anger and irritation, while STATISTICS is also associated with greater indifference (Table C16). Finally, perspective-taking measures show that both REPORTS and STATISTICS spectators report lower ease in imagining the experiences of both agents than spectators in MIXED; these differences are statistically significant at the 1% level (Table C17).

# 5 Fairness Reasoning

The redistribution decisions show whether experience changes the amount spectators allocate to the disadvantaged worker. We next examine whether experience also changes the fairness principles spectators use to justify those decisions. To do so, we classify spectators' open-text explanations into four fairness types and compare their prevalence across treatments. The analysis includes 1,153 explanations, with an average length of 202 characters, or approximately 39 words, and a median length of 172 characters, or 33 words.

[25] Notice that, in STATISTICS, spectators are actually given the answers to the questions as part of their information provision.

We classify explanations into four pre-specified fairness views: libertarians (who argue that the initial allocation should be left unchanged), non-compensating meritocrats (who reward observed performance as it is), compensating meritocrats (who adjust their judgments of merit to account for unequal circumstances), and egalitarians (who justify the allocation by appealing to equality). We use two large language models, GPT-5.5 Thinking and Claude Sonnet5, independently. Each model was given an identical coding protocol and instructed to classify an explanation using only the written text, never the amount redistributed. When an answer contained multiple principles, the model was instructed to assign the answer to the principle most central to the stated justification. When it was not possible to classify an answer, we instructed the model to classify it as 'Other'. The exact prompt and example for each category can be found in Appendix G. The two models assign the same code to 979 of 1,153 explanations (84.9%, Cohen's $\kappa = 0.774$), indicating that the classification is largely robust to the choice of model; we report both classifications in full in Panels A and B of Table 7 and discuss the GPT classification in the text below.

We find that the way spectators reason about unequal opportunities is heterogeneous (see Panel A of Table 7). Overall, we classify 6.9% of spectators as libertarian, 45.4% as non-compensating meritocrats, 32.6% as compensating meritocrats, and 10.1% as egalitarians. Claude's independent classification is close (5.5%, 47.1%, 30.2%, and 13.1%, respectively; Panel B). The two largest groups are meritocrats, together accounting for 78.1% of explanations under GPT (77.3% under Claude). The distribution is broadly comparable to the estimates in Andre (2024). The share of non-compensating meritocrats is somewhat higher in our data (45.4% vs. 40.0% in Andre, 2024), while compensating meritocrats are also slightly more common (32.6% vs. 28.4% in Andre, 2024). These differences may reflect the more concrete opportunity gap in our design, but also differences in measurement: Andre (2024) infer fairness types structurally from repeated allocation choices, whereas we classify the central principle expressed in participants' open-text explanations. Relative to Cappelen et al. (2024), we find a larger combined meritocratic share and fewer libertarian explanations. This comparison should likewise be interpreted cautiously, since Cappelen et al. (2024) recover fairness types from treatment differences in redistribution behavior rather than from individual text responses.

The patterns across treatment arms suggest that experience affects not only how much participants redistribute, but also how they think about fairness. Among participants in `EASY`, we classify 8.0% as libertarian, 57.7% as non-compensating meritocrats, 17.0% as compensating meritocrats, and 11.3% as egalitarians (Table 7, Panel A). Among participants in `HARD`, compensating meritocrats are a larger group (34.9%), while non-compensating meritocrats are fewer (43.5%). The difference is largest in `MIXED`, where participants directly experience both versions of the task. More than half of participants (53.3%) are classified as compensating meritocrats, while non-compensating meritocrats are only 23.0%. This pattern suggests that direct comparative experience makes the opportunity gap easier to perceive and harder to ignore. The information treatments display a more nuanced trend. `REPORTS` generates an equally high share of compensating- and non-compensating-meritocratic explanations (40.6% each), suggesting that qualitative accounts of agents' experiences can make the opportunity gap salient. By contrast, `STATISTICS` looks closer to `EASY`: 51.2% are non-compensating meritocrats and 31.4% are compensating

meritocrats. This pattern is consistent with the idea that quantitative information can communicate the existence of a performance gap, but may be less effective than direct experience or qualitative information at changing the principle participants use to justify redistribution.

We also compare our text-based classification with participants' self-reported fairness views. The two measures are related, but far from identical: text-coded classifications match self-reports in 44.1% of cases under GPT and 46.6% under Claude. This suggests that fairness views may be partly context-dependent, and that self-classifications do not always fully capture how participants reason when faced with a concrete unequal-opportunity allocation: participants who reason in a straightforwardly performance-based way in their own words are especially likely to select a different label when asked to self-categorize.

**Table 7: Fairness views by classification method**

| | Overall | Mean transfer | EASY | HARD | MIXED | REPORTS | STATISTICS |
|---|---|---|---|---|---|---|---|
| *Panel A. Text classification using GPT-5.5 Thinking* | | | | | | | |
| Libertarian | 79 (6.9%) | 0.00 | 27 (8.0%) | 16 (5.1%) | 12 (7.3%) | 7 (4.2%) | 17 (9.9%) |
| Non-compensating meritocrat | 524 (45.4%) | 1.21 | 194 (57.7%) | 137 (43.5%) | 38 (23.0%) | 67 (40.6%) | 88 (51.2%) |
| Compensating meritocrat | 376 (32.6%) | 2.38 | 57 (17.0%) | 110 (34.9%) | 88 (53.3%) | 67 (40.6%) | 54 (31.4%) |
| Egalitarian | 117 (10.1%) | 2.89 | 38 (11.3%) | 35 (11.1%) | 15 (9.1%) | 19 (11.5%) | 10 (5.8%) |
| Other | 57 (4.9%) | 2.16 | 20 (6.0%) | 17 (5.4%) | 12 (7.3%) | 5 (3.0%) | 3 (1.7%) |
| *Panel B. Text classification using Claude* | | | | | | | |
| Libertarian | 63 (5.5%) | 0.08 | 15 (4.5%) | 20 (6.3%) | 6 (3.6%) | 7 (4.2%) | 15 (8.7%) |
| Non-compensating meritocrat | 543 (47.1%) | 1.07 | 200 (59.5%) | 133 (42.2%) | 49 (29.7%) | 70 (42.4%) | 91 (52.9%) |
| Compensating meritocrat | 348 (30.2%) | 2.52 | 52 (15.5%) | 104 (33.0%) | 83 (50.3%) | 61 (37.0%) | 48 (27.9%) |
| Egalitarian | 151 (13.1%) | 2.89 | 51 (15.2%) | 47 (14.9%) | 21 (12.7%) | 21 (12.7%) | 11 (6.4%) |
| Other | 48 (4.2%) | 1.94 | 18 (5.4%) | 11 (3.5%) | 6 (3.6%) | 6 (3.6%) | 7 (4.1%) |
| *Panel C. participants' self-classified fairness views* | | | | | | | |
| Libertarian | 89 (7.7%) | 1.36 | 26 (7.7%) | 26 (8.3%) | 9 (5.5%) | 18 (10.9%) | 10 (5.8%) |
| Non-compensating meritocrat | 278 (24.1%) | 1.00 | 129 (38.4%) | 82 (26.0%) | 11 (6.7%) | 25 (15.2%) | 31 (18.0%) |
| Compensating meritocrat | 495 (42.9%) | 1.83 | 107 (31.8%) | 129 (41.0%) | 89 (53.9%) | 80 (48.5%) | 90 (52.3%) |
| Egalitarian | 224 (19.4%) | 2.59 | 56 (16.7%) | 62 (19.7%) | 47 (28.5%) | 33 (20.0%) | 26 (15.1%) |
| Other | 67 (5.8%) | 1.55 | 18 (5.4%) | 16 (5.1%) | 9 (5.5%) | 9 (5.5%) | 15 (8.7%) |
| Observations | 1,153 | | 336 | 315 | 165 | 165 | 172 |

*Notes:* Panels A and B report the distribution of fairness types based on participants' open-text explanations following the costless redistribution decision. Both text classifications use only the written explanations and do not use the amount redistributed. Panel C reports participants' self-classified fairness views from the endline survey. The self-classified category corresponding to compensating meritocrats was labeled "meritocratic (comparable)" in the survey. Percentages in the treatment columns are shares within each treatment arm. Mean transfer is the average amount assigned to the disadvantaged agent within each classification category and is calculated only after classification. GPT and Claude assign the same classification to 979 of 1,153 explanations (84.9%). Cohen's $\kappa = 0.774$, with a bootstrap 95% confidence interval of [0.743, 0.805].

# 6 Discussion

In this paper, we provide causal evidence that experiencing unequal opportunities reshapes redistributive preferences. Our results speak to a broader question in political economy and behavioral economics: where do fairness views come from? In our setting, experience affects how people interpret the sources of inequality. Before the intervention, most participants viewed performance differences mainly through

effort and ability. Unequal opportunities were present in the environment, but they were not the main lens through which participants initially interpreted outcomes. Direct exposure to disadvantage changes this interpretation. The opportunity gap becomes more salient, the link between performance and merit weakens, and redistribution shifts toward greater compensation for the disadvantaged worker. In our extension, we show that the effects of direct experience are stronger than information provision. Qualitative and quantitative information affect redistribution, but fall consistently short of the impact of experiencing unequal opportunities. We discuss these findings, examining how experience shifts the fairness views and the beliefs underlying them, how it impacts willingness to pay to act on redistributive judgments, what drives the gap between experience and information, and what the results imply for institutional design and future research.

## 6.1 Experience, Beliefs, and Fairness Views

Our first main result is that experience affects what spectators consider to be fair. Spectators who experience the hard version of the task redistribute more to the disadvantaged agent than those who experience the easy version. The treatment effect is economically meaningful. Redistribution is 18% larger among participants who experience disadvantage opportunities. This decision reveals participants' fairness view. Experiencing disadvantage shifts views toward greater compensation for the disadvantaged agent.

Our mechanism exploration points to an important role for beliefs about the determinants of performance. Spectators who experience disadvantage place greater weight on luck and circumstances, reward observed performance less, and judge the initial allocation as less fair. Accounting for these beliefs substantially attenuates the treatment effect on redistribution. These patterns are consistent with experience affecting redistributive judgments partly by changing how spectators interpret the sources of inequality.

This result speaks directly to the literature on fairness under unequal opportunities. Recent studies show that redistribution responds to beliefs and information about the role of circumstances in shaping outcomes (Andre, 2024; Bhattacharya and Mollerstrom, 2025; Brun, 2026; Cappelen et al., 2024; Dong et al., 2024; Pogliano, 2024; Pruess et al., 2025). Our results build on this evidence by showing that these beliefs are shaped by experiences. Experiencing disadvantage makes the role of circumstances more visible and reduces the tendency to interpret performance differences as reflecting only effort or ability.

Results from our extension study reinforce this interpretation. Merit evaluation under unequal opportunities needs counterfactual reasoning about what would have happened under different circumstances (Amemiya et al., 2023; Andre, 2024), and that this process is cognitively demanding (Brun, 2026). Direct experience may facilitate it by making the relevant counterfactual more concrete. Consistent with this idea, we find the strongest belief revisions and redistributive responses when participants can directly compare how advantage and disadvantage affect performance. This suggests that experience not only matters because it reveals that circumstances differ, but also because it helps individuals to understand how unequal opportunities affect the translation of effort into performance. Perspective-taking may

reinforce this process (Jia, 2022), but the main pattern is most consistent with experience correcting attribution bias and shifting the fairness view toward greater compensation.

Importantly, experience increases redistribution without eliminating income differences between agents. Yet equal division is not necessarily what spectators consider fair. Even when the bonus is assigned by a coin toss, spectators leave a substantial share with the winner, with no significant difference across treatments. Relative to this personal benchmark, the shortfall in redistribution is smaller among spectators who experience the hard version than among those who experience the easy version. This treatment difference is substantially attenuated after accounting for spectators' beliefs about the relevance of luck in determining performance. These patterns suggest that experience brings redistributive decisions closer to spectators' own fairness judgments, partly by changing how they interpret the role of opportunities in performance. The inequality that remains may also reflect differences in fairness principles. Consistent with this interpretation, our classification of spectators' written explanations identifies two groups who place relatively little weight on compensating for unequal opportunities. Non-compensating meritocrats, who reward observed performance regardless of opportunities, remain the largest group even after experiencing disadvantage and transfer relatively little. Libertarians, who argue that the initial allocation should remain unchanged, form a smaller group and do not redistribute. Interestingly, spectators who experience both versions redistribute more than they would under their own coin-toss allocation, and compensating meritocrats become the largest group. This suggests that comparative experience may also change how spectators reason about fairness, beyond simply bringing their decisions closer to their luck-based benchmark. However, differences between the allocation settings and experimental waves warrant caution in interpreting this pattern.

The result is also related to a growing literature showing that personal experiences of success and failure affect redistributive preferences and meritocratic beliefs (Amasino et al., 2024; Cassar and Klein, 2019; Deffains et al., 2016; Espinosa et al., 2020; Fehr and Vollmann, 2025). These studies show that individuals' own outcomes can influence how they interpret inequality, often through self-serving beliefs and motivated reasoning. Our setting is similar in showing that experience can shape fairness views. However, it differs in a key respect: the relevant experience is not winning or losing, but facing advantaged or disadvantaged conditions. The belief formation we document is less likely to be motivated by a need to justify one's own outcome, and more likely to reflect learning about how the environment shapes performance. The extension study further supports this interpretation. If the effect were mainly driven by motivated beliefs or identification with an experiential in-group, we would expect it to be strongest among spectators who experienced the same task version as the disadvantaged agent. Instead, belief revision and redistribution are strongest when participants experience both task versions and therefore have no single task-based in-group by design. This pattern suggests that the effect is better understood as learning about the opportunity structure than as self-serving or in-group motivated reasoning.

More broadly, our findings connect to the literature on experience effects in economic beliefs and behavior. A large body of work shows that individuals often extrapolate from their own experiences when forming

beliefs about economic environments (Kuchler and Zafar, 2019; Malmendier and Nagel, 2016; Malmendier, 2021a,b; Malmendier and Wachter, 2024). Experiences may also shape which aspects of an environment become salient and are later retrieved in judgment (Bordalo et al., 2020). Our results extend this logic to fairness views. We show that a short exposure to disadvantage affects not only beliefs about the task, but also the fairness views used to evaluate redistribution between others.

## 6.2 Experience and Willingness to Pay for Redistribution

The second main result is that experience affects not only what spectators consider fair, but also their willingness to pay to implement that fairness view. This distinction is important because the costless treatment effect could be interpreted as a low-cost expressive response: spectators who experience disadvantage may express sympathy when doing so is free. The costly decision addresses this concern directly. Introducing a personal cost substantially reduces redistribution for all spectators, confirming that self-interest constrains redistributive behavior. Yet the treatment gap persists, as spectators who experienced disadvantage continue to redistribute more toward the disadvantaged agent than spectators who experienced advantage. Thus, the effect of experience is not limited to cheap expressions of fairness; it also affects willingness to bear a private cost to correct an allocation perceived as unfair.

The structural estimates reinforce this interpretation. Experiencing advantage is associated with little willingness to bear a personal cost to correct the allocation, whereas experiencing disadvantage gives the unfairness of the opportunity gap greater behavioral force. In the model, this appears as a fairness weight that is close to zero and statistically insignificant among spectators who experience the advantaged opportunity set, but positive and significant among those who experience disadvantage; the difference between the two is statistically significant. The reduced-form analysis points in the same direction. We show that, conditional on the exact amount redistributed when redistribution was costless, spectators who experience disadvantage are more likely to continue redistributing once a cost is introduced compared to those who experienced advantage. In our extension, we show that the willingness to pay is the highest among those who directly experience the opportunity gap. Direct comparative experience therefore affects not only the fairness view, but also the weight attached to acting on that view.

This result contributes to the incipient literature on costly fairness and willingness to pay for redistribution. Most studies using spectator games impose no cost on the decision-makers, preventing researchers from distinguishing fairness views from the willingness to act on them. Recent empirical work shows that redistributive decisions are sensitive to personal costs, but that many individuals remain willing to incur such costs to implement allocations they regard as fair (Dimant et al., 2024; Dwenger et al., 2024; Piraino and Ryan, 2024). Closest to our setting, Dimant et al. (2024) study costly third-party redistribution in a related environment and document willingness-to-pay rates similar to ours. We further show that the willingness to act on fairness views is malleable. This margin is often treated as a stable preference parameter or inferred from cross-sectional differences in costly redistribution. We provide causal evidence that it can be shifted by experience with the opportunity structure that produced the unequal outcome.

## 6.3 The Role of Direct Experience vs. Information

The information treatments show that information alone does not fully replicate the effect of direct experience. Both qualitative and quantitative information make the opportunity gap explicit, but they generate less redistribution than directly experiencing that gap. The difference is smaller for qualitative accounts of the agents' experiences and substantially larger for quantitative information, suggesting that narratives can shift redistributive behavior but still fall short of reproducing the effect of direct comparative experience.

Redistributive decisions in the costly setting point toward the same direction. Relative to spectators who directly experience the opportunity gap, those who receive information redistribute less when it entails a personal cost. This translates into lower willingness to keep redistributing after receiving qualitative information than after direct comparative experience. The corresponding estimate for quantitative information is negative but not statistically significant, possibly because of limited power. Overall, the results suggest that direct comparative experience affects both fairness views and, to some extent, the willingness to pay to implement those views in a different way than information does.

One interpretation for this asymmetry is that experiential learning differs from information transmission. In our setting, quantitative information allows spectators to accurately assess the magnitude of the opportunity gap, yet generates substantially less redistribution than direct experience. This suggests that knowing the size of an opportunity gap does not necessarily imply incorporating it into judgments about merit and fairness. Personal experience may provide a different understanding of how unequal circumstances shape performance. A large literature shows that individuals learn differently from experienced evidence than from described or observed evidence, mostly in the context of risk preferences (Hertwig and Erev, 2009; Hertwig, 2015). Recent work by Gonzalez-Jimenez (2026) similarly shows that beliefs and ambiguity attitudes differ when individuals learn from described rather than experienced signals, while Chen et al. (2026) provides related evidence that experience and observation are processed differently. Our findings extend this distinction to redistributive judgments under unequal opportunities.

The asymmetry between experience and information may also reflect differences in emotional engagement and perspective-taking. Information can reveal the existence of unequal opportunities, and narratives can partly convey their subjective meaning. But direct comparative experience appears to combine belief updating, emotional engagement, and perspective-taking in a way that information alone does not fully reproduce. In our data, directly experiencing the opportunity gap is associated with stronger emotional responses and greater perspective-taking than information provision. This matters because emotions such as moral anger and compassion can motivate costly third-party interventions in fairness settings (Chen et al., 2026). Thus, direct experience may generate stronger redistribution not only because it provides information, but because it makes unfairness more emotionally accessible and salient.

The comparison between different types of information provision is also informative. Narrative information generates more redistribution than statistical information in the costless condition. This is consistent with

evidence that identifiable or narratively described individuals often elicit stronger prosocial responses than statistical information, both in charitable giving and in attitudes toward immigration and redistribution (Slovic, 2007; Small et al., 2007; Alesina et al., 2023). More broadly, it is also consistent with the idea that social information is not absorbed mechanically, as people may underuse, reinterpret, or discount information depending on how it is presented (Morin et al., 2021; Graeber et al., 2024; Thaler et al., 2024). In our context, reports may make it easier for spectators to imagine why the harder version matters: they convey frustration, difficulty, and the subjective experience of facing a constraint. Statistics, by contrast, convey the size of the performance gap more precisely, but may do less to make the mechanism behind the gap vivid. At the same time, the advantage of qualitative over quantitative information narrows when redistribution becomes costly. This suggests that narratives may be more effective at shifting costless fairness views than at increasing willingness to pay to implement them.

## 6.4 Implications

Our findings primarily inform how we understand the formation of redistributive preferences, and only more tentatively the design of institutions.

Our results suggest that support for redistribution depends in part on the conditions individuals have experienced. This has implications for political economy models, in which redistributive preferences are often treated as stable parameters. If beliefs about the role of opportunities are shaped by experience, structural changes in economic sorting, residential segregation, or occupational integration could indirectly affect support for redistribution by changing which constraints individuals face and observe.

The comparison between experience and information also sheds light on how beliefs about opportunities are formed. In our setting, statistics accurately conveyed the size of the opportunity gap but generated little additional redistribution, qualitative reports generated more, and direct comparative experience produced the strongest response. This suggests that knowing the size of an opportunity gap is not necessarily sufficient for incorporating it into fairness judgments. Direct experience may also help individuals understand how opportunities shape performance and the relevant counterfactual. Programs such as civil service rotations, poverty simulations, or immersive experiences designed to convey disadvantage (Andries et al., 2024; Piff et al., 2020) seek to foster a similar understanding. Our results are consistent with the rationale behind such programs, but do not establish their effectiveness in settings where constraints are more complex or difficult to experience.

Finally, our findings have potential implications for how merit is evaluated in organizations. When evaluators underestimate the role of structural conditions in shaping performance, their assessments may disadvantage candidates who faced greater barriers. Many relevant barriers, including those related to gender, ethnicity, disability, or care responsibilities, cannot readily be experienced by evaluators who do not face them. Rather than attempting to reproduce these experiences, institutions could seek to make the relevant circumstances more visible, for instance by comparing candidates who faced similar conditions or evaluating performance relative to the opportunities available. Including decision-makers

with diverse experiences may also help bring otherwise overlooked constraints into consideration. Whether these approaches improve evaluations of merit in collectives remains a question for future research.

## 6.5 Open Questions

Our findings raise several questions for future research. A first question concerns the persistence of experience effects. Our experiment shows that even a short exposure to unequal opportunities can change beliefs, fairness views, and willingness to pay for redistribution. An important next step is to examine whether these effects persist beyond the immediate decision environment, whether repeated exposure amplifies them, and whether individuals return to prior beliefs once the experience becomes less salient.

A second question concerns external validity. Our controlled effort task allows us to isolate the experience of advantage or disadvantage from personal success and failure. However, real-world inequalities of opportunity are often more complex, persistent, and difficult to experience directly. Moreover, individuals approach these inequalities with established beliefs and personal histories that may shape how they interpret new experiences. Although we find no clear evidence that the order of exposure to advantage and disadvantage matters, both task versions are unfamiliar and experienced within minutes of each other. In real-world settings, new experiences often follow years of accumulated experience. Whether similar effects arise in more complex settings, and how new experiences interact with pre-existing beliefs, remain important questions for future research.

A third question concerns which components of experience matter for redistributive judgments. Direct exposure may help individuals recognize constraints they previously underestimated, make the relevant equal-opportunity counterfactual easier to understand, or generate emotional engagement with disadvantaged individuals. Our evidence suggests that belief updating plays an important role, while emotional responses and perspective-taking may reinforce this process. These channels are not independently manipulated in our experiment, and their relative contributions remain uncertain. Future research could further investigate which components of experience need to be reproduced to generate similar changes in redistributive judgments.

A fourth question concerns the relationship between experience and information. Our extension study shows that reports and statistics do not fully reproduce the effect of direct comparative experience. This raises a broader question about what kinds of information can substitute for experience, and when. Narratives may make disadvantage easier to imagine, while statistics may communicate the size of the opportunity gap more precisely. Yet neither fully replicates the effect of direct exposure. Future work could vary the intensity, duration, format, and credibility of information to understand when information affects only beliefs, when it impacts fairness views, and when it also generates willingness to act on those views. A related question is how experience and information interact. Information may be more effective when it is received after experience, because experience can provide a concrete reference point that makes abstract facts easier to interpret. Conversely, information may shape how individuals process subsequent experiences by directing attention to the role of circumstances. Studying these interactions would help

clarify whether experience and information are substitutes, complements, or inputs into a sequential process of belief formation.

# 7 Conclusion

This paper studies whether experience with unequal opportunities shapes redistributive preferences. We implement a spectator experiment in which participants first experience either an advantaged or disadvantaged version of an effort task and then decide how to redistribute earnings between two workers whose unequal opportunities arise from differences in task difficulty. We find that experiencing disadvantage affects how participants interpret unequal outcomes: they become more likely to recognize the role of circumstances, less likely to view performance as a pure signal of merit, and more willing to redistribute toward the disadvantaged worker. This effect persists when redistribution requires paying a personal cost, showing that experience affects not only what spectators regard as fair, but also their willingness to act on that view.

The extension study shows that comparative experience is even more powerful. Participants who experience both the easy and hard versions redistribute more than those who experience only one side of the opportunity structure, suggesting that direct exposure to the contrast between advantage and disadvantage helps individuals understand how circumstances shape performance. Information provision through reports or statistics also conveys unequal opportunities, but produces weaker redistributive responses than direct experience. Taken together, the results suggest that redistributive preferences are shaped not only by ideology or self-interest, but also by the experiences through which individuals learn to interpret inequality. Correcting misperceptions may therefore require more than communicating facts: it may also require making concrete how unequal opportunities shape outcomes, even when the constraints themselves cannot be experienced.

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# A Descriptive Statistics

**Table A1: Survey Variables for Agents**

| Variable | Type | Question |
|---|---|---|
| | (1) | (2) |
| PERFORMANCE | Integer | [Number of tables solved correctly] |
| AGENTAB_SHAREB | 0-6 | How would you split the 6 USD bonus? |
| PERSUASION | Open Text | What would you like to tell the participant before they decide how to split the bonus? |

**Table A2: Survey Variables for Spectators**

| Variable | Type | Question |
|---|---|---|
| | (1) | (2) |
| PRIOR_RANKING | Categorical (9 categories) | What matters more for deciding who wins? |
| PRIOR_ADVANTAGE | Categorical (4 categories) | If there is a difference between the versions, which version do you expect would lead to better performance? |
| PRIOR_INTENSITY | 1-7 score | In a scale from 1 to 7, where 1 is 'Getting Version A is a major advantage' and 7 is 'Getting Version B is a major advantage', Which version do you think provides a greater advantage for performance? |
| ATTENTION_CHECK1 | Categorical (4 categories) | The following question is an attention check. Please select letter C. Select one of the following letters. |
| EXPERIENCE_CHECK | Categorical (2 categories) | Which version of the task did you just complete? |
| SPECTATORS_PERFORMANCE | Integer | # of succesfully completed tables |
| POSTERIOR_PERFORMANCEA | Integer | On average, how many tables do you think Agents A successfully solved in 4 minutes? |
| CERTAINTY_PERFORMANCEA | Ordinal (4 categories) | How sure are you about your answer? |
| POSTERIOR_PERFORMANCEB | Integer | On average, how many tables do you think Agents B successfully solved in 4 minutes? |
| CERTAINTY_PERFORMANCEB | Ordinal (4 categories) | How sure are you about your answer? |

| Variable | Type | Question |
|---|---|---|
| | (1) | (2) |
| POSTERIOR_FREQUENCY | 1-100 | In how many pairs (out of the 100) do you think that the Agent B solves more tables than the Agent A? |
| CERTAINTY_FREQUENCY | Ordinal (4 categories) | How sure are you about your answer? |
| ATTENTION_CHECK2 | Binary | Attention check: Do not select this option. |
| SPECTATORAB_SHAREB | 0-6 | How would you split the 6 USD bonus? |
| POSTERIOR_EXPLANATION | Open Text | Explain, in at least 2 or 3 sentences, the reasoning behind why you made this decision. |
| SPECTATORABCOSTLY_SHAREB | 0-6 | How would you split the 6 USD bonus? (redistribution costs 0.20 USD) |
| POSTERIOR_FAIR | Ordinal (4 categories) | In this pair, how fair was that the agent that solved more correct tables was initially assigned all the bonus? |
| POSTERIOR_HARDWORKGL | Ordinal (5 categories) | Imagine all agents complete the same Version of the task, but are still randomly assigned to be called either Agents A or Agents B. Which group of agents do you think would solve more correct tables? |
| POSTERIOR_HARDWORK | Ordinal (4 categories) | In this pair, if Agent B had worked really hard they could have solved more correct tables than Agent A. |
| POSTERIOR_COUNTERFACTUAL | Categorical (3 categories) | Imagine that both agents in this pair completed the same Version of the task. Who do you think would have solved more tables in that case? |
| POSTERIOR_EFFORT | 0-100 | How much does effort explain which of the agents solves more tables? |
| POSTERIOR_ABILITY | 0-100 | How much does ability explain which of the agents solves more tables? |
| POSTERIOR_LUCK | 0-100 | How much does luck explain which of the agents solves more tables? |
| POSTERIOR_INTENSITY | 1-7 score | Do you think the version of the task assigned to each agent provided any advantage in the competition? |

| Variable | Type | Question |
|---|---|---|
| | (1) | (2) |
| POSTERIOR_EXPERIENCEA | Ordinal (4 categories) | It is easy for me to imagine Agent A's experience with the task. |
| POSTERIOR_EXPERIENCEB | Ordinal (4 categories) | It is easy for me to imagine Agent B's experience with the task. |
| POSTERIOR_ANGER | Ordinal (4 categories) | How much did you feel Anger? |
| POSTERIOR_COMPASSION | Ordinal (4 categories) | How much did you feel Compassion? |
| POSTERIOR_SADNESS | Ordinal (4 categories) | How much did you feel Sadness? |
| POSTERIOR_INDIFFERENCE | Ordinal (4 categories) | How much did you feel Indifference? |
| POSTERIOR_IRRITATION | Ordinal (4 categories) | How much did you feel Irritation? |
| SPECTATORLUCK_SHAREB | 0-6 | How would you split the 6 USD bonus? |
| SPECTATORMERIT_SHAREB | 0-6 | How would you split the 6 USD bonus? |
| SPECTATORUNDERDOG_SHAREB | 0-6 | How would you split the 6 USD bonus? |
| SPECTATORVIGNETTE_SHAREB | 0-6 | How would you split the 6 USD bonus? |
| VIEWS | Categorical (5 categories) | ... |

**Table A3: Balance Test in the Main Study**

| | Total | EASY | HARD | p-value |
|---|---|---|---|---|
| **N** | 651 | 336 | 315 | |
| **Sex** | | | | .714 |
| Male | 302 (46.4%) | 150 (44.6%) | 152 (48.3%) | |
| Female | 340 (52.2%) | 182 (54.2%) | 158 (50.2%) | |
| Other | 3 (0.5%) | 1 (0.3%) | 2 (0.6%) | |
| N/A | 6 (0.9%) | 3 (0.9%) | 3 (1.0%) | |
| **Age** | | | | .771 |
| Years | 43.7 (12.9) | 43.5 (12.5) | 43.8 (13.3) | |
| **Race** | | | | .548 |
| White | 454 (69.7%) | 231 (68.8%) | 223 (70.8%) | |
| Black | 101 (15.5%) | 49 (14.6%) | 52 (16.5%) | |
| Asian | 52 (8.0%) | 30 (8.9%) | 22 (7.0%) | |
| Native | 2 (0.3%) | 2 (0.6%) | 0 (0.0%) | |
| Other | 15 (2.3%) | 8 (2.4%) | 7 (2.2%) | |
| Multiple races | 18 (2.8%) | 12 (3.6%) | 6 (1.9%) | |
| No race selected | 9 (1.4%) | 4 (1.2%) | 5 (1.6%) | |
| **Education** | | | | .176 |
| Less than high school | 3 (0.5%) | 3 (0.9%) | 0 (0.0%) | |
| High school | 69 (10.6%) | 38 (11.3%) | 31 (9.8%) | |
| College, not degree | 87 (13.4%) | 47 (14.0%) | 40 (12.7%) | |
| Associate degree | 55 (8.4%) | 20 (6.0%) | 35 (11.1%) | |
| Bachelor degree | 266 (40.9%) | 137 (40.8%) | 129 (41.0%) | |
| Master degree | 140 (21.5%) | 75 (22.3%) | 65 (20.6%) | |
| Professional degree | 13 (2.0%) | 5 (1.5%) | 8 (2.5%) | |
| Doctorate degree | 18 (2.8%) | 11 (3.3%) | 7 (2.2%) | |
| **Employment** | | | | .144 |
| Full-time | 373 (57.3%) | 203 (60.4%) | 170 (54.0%) | |
| Part-time | 98 (15.1%) | 46 (13.7%) | 52 (16.5%) | |
| Self-employed | 69 (10.6%) | 42 (12.5%) | 27 (8.6%) | |
| Unemployed and looking | 41 (6.3%) | 15 (4.5%) | 26 (8.3%) | |
| Unemployed and non-looking | 12 (1.8%) | 4 (1.2%) | 8 (2.5%) | |
| Student | 8 (1.2%) | 4 (1.2%) | 4 (1.3%) | |
| Retired | 32 (4.9%) | 14 (4.2%) | 18 (5.7%) | |
| Homecare | 18 (2.8%) | 8 (2.4%) | 10 (3.2%) | |

*Notes:* This table reports demographic balance in the main study. Columns (1)–(3) report statistics for the full sample and by treatment arm. For categorical variables, cells report frequencies and column shares in parentheses. Age reports means and standard deviations in parentheses. Race is coded as mutually exclusive: participants selecting more than one race are classified as Multiple races, and participants selecting no race are classified as No race selected. p-values are from Pearson $\chi^2$ tests for categorical variables and Welch two-sample t-tests for age.

**Table A4: Balance Test in the Main Study (ctd.)**

| | Total | EASY | HARD | p-value |
|---|---|---|---|---|
| **N** | 651 | 336 | 315 | |
| **Income (individual)** | | | | .002 |
| 0 | 11 (1.7%) | 10 (3.0%) | 1 (0.3%) | |
| <10k | 61 (9.4%) | 19 (5.7%) | 42 (13.3%) | |
| 10k–15k | 34 (5.2%) | 17 (5.1%) | 17 (5.4%) | |
| 15k–20k | 34 (5.2%) | 12 (3.6%) | 22 (7.0%) | |
| 20k–25k | 36 (5.5%) | 21 (6.2%) | 15 (4.8%) | |
| 25k–50k | 143 (22.0%) | 71 (21.1%) | 72 (22.9%) | |
| 50k–100k | 228 (35.0%) | 127 (37.8%) | 101 (32.1%) | |
| >100k | 94 (14.4%) | 54 (16.1%) | 40 (12.7%) | |
| N/A | 10 (1.5%) | 5 (1.5%) | 5 (1.6%) | |
| **Household income** | | | | .668 |
| <20k | 48 (7.4%) | 22 (6.5%) | 26 (8.3%) | |
| 20k–35k | 71 (10.9%) | 39 (11.6%) | 32 (10.2%) | |
| 35k–50k | 78 (12.0%) | 40 (11.9%) | 38 (12.1%) | |
| 50k–75k | 100 (15.4%) | 44 (13.1%) | 56 (17.8%) | |
| 75k–100k | 113 (17.4%) | 60 (17.9%) | 53 (16.8%) | |
| 100k–150k | 121 (18.6%) | 62 (18.5%) | 59 (18.7%) | |
| 150k–200k | 66 (10.1%) | 39 (11.6%) | 27 (8.6%) | |
| 200k–250k | 22 (3.4%) | 14 (4.2%) | 8 (2.5%) | |
| 250k–300k | 9 (1.4%) | 6 (1.8%) | 3 (1.0%) | |
| >300k | 11 (1.7%) | 5 (1.5%) | 6 (1.9%) | |
| N/A | 12 (1.8%) | 5 (1.5%) | 7 (2.2%) | |
| **Political party** | | | | .193 |
| Democrat | 261 (40.1%) | 125 (37.2%) | 136 (43.2%) | |
| Republican | 217 (33.3%) | 122 (36.3%) | 95 (30.2%) | |
| Independent | 173 (26.6%) | 89 (26.5%) | 84 (26.7%) | |

*Notes:* This table reports socio-economic and political balance in the main study. Columns (1)–(3) report statistics for the full sample and by treatment arm. Cells report frequencies and column shares in parentheses. p-values are from Pearson $\chi^2$ tests.

**Table A5: Balance Test in the Extension Sample**

| | Total | MIXED | REPORTS | STATISTICS | p-value |
|---|---|---|---|---|---|
| **N** | 502 | 165 | 165 | 172 | |
| **Sex** | | | | | .016 |
| Male | 226 (45.0%) | 80 (48.5%) | 82 (49.7%) | 64 (37.2%) | |
| Female | 267 (53.2%) | 82 (49.7%) | 77 (46.7%) | 108 (62.8%) | |
| Other | 5 (1.0%) | 1 (0.6%) | 4 (2.4%) | 0 (0.0%) | |
| N/A | 4 (0.8%) | 2 (1.2%) | 2 (1.2%) | 0 (0.0%) | |
| **Age** | | | | | .998 |
| Years | 38.4 (13.2) | 38.4 (13.8) | 38.4 (12.8) | 38.4 (13.0) | |
| **Race** | | | | | .367 |
| White | 305 (60.8%) | 100 (60.6%) | 107 (64.8%) | 98 (57.0%) | |
| Black | 105 (20.9%) | 37 (22.4%) | 33 (20.0%) | 35 (20.3%) | |
| Asian | 43 (8.6%) | 12 (7.3%) | 12 (7.3%) | 19 (11.0%) | |
| Native | 3 (0.6%) | 2 (1.2%) | 0 (0.0%) | 1 (0.6%) | |
| Other | 15 (3.0%) | 8 (4.8%) | 3 (1.8%) | 4 (2.3%) | |
| Multiple races | 20 (4.0%) | 3 (1.8%) | 8 (4.8%) | 9 (5.2%) | |
| No race selected | 11 (2.2%) | 3 (1.8%) | 2 (1.2%) | 6 (3.5%) | |
| **Education** | | | | | .809 |
| Less than high school | 4 (0.8%) | 2 (1.2%) | 1 (0.6%) | 1 (0.6%) | |
| High school | 60 (12.0%) | 19 (11.5%) | 24 (14.5%) | 17 (9.9%) | |
| College, not degree | 95 (18.9%) | 30 (18.2%) | 29 (17.6%) | 36 (20.9%) | |
| Associate degree | 46 (9.2%) | 15 (9.1%) | 14 (8.5%) | 17 (9.9%) | |
| Bachelor degree | 189 (37.6%) | 66 (40.0%) | 57 (34.5%) | 66 (38.4%) | |
| Master degree | 81 (16.1%) | 26 (15.8%) | 29 (17.6%) | 26 (15.1%) | |
| Professional degree | 11 (2.2%) | 2 (1.2%) | 7 (4.2%) | 2 (1.2%) | |
| Doctorate degree | 16 (3.2%) | 5 (3.0%) | 4 (2.4%) | 7 (4.1%) | |

*Notes:* This table reports demographic balance in the extension sample. Columns (1)–(4) report statistics for the full sample and by treatment arm. For categorical variables, cells report frequencies and column shares in parentheses. Age reports means and standard deviations in parentheses. Race is coded as mutually exclusive: participants selecting more than one race are classified as Multiple races, and participants selecting no race are classified as No race selected. p-values are from Pearson $\chi^2$ tests for categorical variables and one-way ANOVA for age.

**Table A6: Balance Test in the Extension Sample (ctd.)**

| | Total | MIXED | REPORTS | STATISTICS | p-value |
|---|---|---|---|---|---|
| **N** | 502 | 165 | 165 | 172 | |
| **Employment** | | | | | .333 |
| Full-time | 246 (49.0%) | 79 (47.9%) | 82 (49.7%) | 85 (49.4%) | |
| Part-time | 73 (14.5%) | 25 (15.2%) | 21 (12.7%) | 27 (15.7%) | |
| Self-employed | 52 (10.4%) | 17 (10.3%) | 17 (10.3%) | 18 (10.5%) | |
| Unemployed and looking | 52 (10.4%) | 18 (10.9%) | 23 (13.9%) | 11 (6.4%) | |
| Unemployed and non-looking | 13 (2.6%) | 3 (1.8%) | 3 (1.8%) | 7 (4.1%) | |
| Student | 28 (5.6%) | 6 (3.6%) | 7 (4.2%) | 15 (8.7%) | |
| Retired | 18 (3.6%) | 7 (4.2%) | 6 (3.6%) | 5 (2.9%) | |
| Homecare | 20 (4.0%) | 10 (6.1%) | 6 (3.6%) | 4 (2.3%) | |
| **Income (individual)** | | | | | .987 |
| 0 | 14 (2.8%) | 5 (3.0%) | 2 (1.2%) | 7 (4.1%) | |
| <10k | 67 (13.3%) | 23 (13.9%) | 24 (14.5%) | 20 (11.6%) | |
| 10k–15k | 38 (7.6%) | 12 (7.3%) | 14 (8.5%) | 12 (7.0%) | |
| 15k–20k | 20 (4.0%) | 7 (4.2%) | 5 (3.0%) | 8 (4.7%) | |
| 20k–25k | 27 (5.4%) | 9 (5.5%) | 10 (6.1%) | 8 (4.7%) | |
| 25k–50k | 101 (20.1%) | 34 (20.6%) | 31 (18.8%) | 36 (20.9%) | |
| 50k–100k | 156 (31.1%) | 51 (30.9%) | 52 (31.5%) | 53 (30.8%) | |
| >100k | 59 (11.8%) | 17 (10.3%) | 19 (11.5%) | 23 (13.4%) | |
| N/A | 20 (4.0%) | 7 (4.2%) | 8 (4.8%) | 5 (2.9%) | |
| **Household income** | | | | | .715 |
| <20k | 42 (8.4%) | 15 (9.1%) | 14 (8.5%) | 13 (7.6%) | |
| 20k–35k | 41 (8.2%) | 14 (8.5%) | 12 (7.3%) | 15 (8.7%) | |
| 35k–50k | 56 (11.2%) | 22 (13.3%) | 17 (10.3%) | 17 (9.9%) | |
| 50k–75k | 81 (16.1%) | 31 (18.8%) | 24 (14.5%) | 26 (15.1%) | |
| 75k–100k | 77 (15.3%) | 23 (13.9%) | 25 (15.2%) | 29 (16.9%) | |
| 100k–150k | 113 (22.5%) | 33 (20.0%) | 39 (23.6%) | 41 (23.8%) | |
| 150k–200k | 38 (7.6%) | 12 (7.3%) | 12 (7.3%) | 14 (8.1%) | |
| 200k–250k | 18 (3.6%) | 3 (1.8%) | 11 (6.7%) | 4 (2.3%) | |
| 250k–300k | 3 (0.6%) | 0 (0.0%) | 2 (1.2%) | 1 (0.6%) | |
| >300k | 13 (2.6%) | 4 (2.4%) | 2 (1.2%) | 7 (4.1%) | |
| N/A | 20 (4.0%) | 8 (4.8%) | 7 (4.2%) | 5 (2.9%) | |
| **Political party** | | | | | .234 |
| Democrat | 221 (44.0%) | 63 (38.2%) | 71 (43.0%) | 87 (50.6%) | |
| Republican | 135 (26.9%) | 50 (30.3%) | 46 (27.9%) | 39 (22.7%) | |
| Independent | 146 (29.1%) | 52 (31.5%) | 48 (29.1%) | 46 (26.7%) | |

*Notes:* This table reports socio-economic and political balance in the extension sample. Columns (1)–(4) report statistics for the full sample and by treatment arm. Cells report frequencies and column shares in parentheses. p-values are from Pearson $\chi^2$ tests.

**Table A7: Comparison between the Main Study and Extension Sample**

| | Total | Main | Extension | p-value |
|---|---|---|---|---|
| **N** | 1153 | 651 | 502 | |
| **Sex** | | | | .709 |
| Male | 528 (45.8%) | 302 (46.4%) | 226 (45.0%) | |
| Female | 607 (52.6%) | 340 (52.2%) | 267 (53.2%) | |
| Other | 8 (0.7%) | 3 (0.5%) | 5 (1.0%) | |
| N/A | 10 (0.9%) | 6 (0.9%) | 4 (0.8%) | |
| **Age** | | | | $< .001$ |
| Years | 41.4 (13.3) | 43.7 (12.9) | 38.4 (13.2) | |
| **Race** | | | | .071 |
| White | 759 (65.8%) | 454 (69.7%) | 305 (60.8%) | |
| Black | 206 (17.9%) | 101 (15.5%) | 105 (20.9%) | |
| Asian | 95 (8.2%) | 52 (8.0%) | 43 (8.6%) | |
| Native | 5 (0.4%) | 2 (0.3%) | 3 (0.6%) | |
| Other | 30 (2.6%) | 15 (2.3%) | 15 (3.0%) | |
| Multiple races | 38 (3.3%) | 18 (2.8%) | 20 (4.0%) | |
| No race selected | 20 (1.7%) | 9 (1.4%) | 11 (2.2%) | |
| **Education** | | | | .103 |
| Less than high school | 7 (0.6%) | 3 (0.5%) | 4 (0.8%) | |
| High school | 129 (11.2%) | 69 (10.6%) | 60 (12.0%) | |
| College, not degree | 182 (15.8%) | 87 (13.4%) | 95 (18.9%) | |
| Associate degree | 101 (8.8%) | 55 (8.4%) | 46 (9.2%) | |
| Bachelor degree | 455 (39.5%) | 266 (40.9%) | 189 (37.6%) | |
| Master degree | 221 (19.2%) | 140 (21.5%) | 81 (16.1%) | |
| Professional degree | 24 (2.1%) | 13 (2.0%) | 11 (2.2%) | |
| Doctorate degree | 34 (2.9%) | 18 (2.8%) | 16 (3.2%) | |

*Notes:* This table compares demographic composition between the main study and the extension sample. Columns (1)–(3) report statistics for the full sample and by sample. For categorical variables, cells report frequencies and column shares in parentheses. Age reports means and standard deviations in parentheses. Race is coded as mutually exclusive: participants selecting more than one race are classified as Multiple races, and participants selecting no race are classified as No race selected. p-values are from Pearson $\chi^2$ tests for categorical variables and Welch two-sample t-tests for age.

**Table A8: Comparison between the Main Study and Extension Sample (ctd.)**

| | Total | Main | Extension | p-value |
|---|---|---|---|---|
| **N** | 1153 | 651 | 502 | |
| **Employment** | | | | < .001 |
| Full-time | 619 (53.7%) | 373 (57.3%) | 246 (49.0%) | |
| Part-time | 171 (14.8%) | 98 (15.1%) | 73 (14.5%) | |
| Self-employed | 121 (10.5%) | 69 (10.6%) | 52 (10.4%) | |
| Unemployed and looking | 93 (8.1%) | 41 (6.3%) | 52 (10.4%) | |
| Unemployed and non-looking | 25 (2.2%) | 12 (1.8%) | 13 (2.6%) | |
| Student | 36 (3.1%) | 8 (1.2%) | 28 (5.6%) | |
| Retired | 50 (4.3%) | 32 (4.9%) | 18 (3.6%) | |
| Homecare | 38 (3.3%) | 18 (2.8%) | 20 (4.0%) | |
| **Income (individual)** | | | | .015 |
| 0 | 25 (2.2%) | 11 (1.7%) | 14 (2.8%) | |
| <10k | 128 (11.1%) | 61 (9.4%) | 67 (13.3%) | |
| 10k–15k | 72 (6.2%) | 34 (5.2%) | 38 (7.6%) | |
| 15k–20k | 54 (4.7%) | 34 (5.2%) | 20 (4.0%) | |
| 20k–25k | 63 (5.5%) | 36 (5.5%) | 27 (5.4%) | |
| 25k–50k | 244 (21.2%) | 143 (22.0%) | 101 (20.1%) | |
| 50k–100k | 384 (33.3%) | 228 (35.0%) | 156 (31.1%) | |
| >100k | 153 (13.3%) | 94 (14.4%) | 59 (11.8%) | |
| N/A | 30 (2.6%) | 10 (1.5%) | 20 (4.0%) | |
| **Household income** | | | | .124 |
| <20k | 90 (7.8%) | 48 (7.4%) | 42 (8.4%) | |
| 20k–35k | 112 (9.7%) | 71 (10.9%) | 41 (8.2%) | |
| 35k–50k | 134 (11.6%) | 78 (12.0%) | 56 (11.2%) | |
| 50k–75k | 181 (15.7%) | 100 (15.4%) | 81 (16.1%) | |
| 75k–100k | 190 (16.5%) | 113 (17.4%) | 77 (15.3%) | |
| 100k–150k | 234 (20.3%) | 121 (18.6%) | 113 (22.5%) | |
| 150k–200k | 104 (9.0%) | 66 (10.1%) | 38 (7.6%) | |
| 200k–250k | 40 (3.5%) | 22 (3.4%) | 18 (3.6%) | |
| 250k–300k | 12 (1.0%) | 9 (1.4%) | 3 (0.6%) | |
| >300k | 24 (2.1%) | 11 (1.7%) | 13 (2.6%) | |
| N/A | 32 (2.8%) | 12 (1.8%) | 20 (4.0%) | |
| **Political party** | | | | .063 |
| Democrat | 482 (41.8%) | 261 (40.1%) | 221 (44.0%) | |
| Republican | 352 (30.5%) | 217 (33.3%) | 135 (26.9%) | |
| Independent | 319 (27.7%) | 173 (26.6%) | 146 (29.1%) | |

*Notes:* This table compares socio-economic and political characteristics between the main study and the extension sample. Columns (1)–(3) report statistics for the full sample and by sample. Cells report frequencies and column shares in parentheses. p-values are from Pearson $\chi^2$ tests.

# B Additional Results in Main Study

**Table B1: Beliefs after the intervention**

| | Total | EASY | HARD | Diff. |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| **Panel A. Individual-level** | | | | |
| Initial assignment is fair | .736 | .764 | .706 | -.058*** |
| | | | | (.019) |
| Agent B needs more effort | .656 | .702 | .606 | -.097*** |
| | | | | (.019) |
| Agent A would win if assigned same version | .398 | .494 | .295 | -.199*** |
| | | | | (.038) |
| **Panel B. Group-level** | | | | |
| Agents A would win if assigned same version | .339 | .402 | .273 | -.129*** |
| | | | | (.037) |
| Improved performance with easy version | .237 | .164 | .316 | .152*** |
| | | | | (.045) |
| Probability that Agents A outperform Agents B | .654 | .648 | .661 | .013 |
| | | | | (.016) |
| Tables completed with easy version | 7.46 | 9.55 | 5.24 | -4.30*** |
| | | | | (.33) |
| Tables completed with hard version | 6.48 | 8.41 | 4.41 | -4.00*** |
| | | | | (.29) |

*Notes:* This table reports beliefs collected after the intervention. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (EASY) is exposed to the easy version of the task. The second group (HARD) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. $^{*}$ $p < .10$, $^{**}$ $p < .05$, $^{***}$ $p < .01$

**Table B2: Heterogeneities in Prior Beliefs**

| | Total | EASY | HARD | Diff. |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| **Panel A. Demographics** | | | | |
| Men | .831 | .807 | .855 | .049 |
| | | | | (.043) |
| Women | .888 | .896 | .880 | -.016 |
| | | | | (.035) |
| Age <40 | .855 | .857 | .853 | -.004 |
| | | | | (.043) |
| Age ≥40 | .867 | .852 | .883 | .031 |
| | | | | (.035) |
| White | .860 | .847 | .873 | .026 |
| | | | | (.032) |
| Non-White | .867 | .872 | .862 | -.010 |
| | | | | (.051) |
| **Panel B. Education, Employment and Income** | | | | |
| with Bachelor Degree | .865 | .842 | .890 | .048 |
| | | | | (.033) |
| w/o Bachelor Degree | .855 | .880 | .830 | -.049 |
| | | | | (.048) |
| Employed full-time | .863 | .857 | .871 | .013 |
| | | | | (.036) |
| not Employed full-time | .860 | .850 | .869 | .019 |
| | | | | (.042) |
| Annual Income ≥25k | .869 | .886 | .856 | -.030 |
| | | | | (.051) |
| Annual Income <25k | .883 | .879 | .887 | .008 |
| | | | | (.049) |
| **Panel C. Political Party** | | | | |
| Democratic | .874 | .864 | .882 | .018 |
| | | | | (.041) |
| Republican | .876 | .861 | .895 | .034 |
| | | | | (.045) |
| Independent | .827 | .831 | .821 | -.010 |
| | | | | (.058) |

*Notes:* This table reports prior beliefs on whether the versions of the task are similar. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (EASY) is exposed to the easy version of the task. The second group (HARD) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. $^{*}$ $p < .10$, $^{**}$ $p < .05$, $^{***}$ $p < .01$

**Table B3: Heterogeneities in Posterior Beliefs**

| | Total | EASY | HARD | Diff. |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| **Panel A. Demographics** | | | | |
| Men | .570 | .627 | .513 | -.114** |
| | | | | (.057) |
| Women | .562 | .659 | .449 | -.210*** |
| | | | | (.053) |
| Age <40 | .551 | .564 | .537 | -.028 |
| | | | | (.060) |
| Age ≥40 | .571 | .689 | .441 | -.247*** |
| | | | | (.050) |
| White | .564 | .636 | .487 | -.150*** |
| | | | | (.045) |
| Non-White | .558 | .638 | .471 | -.167** |
| | | | | (.073) |
| **Panel B. Education, Employment and Income** | | | | |
| with Bachelor Degree | .586 | .636 | .531 | -.105** |
| | | | | (.047) |
| w/o Bachelor Degree | .514 | .639 | .387 | -.252*** |
| | | | | (.066) |
| Employed full-time | .606 | .640 | .565 | -.076 |
| | | | | (.051) |
| not Employed full-time | .504 | .632 | .386 | -.245*** |
| | | | | (.058) |
| Annual Income ≥25k | .500 | .608 | .412 | -.195*** |
| | | | | (.075) |
| Annual Income <25k | .608 | .637 | .575 | -.062 |
| | | | | (.075) |
| **Panel C. Political Party** | | | | |
| Democratic | .582 | .680 | .493 | -.187*** |
| | | | | (.060) |
| Republican | .599 | .672 | .505 | -.167** |
| | | | | (.067) |
| Independent | .486 | .528 | .440 | -.088 |
| | | | | (.076) |

*Notes:* This table reports posterior beliefs on whether the versions of the task are similar. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (EASY) is exposed to the easy version of the task. The second group (HARD) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

**Table B4: Heterogeneities in Redistribution Decisions**

| | Total | EASY | HARD | Diff. |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| **Panel A. Demographics** | | | | |
| Men | 1.60 | 1.38 | 1.81 | .43*** |
| | | | | (.14) |
| Women | 1.71 | 1.64 | 1.79 | .15 |
| | | | | (.13) |
| Age <40 | 1.65 | 1.57 | 1.73 | .16 |
| | | | | (.14) |
| Age ≥40 | 1.68 | 1.50 | 1.86 | .35*** |
| | | | | (.12) |
| White | 1.68 | 1.52 | 1.84 | .31*** |
| | | | | (.11) |
| Non-White | 1.63 | 1.55 | 1.71 | .16 |
| | | | | (.18) |
| **Panel B. Education, Employment and Income** | | | | |
| with Bachelor Degree | 1.60 | 1.47 | 1.74 | .27** |
| | | | | (.12) |
| w/o Bachelor Degree | 1.79 | 1.67 | 1.93 | .26 |
| | | | | (.16) |
| Employed full-time | 1.60 | 1.50 | 1.72 | .21* |
| | | | | (.13) |
| not Employed full-time | 1.75 | 1.58 | 1.90 | .33** |
| | | | | (.14) |
| Annual Income ≥25k | 1.65 | 1.63 | 1.67 | .04 |
| | | | | (.17) |
| Annual Income <25k | 1.64 | 1.54 | 1.76 | .22 |
| | | | | (.20) |
| **Panel C. Political Party** | | | | |
| Democratic | 1.75 | 1.63 | 1.85 | .22 |
| | | | | (.14) |
| Republican | 1.54 | 1.41 | 1.72 | .31* |
| | | | | (.17) |
| Independent | 1.69 | 1.56 | 1.82 | .26 |
| | | | | (.19) |

*Notes:* This table reports redistribution decisions. Columns (1) refers to the whole sample. Columns (2) and (3) distinguishes by treatment arm. The first group (EASY) is exposed to the easy version of the task. The second group (HARD) is exposed to the hard version of the task. Columns (2) and (3) report statistics in each group. Columns (4) reports the difference between the two groups. Each row reports estimates from a linear model. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

**Table B5: Redistribution Decision by Context**

| | Same Context | | Different Context | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| HARD | .270*** | .270*** | .202** | .170* |
| | (.095) | (.100) | (.096) | (.100) |
| | [.004] | [.007] | [.036] | [.088] |
| Covariates | No | Yes | No | Yes |
| Effect magn. | 18% | 18% | 10% | 8% |
| Observations | 651 | 651 | 651 | 651 |
| $R^2$ | .012 | .071 | .007 | .076 |

*Notes:* This table reports treatment effects redistribution decisions for same and different context. The independent variable is spectator's assignment to HARD. Columns (1) and (2) report redistribution decisions for same context, our main dependent variable. Columns (3) and (4) report redistribution decisions for a different context. The context features unequal opportunities, but under different tasks than those in the experience. The dependent variable in all cases is the dollar transfer to the disadvantaged agent. Each row reports estimates from a linear model. Columns (2) and (4) include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}$ $p < .10$, $^{**}$ $p < .05$, $^{***}$ $p < .01$

**Table B6: Treatment Effects on Reported Emotions**

| | Compassion | Indifference | Sadness | Irritation | Anger |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| HARD | .026 | -.033 | .027 | .064*** | .051*** |
| | (.027) | (.030) | (.017) | (.017) | (.016) |
| | [.328] | [.273] | [.106] | [<.001] | [.001] |
| Mean dep. var. | .423 | .371 | .093 | .073 | .052 |
| Effect magn. | 6% | -9% | 30% | 88% | 98% |
| Observations | 651 | 651 | 651 | 651 | 651 |
| $R^2$ | .085 | .068 | .066 | .126 | .104 |

*Notes:* This table reports treatment effects on emotions reported by spectators. The dependent variable is an intensity value, ranging from 0 to 1. The independent variable is spectator's assignment to HARD. Each row reports estimates from a linear model. All columns include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

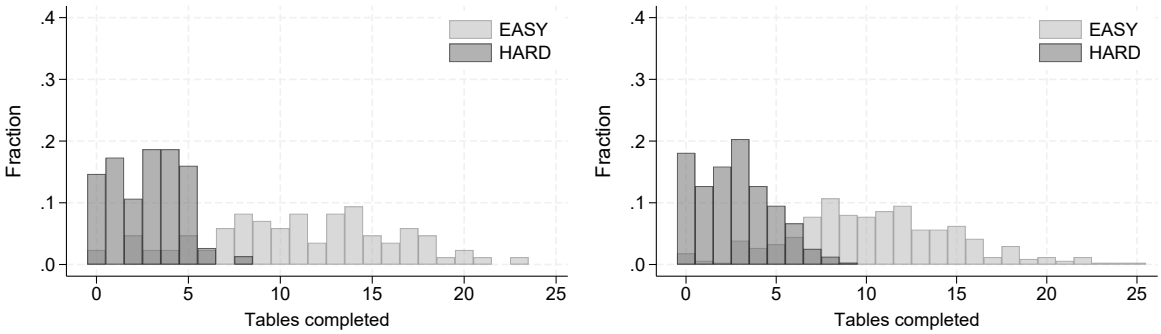
**Table B7: Mechanism Exploration**

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| HARD | .270*** | .244** | .243** | .240** | .167* | .197** | .151 |
| | (.100) | (.096) | (.098) | (.098) | (.099) | (.099) | (.096) |
| | [.007] | [.011] | [.013] | [.015] | [.092] | [.047] | [.117] |
| Compassion | | .987*** | | | | | .625*** |
| | | (.149) | | | | | (.166) |
| Indifference | | | -.816*** | | | | -.493*** |
| | | | (.136) | | | | (.147) |
| Sadness | | | | 1.083*** | | | -.001 |
| | | | | (.214) | | | (.253) |
| Irritation | | | | | 1.610*** | | 1.077*** |
| | | | | | (.227) | | (.286) |
| Anger | | | | | | 1.438*** | .347 |
| | | | | | | (.271) | (.323) |
| Covariates | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Reduction in TE | - | -10% | -10% | -11% | -38% | -27% | -44% |
| Observations | 651 | 651 | 651 | 651 | 651 | 651 | 651 |
| $R^2$ | .071 | .134 | .127 | .102 | .137 | .111 | .199 |

*Notes:* This table reports treatment effects on redistribution decisions. The dependent variable is the dollar transfer to the disadvantaged agent. The independent variable is spectator's assignment to HARD. Each row reports estimates from a linear model. All columns include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$

## Figure B1: Performance in Task

**(a) Agent's Sample** **(b) Spectator's Sample**




*Notes:* These figures show performance in the effort task by treatment arm. Subfigure (a) focuses in the agent's sample. Subfigure (b) focuses in the spectator's sample. Both subfigures plot an histogram of the tables completed. Each bar plots the fraction of participants completing a discrete number of tables. The light bars plot completion among participants in `EASY`. The dark bars plot completion among participants in `HARD`.

**Table B8: Mechanism Exploration: Willingness to Keep Redistributing**

| | Willingness to keep redistributing | | | | | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| HARD | .103** | .089* | .056 | .081* | .084 | .066 |
| | (.046) | (.049) | (.051) | (.048) | (.052) | (.049) |
| Relevance of Luck | | | .252*** | | | |
| | | | (.091) | | | |
| Assignment when Equal Opp. | | | | .116*** | | |
| | | | | (.022) | | |
| Understanding of Agent B | | | | | .026 | |
| | | | | | (.095) | |
| Anger | | | | | | .496*** |
| | | | | | | (.124) |
| Costless transfer FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Covariates | No | Yes | Yes | Yes | Yes | Yes |
| Reduction in TE | – | 0% | -37% | -9% | -6% | -30% |
| Observations | 475 | 475 | 475 | 475 | 475 | 475 |
| $R^2$ | .037 | .134 | .149 | .184 | .134 | .173 |

*Notes:* This table reports treatment effects on willingness to keep redistributing once redistribution becomes costly. The dependent variable is an indicator equal to one if the spectator assigns a positive amount to the disadvantaged agent in the costly redistribution decision. The sample is restricted to spectators who assigned a positive amount to the disadvantaged agent in the costless redistribution decision. The independent variable is spectator's assignment to HARD. All columns include fixed effects for the exact amount redistributed in the costless decision. Column (1) includes no additional covariates. Columns (2) to (6) include socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Race enters as a set of dummy variables. Columns (3) to (6) add, respectively, relevance of luck, assignment when equal opportunities, understanding of Agent B, and anger. Understanding of Agent B, anger, and relevance of luck are scaled from 0 to 1. Assignment when Equal Opp. is the dollar transfer to Agent B in the equal-opportunity merit decision. Reduction in TE reports the percentage reduction in the estimated HARD coefficient relative to column (2). Robust standard errors are reported in parentheses. $^{*}$ $p < .10$, $^{**}$ $p < .05$, $^{***}$ $p < .01$.

**Table B9: Emotional Responses and Willingness to Keep Redistributing**

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| HARD | .094* | .095* | .086* | .090* | .082* | .066 | .075 |
| | (.049) | (.048) | (.048) | (.049) | (.050) | (.049) | (.048) |
| Compassion | | .304*** | | | | | .155* |
| | | (.075) | | | | | (.083) |
| Indifference | | | -.329*** | | | | -.269*** |
| | | | (.067) | | | | (.072) |
| Sadness | | | | .339*** | | | .107 |
| | | | | (.104) | | | (.124) |
| Irritation | | | | | .171 | | -.219 |
| | | | | | (.125) | | (.137) |
| Anger | | | | | | .496*** | .478*** |
| | | | | | | (.124) | (.147) |
| Costless transfer FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Covariates | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Reduction in TE | - | +1% | -8% | -4% | -13% | -30% | -21% |
| p-value | .057 | .051 | .074 | .065 | .099 | .180 | .118 |
| Observations | 475 | 475 | 475 | 475 | 475 | 475 | 475 |
| $R^2$ | .142 | .174 | .186 | .162 | .147 | .173 | .224 |

*Notes:* This table reports treatment effects on willingness to keep redistributing once redistribution becomes costly. The dependent variable is an indicator equal to one if the spectator assigns a positive amount to the disadvantaged agent in the costly redistribution decision. The sample is restricted to spectators who assigned a positive amount to the disadvantaged agent in the costless redistribution decision. The independent variable is spectator's assignment to HARD. All columns include fixed effects for the exact amount redistributed in the costless decision. All columns include a set of socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Race enters as a set of dummy variables. Compassion, indifference, sadness, irritation, and anger are scaled from 0 to 1. Reduction in TE reports the percentage change in the estimated treatment effect relative to column (1). Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

# C Additional Results in Extensions

**Table C1: Prior Beliefs in the Extension Sample**

| | Total | MIXED | REPORTS | STATISTICS | p-value |
|---|---|---|---|---|---|
| *Panel A. Perceived differences between task versions* | | | | | |
| Versions are approximately the same | 82.9% | 80.6% | 83.6% | 84.3% | .634 |
| EASY version expected to be better | 12.2% | 13.3% | 11.5% | 11.6% | .851 |
| HARD version expected to be better | 5.0% | 6.1% | 4.8% | 4.1% | .700 |
| *Panel B. Perceived determinants of success* | | | | | |
| Version ranked least relevant | 76.9% | 72.1% | 75.8% | 82.6% | .069 |
| Observations | 502 | 165 | 165 | 172 | |

*Notes:* This table reports prior beliefs about the task among spectators in the extension sample. The sample includes spectators assigned to MIXED, REPORTS, and STATISTICS. Percentages are column shares. Panel A reports beliefs about whether the two task versions differ in expected performance. Panel B reports rankings of the perceived determinants of success. p-values are from chi-squared tests comparing the three extension treatment arms.

**Table C2: Beliefs in EASY, HARD, and MIXED**

| | EASY | HARD | MIXED | MIXED−EASY | MIXED−HARD |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| **Panel A. Prior Beliefs** | | | | | |
| Versions are similar | .854 | .870 | .806 | -.048 | -.064* |
| | | | | (.036) | (.036) |
| Easy version is better | .107 | .111 | .133 | .026 | .022 |
| | | | | (.031) | (.032) |
| Version ranked least important | .821 | .816 | .721 | -.100** | -.095** |
| | | | | (.041) | (.041) |
| **Panel B. Posterior Beliefs** | | | | | |
| Versions are similar | .637 | .483 | .200 | -.437*** | -.283*** |
| | | | | (.041) | (.042) |
| Easy version is better | .345 | .489 | .764 | .418*** | .275*** |
| | | | | (.042) | (.044) |
| Luck weighs less than ability and effort | .690 | .511 | .218 | -.472*** | -.293*** |
| | | | | (.041) | (.043) |

*Notes:* This table reports spectators' beliefs about the two versions of the task in EASY, HARD, and MIXED. Columns (1) to (3) report the share of spectators in each treatment arm. Spectators in EASY and HARD experience the easy and the hard version of the task, respectively; spectators in MIXED experience both versions, for two minutes each and in randomized order. Columns (4) and (5) report the difference between MIXED and EASY and between MIXED and HARD, estimated from linear regressions of each outcome on a treatment indicator, with robust standard errors in parentheses. MIXED was collected in a separate wave. Panel A reports beliefs elicited before the experience and Panel B beliefs elicited after the redistribution decisions. Variables are defined as in Table 1. "Version ranked least important" and "Luck weighs less than ability and effort" are elicited differently and are not directly comparable. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C3: Beliefs in MIXED, REPORTS, and STATISTICS**

| | Total | MIXED | REPORTS | STATS | MIXED− REPORTS | MIXED− STATS |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| **Panel A. Prior Beliefs** | | | | | | |
| Versions are similar | .829 | .806 | .836 | .843 | -.030 | -.037 |
| | | | | | (.042) | (.042) |
| Easy version is better | .122 | .133 | .115 | .116 | .018 | .017 |
| | | | | | (.036) | (.036) |
| Version ranked least important | .769 | .721 | .758 | .826 | -.036 | -.104** |
| | | | | | (.048) | (.045) |
| **Panel B. Posterior Beliefs** | | | | | | |
| Versions are similar | .333 | .200 | .333 | .459 | -.133*** | -.259*** |
| | | | | | (.048) | (.049) |
| Easy version is better | .639 | .764 | .642 | .517 | .121** | .246*** |
| | | | | | (.050) | (.051) |
| Luck weighs less than ability and effort | .396 | .218 | .448 | .517 | -.230*** | -.299*** |
| | | | | | (.050) | (.050) |

*Notes:* This table reports spectators' beliefs about the two versions of the task in the extension. Column (1) refers to all spectators in the extension; columns (2) to (4) report the share of spectators in each treatment arm. Spectators in MIXED experience both versions of the task; spectators in REPORTS receive qualitative accounts of agents' experiences; spectators in STATS (STATISTICS) receive quantitative information about agents' performance in each version. Columns (5) and (6) report the difference between MIXED and REPORTS and between MIXED and STATISTICS, estimated from linear regressions of each outcome on a treatment indicator, with robust standard errors in parentheses. Panel A reports beliefs elicited before the intervention and Panel B beliefs elicited after the redistribution decisions. Variables are defined as in Table 1. "Version ranked least important" and "Luck weighs less than ability and effort" are elicited differently and are not directly comparable. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C4: Redistribution in EASY. HARD, MIXED**

| | Costless | | Costly | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| HARD | .270*** | .269*** | .345*** | .336*** |
| | (.095) | (.099) | (.099) | (.103) |
| | [.004] | [.007] | [< .001] | [.001] |
| MIXED | .595*** | .603*** | .717*** | .746*** |
| | (.114) | (.115) | (.126) | (.130) |
| | [< .001] | [< .001] | [< .001] | [< .001] |
| MIXED - HARD | .324*** | .334*** | .371*** | .409*** |
| | (.117) | (.119) | (.132) | (.135) |
| | [.006] | [.005] | [.005] | [.002] |
| Dep. var. mean | 1.533 | 1.533 | .750 | .750 |
| Effect magn. HARD vs EASY | 18% | 18% | 46% | 45% |
| Effect magn. MIXED vs EASY | 39% | 39% | 96% | 99% |
| Effect magn. MIXED vs HARD | 18% | 19% | 34% | 38% |
| Controls | NO | YES | NO | YES |
| Observations | 816 | 816 | 816 | 816 |
| $R^2$ | .033 | .088 | .042 | .106 |

*Notes:* The dependent variable is the amount allocated to Agent B. EASY is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C5: Main Beliefs in EASY. HARD, MIXED**

| | Posterior comparison | | Posterior luck | | Posterior fairness | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| HARD | .389*** | .341*** | 13.585*** | 12.371*** | -.231*** | -.195** |
| | (.097) | (.099) | (2.228) | (2.316) | (.076) | (.078) |
| MIXED | 1.381*** | 1.292*** | 34.946*** | 32.757*** | -.699*** | -.666*** |
| | (.118) | (.121) | (2.700) | (2.813) | (.092) | (.094) |
| MIXED - HARD | .992*** | .951*** | 21.362*** | 20.386*** | -.468*** | -.471*** |
| | (.119) | (.121) | (2.730) | (2.808) | (.093) | (.094) |
| Controls | NO | YES | NO | YES | NO | YES |
| Observations | 816 | 816 | 816 | 816 | 816 | 816 |
| $R^2$ | .146 | .219 | .172 | .221 | .066 | .152 |

*Notes:* The dependent variables are posterior beliefs about the advantage provided by the task version, the perceived role of luck in determining which agent solved more tables, and the perceived fairness of the initial bonus allocation. EASY is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C6: Other Beliefs in EASY. HARD, MIXED**

| | Estimation of A | | Estimation of B | | (A-B)/B | | Probability | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| HARD | -4.303*** | -4.139*** | -4.004*** | -3.787*** | .152*** | .132** | -1.300 | -.032 |
| | (.331) | (.347) | (.291) | (.313) | (.045) | (.052) | (1.650) | (1.641) |
| MIXED | -2.348*** | -2.230*** | -4.250*** | -4.035*** | 1.029*** | .998*** | -10.354*** | -9.124*** |
| | (.417) | (.431) | (.361) | (.386) | (.111) | (.112) | (2.131) | (2.140) |
| MIXED - HARD | 1.956*** | 1.908*** | -.246 | -.247 | .876*** | .865*** | -9.053*** | -9.091*** |
| | (.398) | (.411) | (.354) | (.367) | (.117) | (.117) | (2.108) | (2.084) |
| Controls | NO | YES | NO | YES | NO | YES | NO | YES |
| Observations | 816 | 816 | 816 | 816 | 805 | 805 | 816 | 816 |
| $R^2$ | .170 | .224 | .226 | .276 | .192 | .239 | .033 | .134 |

*Notes:* The dependent variables are posterior beliefs about average performance in the easy and hard versions, the perceived proportional performance gap between the two versions, and the perceived probability that Agent B outperforms Agent A. EASY is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}$ $p < .10$, $^{**}$ $p < .05$, $^{***}$ $p < .01$.

**Table C7: Counterfactual Beliefs in EASY. HARD, MIXED**

| | Hard Work (inv.) | | Counterfactual GL | | Counterfactual IL | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| HARD | .387*** | .316*** | .209*** | .209*** | .199*** | .188*** |
| | (.076) | (.075) | (.060) | (.062) | (.037) | (.037) |
| MIXED | .761*** | .728*** | .145** | .153** | .282*** | .262*** |
| | (.092) | (.092) | (.073) | (.076) | (.044) | (.045) |
| MIXED - HARD | .374*** | .412*** | -.064 | -.056 | .083* | .074 |
| | (.093) | (.091) | (.074) | (.076) | (.045) | (.045) |
| Controls | NO | YES | NO | YES | NO | YES |
| Observations | 816 | 816 | 816 | 816 | 816 | 816 |
| $R^2$ | .082 | .205 | .015 | .080 | .058 | .145 |

*Notes:* The dependent variables are posterior beliefs about whether Agent B could have solved more tables through greater effort, group-level counterfactual beliefs about which agents would solve more tables if assigned the same version, and individual-level counterfactual beliefs about which agent in the pair would have solved more tables if assigned the same version. EASY is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C8: Emotions in EASY. HARD, MIXED**

| | Anger | | Compassion | | Sadness | | Indifference | | Irritation | |
|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| HARD | .140*** | .159*** | .056 | .074 | .057 | .080 | -.085 | -.094 | .202*** | .195*** |
| | (.046) | (.048) | (.077) | (.080) | (.048) | (.050) | (.083) | (.086) | (.051) | (.053) |
| MIXED | .336*** | .330*** | .241*** | .272*** | .060 | .067 | -.289*** | -.307*** | .368*** | .351*** |
| | (.056) | (.058) | (.093) | (.097) | (.058) | (.060) | (.101) | (.105) | (.062) | (.064) |
| MIXED - HARD | .196*** | .172*** | .185** | .198** | .003 | -.013 | -.204** | -.213** | .166*** | .156** |
| | (.056) | (.058) | (.094) | (.096) | (.058) | (.060) | (.102) | (.105) | (.063) | (.064) |
| Controls | NO | YES | NO | YES | NO | YES | NO | YES | NO | YES |
| Observations | 816 | 816 | 816 | 816 | 816 | 816 | 816 | 816 | 816 | 816 |
| $R^2$ | .043 | .111 | .008 | .073 | .002 | .055 | .010 | .074 | .044 | .118 |

*Notes:* The dependent variables are self-reported emotional responses after learning about the initial bonus allocation, including anger, compassion, sadness, indifference, and irritation. EASY is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}$ $p < .10$, $^{**}$ $p < .05$, $^{***}$ $p < .01$.

**Table C9: Perspective-taking in EASY. HARD, MIXED**

| | Posterior empathy A | | Posterior empathy B | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| HARD | -.439*** | -.420*** | .510*** | .530*** |
| | (.060) | (.063) | (.064) | (.067) |
| MIXED | .013 | .048 | .443*** | .479*** |
| | (.068) | (.073) | (.078) | (.082) |
| MIXED - HARD | .452*** | .468*** | -.067 | -.052 |
| | (.073) | (.074) | (.074) | (.075) |
| Controls | NO | YES | NO | YES |
| Observations | 816 | 816 | 816 | 816 |
| $R^2$ | .076 | .135 | .081 | .148 |

*Notes:* The dependent variables are self-reported measures of how easy it was for spectators to imagine Agent A's and Agent B's experience with the task. EASY is the omitted category. Control specifications include sex, age, race, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C10: Staying in redistribution once a cost is introduced: EASY, HARD, and MIXED**

| | (1) | (2) |
|---|---|---|
| HARD | .106** | .101** |
| | (.046) | (.048) |
| | [.021] | [.035] |
| MIXED | .192*** | .208*** |
| | (.053) | (.056) |
| | [< .001] | [< .001] |
| MIXED - HARD | .085 | .107* |
| | (.052) | (.055) |
| | [.102] | [.052] |
| Dep. var. mean | .548 | .548 |
| sg_b fixed effects | YES | YES |
| Controls | NO | YES |
| Observations | 611 | 611 |
| $R^2$ | .047 | .120 |

*Notes:* The dependent variable is an indicator equal to one if a spectator chose a positive transfer in the costless decision (sg_b $> 0$) and also chose a positive transfer in the costly decision (gamma_b $> 0$). The sample is restricted to EASY, HARD, and MIXED spectators and keeps only spectators with sg_b $> 0$. EASY is the omitted category. All specifications include fixed effects for the exact amount redistributed in the costless decision. Column (2) includes sex, age, race, education, employment status, individual income, household income, and political party affiliation, with race entered through all variables matching race*. Robust standard errors are reported in parentheses. Two-sided p-values are reported in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C11: Mechanism Exploration: Willingness to Keep Redistributing in MIXED**

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| *Panel A. MIXED vs. EASY* | | | | | | |
| MIXED | .198*** | .212*** | .196*** | .213*** | .193*** | .177*** |
| | (.054) | (.058) | (.066) | (.057) | (.062) | (.059) |
| Relevance of Luck | | | .055 | | | |
| | | | (.108) | | | |
| Assignment when Equal Opp. | | | | .093*** | | |
| | | | | (.024) | | |
| Understanding of Agent B | | | | | .105 | |
| | | | | | (.103) | |
| Anger | | | | | | .359*** |
| | | | | | | (.133) |
| Change in TE | – | – | -7% | +0% | -9% | -17% |
| Observations | 373 | 373 | 373 | 373 | 373 | 373 |
| $R^2$ | .058 | .150 | .151 | .189 | .153 | .168 |
| *Panel B. MIXED vs. HARD* | | | | | | |
| MIXED | .085 | .081 | .077 | .088 | .082 | .072 |
| | (.052) | (.057) | (.059) | (.055) | (.057) | (.057) |
| Relevance of Luck | | | .022 | | | |
| | | | (.106) | | | |
| Assignment when Equal Opp. | | | | .115*** | | |
| | | | | (.023) | | |
| Understanding of Agent B | | | | | .080 | |
| | | | | | (.116) | |
| Anger | | | | | | .250** |
| | | | | | | (.119) |
| Change in TE | – | – | -4% | +10% | +1% | -11% |
| Observations | 374 | 374 | 374 | 374 | 374 | 374 |
| $R^2$ | .029 | .147 | .147 | .208 | .148 | .158 |
| Costless transfer FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Covariates | No | Yes | Yes | Yes | Yes | Yes |

*Notes:* This table reports treatment effects on willingness to keep redistributing once redistribution becomes costly. The dependent variable is an indicator equal to one if the spectator assigns a positive amount to the disadvantaged agent in the costly redistribution decision. The sample is restricted to spectators who assigned a positive amount to the disadvantaged agent in the costless redistribution decision. Panel A compares spectators assigned to MIXED to spectators assigned to EASY. Panel B compares spectators assigned to MIXED to spectators assigned to HARD. In each panel, the reported treatment coefficient is the effect of assignment to MIXED relative to the omitted group. All columns include fixed effects for the exact amount redistributed in the costless decision. Column (1) includes no additional covariates. Columns (2) to (6) include socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Race enters as a set of dummy variables. Columns (3) to (6) add, respectively, relevance of luck, assignment when equal opportunities, understanding of Agent B, and anger. Understanding of Agent B, anger, and relevance of luck are scaled from 0 to 1. Assignment when Equal Opp. is the dollar transfer to Agent B in the equal-opportunity merit decision. Change in TE reports the percentage change in the estimated MIXED coefficient relative to column (2) within the same panel. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C12: Redistribution in MIXED, REPORTS, and STATISTICS**

| | Costless | | Costly | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| REPORTS | -.236* | -.230* | -.345** | -.294* |
| | (.127) | (.129) | (.150) | (.155) |
| | [.063] | [.075] | [.021] | [.058] |
| STAT | -.714*** | -.717*** | -.554*** | -.563*** |
| | (.128) | (.130) | (.141) | (.144) |
| | [< .001] | [< .001] | [< .001] | [< .001] |
| STAT - REPORTS | -.478*** | -.487*** | -.208 | -.269* |
| | (.121) | (.130) | (.136) | (.142) |
| | [< .001] | [< .001] | [.126] | [.058] |
| Dep. var. mean | 2.127 | 2.127 | 1.467 | 1.467 |
| Effect magn. REPORTS vs MIXED | -11% | -11% | -24% | -20% |
| Effect magn. STAT vs MIXED | -34% | -34% | -38% | -38% |
| Effect magn. STAT vs REPORTS | -22% | -23% | -14% | -18% |
| Controls | NO | YES | NO | YES |
| Observations | 502 | 502 | 502 | 502 |
| $R^2$ | .064 | .162 | .030 | .147 |

*Notes:* The dependent variable is the amount allocated to Agent B. MIXED is the omitted category. Specifications with controls include sex, age, race indicators, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

## Table C13: Main Beliefs in MIXED, REPORTS, and STATISTICS

| | Posterior comparison | | Posterior luck | | Posterior fairness | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| STAT | -.809*** | -.857*** | -20.754*** | -21.789*** | .538*** | .602*** |
| | (.152) | (.150) | (3.249) | (3.246) | (.111) | (.109) |
| REPORTS | -.806*** | -.785*** | -15.618*** | -15.695*** | .442*** | .440*** |
| | (.145) | (.148) | (3.019) | (3.104) | (.109) | (.112) |
| REPORTS - STAT | .003 | .072 | 5.136 | 6.094* | -.095 | -.162 |
| | (.144) | (.154) | (3.254) | (3.349) | (.106) | (.112) |
| Controls | NO | YES | NO | YES | NO | YES |
| Observations | 502 | 502 | 502 | 502 | 502 | 502 |
| $R^2$ | .074 | .207 | .084 | .208 | .053 | .147 |

*Notes:* The dependent variables are posterior beliefs about the advantage provided by the task version, the perceived role of luck in determining which agent solved more tables, and the perceived fairness of the initial bonus allocation. MIXED is the omitted category. Specifications with controls include sex, age, race indicators, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

**Table C14: Other Beliefs in MIXED, REPORTS, and STATISTICS**

| | Estimation of A | | Estimation of B | | (A-B)/B | | Probability | |
|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| STAT | 3.416*** | 3.443*** | .017 | .193 | .739*** | .690*** | -2.843 | -2.693 |
| | (.394) | (.459) | (.342) | (.419) | (.136) | (.143) | (2.410) | (2.432) |
| REPORTS | 2.776*** | 2.670*** | 3.048*** | 3.085*** | -.624*** | -.645*** | 4.297* | 4.048* |
| | (.723) | (.748) | (.638) | (.662) | (.118) | (.120) | (2.251) | (2.349) |
| REPORTS - STAT | -.641 | -.773 | 3.032*** | 2.892*** | -1.364*** | -1.335*** | 7.140*** | 6.742*** |
| | (.675) | (.682) | (.594) | (.598) | (.092) | (.105) | (2.144) | (2.390) |
| Controls | NO | YES | NO | YES | NO | YES | NO | YES |
| Observations | 502 | 502 | 502 | 502 | 497 | 497 | 502 | 502 |
| $R^2$ | .067 | .153 | .079 | .176 | .221 | .287 | .020 | .104 |

*Notes:* The dependent variables are posterior beliefs about average performance in the EASY and HARD versions, the perceived proportional performance gap between the two versions, and the perceived probability that Agent B outperforms Agent A. MIXED is the omitted category. Specifications with controls include sex, age, race indicators, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

**Table C15: Counterfactual Beliefs in MIXED, REPORTS, and STATISTICS**

| | Hard Work (inv.) | | Counterfactual GL | | Counterfactual IL | |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| STAT | -.463*** | -.542*** | -.071 | -.106 | -.224*** | -.270*** |
| | (.109) | (.109) | (.090) | (.091) | (.050) | (.049) |
| REPORTS | -.406*** | -.427*** | .109 | .076 | -.067 | -.082* |
| | (.109) | (.110) | (.083) | (.087) | (.047) | (.049) |
| REPORTS - STAT | .057 | .115 | .181** | .182** | .157*** | .188*** |
| | (.108) | (.112) | (.083) | (.090) | (.052) | (.052) |
| Controls | NO | YES | NO | YES | NO | YES |
| Observations | 502 | 502 | 502 | 502 | 502 | 502 |
| $R^2$ | .041 | .192 | .009 | .113 | .042 | .201 |

*Notes:* The dependent variables are posterior beliefs about whether Agent B could have solved more tables through greater effort, group-level counterfactual beliefs about which agents would solve more tables if assigned the same version, and individual-level counterfactual beliefs about which agent in the pair would have solved more tables if assigned the same version. MIXED is the omitted category. Specifications with controls include sex, age, race indicators, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

## Table C16: Emotions in MIXED, REPORTS, and STATISTICS

| | Anger | | Compassion | | Sadness | | Indifference | | Irritation | |
|---|---|---|---|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| STAT | -.183** | -.201*** | -.108 | -.154 | .050 | .035 | .344*** | .417*** | -.146* | -.194** |
| | (.075) | (.078) | (.111) | (.117) | (.067) | (.070) | (.113) | (.116) | (.086) | (.090) |
| REPORTS | -.170** | -.165** | -.055 | -.045 | .000 | .002 | .200* | .166 | -.230*** | -.239*** |
| | (.077) | (.078) | (.103) | (.107) | (.068) | (.071) | (.106) | (.109) | (.079) | (.080) |
| REPORTS - STAT | .013 | .037 | .053 | .109 | -.050 | -.033 | -.144 | -.251** | -.084 | -.044 |
| | (.067) | (.071) | (.102) | (.106) | (.070) | (.075) | (.115) | (.120) | (.076) | (.083) |
| Controls | NO | YES | NO | YES | NO | YES | NO | YES | NO | YES |
| Observations | 502 | 502 | 502 | 502 | 502 | 502 | 502 | 502 | 502 | 502 |
| $R^2$ | .015 | .124 | .002 | .098 | .001 | .057 | .019 | .144 | .016 | .100 |

*Notes:* The dependent variables are self-reported emotional responses after learning about the initial bonus allocation, including anger, compassion, sadness, indifference, and irritation. MIXED is the omitted category. Specifications with controls include sex, age, race indicators, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

## Table C17: Perspective-Taking in MIXED, REPORTS, and STATISTICS

| | Posterior empathy A | | Posterior empathy B | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| STAT | -.477*** | -.475*** | -.542*** | -.539*** |
| | (.085) | (.088) | (.091) | (.094) |
| REPORTS | -.400*** | -.425*** | -.430*** | -.456*** |
| | (.083) | (.087) | (.091) | (.094) |
| REPORTS - STAT | .077 | .050 | .112 | .083 |
| | (.088) | (.092) | (.096) | (.100) |
| Controls | NO | YES | NO | YES |
| Observations | 502 | 502 | 502 | 502 |
| $R^2$ | .067 | .143 | .071 | .159 |

*Notes:* The dependent variables are self-reported measures of how easy it was for spectators to imagine Agent A's and Agent B's experience with the task. MIXED is the omitted category. Specifications with controls include sex, age, race indicators, education, employment status, individual income, household income, and political party affiliation. Robust standard errors are reported in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

**Table C18: Staying in redistribution once a cost is introduced: MIXED, REPORTS, and STAT**

| | (1) | (2) |
|---|---|---|
| REPORTS | -.113* | -.103* |
| | (.060) | (.062) |
| | [.060] | [.097] |
| STAT | -.059 | -.092 |
| | (.063) | (.068) |
| | [.349] | [.176] |
| STAT - REPORTS | .054 | .011 |
| | (.061) | (.067) |
| | [.376] | [.870] |
| Dep. var. mean | .593 | .593 |
| sg_b fixed effects | YES | YES |
| Controls | NO | YES |
| Observations | 391 | 391 |
| $R^2$ | .052 | .149 |

*Notes:* The dependent variable is an indicator equal to one if a spectator chose a positive transfer in the costless decision (sg_b $> 0$) and also chose a positive transfer in the costly decision (gamma_b $> 0$). The sample is restricted to MIXED, REPORTS, and STAT spectators and keeps only spectators with sg_b $> 0$. MIXED is the omitted category. All specifications include fixed effects for the exact amount redistributed in the costless decision. Column (2) includes sex, age, race, education, employment status, individual income, household income, and political party affiliation. Race enters through all variables matching race*. Robust standard errors are reported in parentheses. Two-sided p-values are reported in square brackets. $^{*}$ $p < .10$, $^{**}$ $p < .05$, $^{***}$ $p < .01$.

**Table C19: Mechanism Exploration: Willingness to Keep Redistributing in Information Treatments**

| | (1) | (2) | (3) | (4) | (5) | (6) |
|---|---|---|---|---|---|---|
| *Panel A. REPORTS vs. MIXED* | | | | | | |
| REPORTS | -.115* | -.078 | -.078 | -.089 | -.054 | -.084 |
| | (.060) | (.064) | (.066) | (.062) | (.066) | (.064) |
| Relevance of Luck | | | .001 | | | |
| | | | (.127) | | | |
| Assignment when Equal Opp. | | | | .066** | | |
| | | | | (.028) | | |
| Understanding of Agent B | | | | | .179 | |
| | | | | | (.110) | |
| Anger | | | | | | -.127 |
| | | | | | | (.151) |
| Change in TE | – | – | 0% | +13% | -31% | +7% |
| Observations | 274 | 274 | 274 | 274 | 274 | 274 |
| $R^2$ | .051 | .224 | .224 | .247 | .232 | .226 |
| *Panel B. STAT vs. MIXED* | | | | | | |
| STAT | -.067 | -.094 | -.144* | -.115 | -.077 | -.093 |
| | (.065) | (.071) | (.073) | (.071) | (.075) | (.072) |
| Relevance of Luck | | | -.355*** | | | |
| | | | (.121) | | | |
| Assignment when Equal Opp. | | | | .100*** | | |
| | | | | (.027) | | |
| Understanding of Agent B | | | | | .090 | |
| | | | | | (.126) | |
| Anger | | | | | | .013 |
| | | | | | | (.147) |
| Change in TE | – | – | +54% | +22% | -18% | 0% |
| Observations | 253 | 253 | 253 | 253 | 253 | 253 |
| $R^2$ | .036 | .189 | .220 | .240 | .191 | .189 |
| Costless transfer FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Covariates | No | Yes | Yes | Yes | Yes | Yes |

*Notes:* This table reports treatment effects on willingness to keep redistributing once redistribution becomes costly. The dependent variable is an indicator equal to one if the spectator assigns a positive amount to the disadvantaged agent in the costly redistribution decision. The sample is restricted to spectators who assigned a positive amount to the disadvantaged agent in the costless redistribution decision. Panel A compares spectators assigned to REPORTS to spectators assigned to MIXED. Panel B compares spectators assigned to STAT to spectators assigned to MIXED. In each panel, MIXED is the omitted category. All columns include fixed effects for the exact amount redistributed in the costless decision. Column (1) includes no additional covariates. Columns (2) to (6) include socio-demographic covariates: sex, age, race, education, employment status, individual and household income, and political party affiliation. Race enters as a set of dummy variables. Columns (3) to (6) add, respectively, relevance of luck, assignment when equal opportunities, understanding of Agent B, and anger. Understanding of Agent B, anger, and relevance of luck are scaled from 0 to 1. Assignment when Equal Opp. is the dollar transfer to Agent B in the equal-opportunity merit decision. Change in TE reports the percentage change in the absolute value of the treatment coefficient relative to column (2) within the same panel. Negative values indicate attenuation toward zero; positive values indicate that the coefficient becomes larger in absolute value. Robust standard errors are reported in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C20: Redistribution in HARD, REPORTS, and STATISTICS**

| | Costless | | Costly | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| REPORTS | .088 | .142 | .026 | .091 |
| | (.110) | (.113) | (.128) | (.131) |
| | [.423] | [.209] | [.839] | [.489] |
| STAT | -.390*** | -.329*** | -.182 | -.167 |
| | (.111) | (.115) | (.117) | (.124) |
| | [< .001] | [.004] | [.118] | [.179] |
| Dep. var. mean | 1.803 | 1.803 | 1.095 | 1.095 |
| Effect magn. REPORTS vs HARD | 5% | 8% | 2% | 8% |
| Effect magn. STAT vs HARD | -22% | -18% | -17% | -15% |
| Controls | NO | YES | NO | YES |
| Observations | 652 | 652 | 652 | 652 |
| $R^2$ | .025 | .084 | .004 | .097 |

*Notes:* The dependent variable is the amount allocated to Agent B. HARD is the omitted category. Robust standard errors are reported in parentheses. Two-sided p-values are reported in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

**Table C21: Order Effects within MIXED**

| | Redistribution | | Posterior beliefs | | | | |
|---|---|---|---|---|---|---|---|
| | Costless | Costly | Comparison | Luck | Fairness | (A-B)/B | Probability |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| *Hard first* | -.090 | -.199 | -.171 | 5.408 | .193 | -.235 | 3.740 |
| | (.188) | (.218) | (.217) | (4.269) | (.161) | (.219) | (3.545) |
| | [.635] | [.362] | [.431] | [.205] | [.230] | [.285] | [.291] |
| Mean (*Easy first*) | 2.173 | 1.568 | 6.111 | 60.914 | 2.259 | 1.311 | 22.951 |
| Observations | 165 | 165 | 165 | 165 | 165 | 161 | 165 |
| $R^2$ | .001 | .005 | .004 | .010 | .009 | .007 | .007 |

*Notes:* This table compares spectators in MIXED by the order in which they completed the two versions of the task. The independent variable equals one if the spectator completed the hard version first; spectators who completed the easy version first are the omitted category. The dependent variables are the amount allocated to Agent B when redistribution is costless (column 1) and costly (column 2), posterior beliefs about the advantage provided by the task version (column 3), the perceived role of luck in determining which agent solved more tables (column 4), the perceived fairness of the initial bonus allocation (column 5), the perceived proportional performance gap between the two versions (column 6), and the perceived probability that Agent B outperforms Agent A (column 7). Column 6 excludes spectators who estimated zero tables solved in the hard version. Robust standard errors are reported in parentheses and two-sided p-values in square brackets. * $p < .10$, ** $p < .05$, *** $p < .01$.

# D Multiple-Testing Adjustments

Tables D1–D3 report Benjamini–Hochberg adjusted p-values for the preregistered multiple-testing families. In the main experiment, the adjustment is applied to the preregistered belief outcomes and to the exploratory associations between posterior beliefs and redistribution. In the extension, the adjustment is applied across the three treatment comparisons relative to `EASY` for each preregistered outcome family. The results are largely unchanged after adjustment.

## Table D1: Multiple-testing adjustments for preregistered outcomes in the main experiment

| Outcome | Comparison | Estimate | SE | $p$ | $q$ | $N$ |
|---|---|---|---|---|---|---|
| *Panel A. Preregistered H2: group-level belief outcomes* | | | | | | |
| Performance gap (A-B) | HARD vs EASY | -0.342* | (0.180) | 0.058 | 0.073 | 651 |
| Role of luck | HARD vs EASY | 12.553*** | (2.357) | <0.001 | <0.001 | 651 |
| Probability B outperforms A | HARD vs EASY | -0.110 | (1.647) | 0.947 | 0.947 | 651 |
| Group-level counterfactual / hard-work belief | HARD vs EASY | 0.232*** | (0.062) | <0.001 | <0.001 | 651 |
| Version advantage/intensity | HARD vs EASY | 0.337*** | (0.099) | <0.001 | 0.001 | 651 |
| *Panel B. Secondary individual-level belief outcomes* | | | | | | |
| Individual counterfactual: not Agent A/easy would win | HARD vs EASY | 0.192*** | (0.039) | <0.001 | <0.001 | 651 |
| Hard-work belief (inverse) | HARD vs EASY | 0.315*** | (0.075) | <0.001 | <0.001 | 651 |
| *Panel C. Exploratory associations between posterior beliefs and redistribution* | | | | | | |
| Role of luck | association with costless redistribution | 0.016*** | (0.002) | <0.001 | <0.001 | 651 |
| Performance gap (A-B) | association with costless redistribution | 0.065** | (0.028) | 0.018 | 0.023 | 651 |
| Frequency inverse | association with costless redistribution | 0.003 | (0.003) | 0.202 | 0.202 | 651 |
| Unfairness of initial allocation | association with costless redistribution | 1.947*** | (0.136) | <0.001 | <0.001 | 651 |
| Hard-work belief inverse | association with costless redistribution | 0.411*** | (0.053) | <0.001 | <0.001 | 651 |

*Notes:* This table reports raw and Benjamini–Hochberg adjusted p-values for the preregistered multiple-testing families in the main experiment. Panel A adjusts across the five group-level belief outcomes associated with the belief-updating hypothesis. Panel B adjusts across the two secondary individual-level belief outcomes. Panel C adjusts across the exploratory associations between posterior beliefs and costless redistribution. All estimates use the preregistered analysis sample and include socio-demographic controls: sex, age, race, education, employment status, individual income, household income, and political party affiliation. Race is coded as a categorical variable distinguishing White only, Black only, Asian only, Native only, Pacific only, Other only, multiple races, and no race selected. Robust standard errors are reported in parentheses. Stars are based on BH-adjusted p-values: $^{*}q < .10$, $^{**}q < .05$, $^{***}q < .01$.

**Table D2: Multiple-testing adjustments for preregistered redistribution outcomes in the extension**

| Outcome | Comparison | Estimate | SE | $p$ | $q$ | $N$ |
|---|---|---|---|---|---|---|
| *Panel A. Preregistered H1: costless redistribution* | | | | | | |
| Costless redistribution | `MIXED` vs `EASY` | 0.597*** | (0.116) | <0.001 | <0.001 | 501 |
| Costless redistribution | `REPORTS` vs `EASY` | 0.372*** | (0.117) | 0.002 | 0.002 | 501 |
| Costless redistribution | `STATISTICS` vs `EASY` | -0.104 | (0.112) | 0.353 | 0.353 | 508 |
| *Panel B. Preregistered H3: costly redistribution* | | | | | | |
| Costly redistribution | `MIXED` vs `EASY` | 0.735*** | (0.130) | <0.001 | <0.001 | 501 |
| Costly redistribution | `REPORTS` vs `EASY` | 0.455*** | (0.137) | <0.001 | 0.001 | 501 |
| Costly redistribution | `STATISTICS` vs `EASY` | 0.223* | (0.118) | 0.060 | 0.060 | 508 |
| *Panel C. Complementary H3 measure: staying in redistribution when costly* | | | | | | |
| Stay in costly redistribution | `MIXED` vs `EASY` | 0.213*** | (0.058) | <0.001 | <0.001 | 373 |
| Stay in costly redistribution | `REPORTS` vs `EASY` | 0.091 | (0.058) | 0.120 | 0.120 | 375 |
| Stay in costly redistribution | `STATISTICS` vs `EASY` | 0.132** | (0.059) | 0.025 | 0.038 | 354 |

*Notes:* This table reports raw and Benjamini–Hochberg adjusted p-values for the primary redistribution outcomes in the extension experiment. Panels A and B adjust across the three extension treatment comparisons relative to `EASY`. Panel C reports the same adjustment for the complementary persistence measure, restricted to spectators who redistributed a positive amount when redistribution was costless and controlling for fixed effects for the exact costless transfer. All estimates use the preregistered analysis sample and include socio-demographic controls. Robust standard errors are reported in parentheses. Stars are based on BH-adjusted p-values: $^{*}q < .10$, $^{**}q < .05$, $^{***}q < .01$.

**Table D3: Multiple-testing adjustments for preregistered belief outcomes in the extension**

| Outcome | Comparison | Estimate | SE | $p$ | $q$ | $N$ |
|---|---|---|---|---|---|---|
| Proportional performance gap | MIXED vs EASY | 0.970*** | (0.117) | <0.001 | <0.001 | 496 |
| Role of luck | MIXED vs EASY | 31.840*** | (2.829) | <0.001 | <0.001 | 501 |
| Probability B outperforms A | MIXED vs EASY | -7.891*** | (2.231) | <0.001 | <0.001 | 501 |
| Group-level counterfactual / hard-work belief | MIXED vs EASY | 0.128 | (0.081) | 0.117 | 0.126 | 501 |
| Version advantage/intensity | MIXED vs EASY | 1.253*** | (0.129) | <0.001 | <0.001 | 501 |
| Proportional performance gap | REPORTS vs EASY | 0.370*** | (0.049) | <0.001 | <0.001 | 499 |
| Role of luck | REPORTS vs EASY | 19.255*** | (2.805) | <0.001 | <0.001 | 501 |
| Probability B outperforms A | REPORTS vs EASY | -5.354** | (2.044) | 0.009 | 0.010 | 501 |
| Group-level counterfactual / hard-work belief | REPORTS vs EASY | 0.247*** | (0.076) | 0.001 | 0.002 | 501 |
| Version advantage/intensity | REPORTS vs EASY | 0.545*** | (0.125) | <0.001 | <0.001 | 501 |
| Proportional performance gap | STATISTICS vs EASY | 1.727*** | (0.086) | <0.001 | <0.001 | 507 |
| Role of luck | STATISTICS vs EASY | 11.732*** | (3.012) | <0.001 | <0.001 | 508 |
| Probability B outperforms A | STATISTICS vs EASY | -11.252*** | (2.151) | <0.001 | <0.001 | 508 |
| Group-level counterfactual / hard-work belief | STATISTICS vs EASY | 0.091 | (0.083) | 0.270 | 0.270 | 508 |
| Version advantage/intensity | STATISTICS vs EASY | 0.492*** | (0.129) | <0.001 | <0.001 | 508 |

*Notes:* This table reports raw and Benjamini–Hochberg adjusted p-values for the belief outcomes in the extension experiment. The adjustment is applied across the fifteen tests defined by the three extension treatment comparisons relative to EASY and the five preregistered belief outcomes. The performance-gap outcome is the proportional difference between the estimated performance of Agents A and Agents B, as specified in the extension pre-analysis plan. All estimates use the preregistered analysis sample and include socio-demographic controls. Robust standard errors are reported in parentheses. Stars are based on BH-adjusted p-values: $^{*}q < .10$, $^{**}q < .05$, $^{***}q < .01$.

# E Deviations from the Pre-Analysis Plans

This appendix summarizes how the analyses reported in the paper relate to the two pre-analysis plans. The first pre-analysis plan covers the main experiment, which compares spectators assigned to experience the hard version of the task with spectators assigned to experience the easy version of the task. The second pre-analysis plan covers the extension experiment, which adds the `MIXED`, `REPORTS`, and `STATISTICS` treatment arms. The main behavioral outcomes remain those specified in the pre-analysis plans: redistribution toward the disadvantaged agent, beliefs about the role of task version in determining performance, and willingness to pay a cost to implement redistribution.

First, the paper reorganizes the presentation of the hypotheses. In the main pre-analysis plan, Hypothesis 1 concerned costless redistribution, Hypothesis 2 concerned beliefs about the importance of the task version, and Hypothesis 3 concerned costly redistribution. In the paper, we present the hypotheses through the lens of a simple fairness model. The first hypothesis concerns what participants consider to be fair, measured by costless redistribution. The second hypothesis concerns the weight attached to what they consider fair, measured by willingness to redistribute when redistribution is costly. Belief updating is therefore presented as a central mechanism rather than as a separately numbered main hypothesis. This change affects the exposition, but not the underlying empirical tests.

Second, the analysis of costly redistribution is refined relative to the pre-analysis plan. The pre-analysis plan proposed studying costly redistribution using both a constructed indicator for paying to implement redistribution and a regression of costly redistribution on treatment assignment controlling for costless redistribution. In the paper, we retain this logic but implement it in two complementary ways. We estimate a reduced-form persistence specification that conditions on the exact amount redistributed in the costless decision, and we estimate a structural threshold model that maps directly onto the fixed-cost design. This refinement separates changes in the what participant consider to be fair from changes in their willingness to act on it.

Third, the pre-analysis plan specified Benjamini–Hochberg adjustments for multiple testing in the belief outcomes associated with the belief-updating hypothesis, and for exploratory correlations between posterior beliefs and redistribution. We therefore report conventional p-values in the main tables and provide Benjamini–Hochberg adjusted p-values for these pre-specified belief families in the appendix. We do not adjust the primary redistribution outcome or the primary costly-redistribution test as multiple-outcome families, because each corresponds to a single behavioral hypothesis.

Fourth, the mechanism analyses are reported as exploratory. The pre-analysis plan proposed examining whether belief variables and emotions account for the treatment effect on redistribution by sequentially adding these variables to the main regression. The paper follows this approach but interprets these estimates as descriptive evidence on plausible mechanisms rather than as causal mediation estimates.

Fifth, the extension analyses follow the second pre-analysis plan, which introduced the `MIXED`, `REPORTS`,

and STATISTICS treatment arms. The extension pre-analysis plan specified tests of whether each of these treatments increases redistribution, belief updating, and willingness to pay relative to EASY. In the paper, we also emphasize comparisons among MIXED, REPORTS, and STATISTICS because the purpose of the extension is to distinguish direct comparative experience from qualitative and quantitative information provision. These comparisons are therefore aligned with the motivation of the extension pre-analysis plan, although not all were stated as separate numbered hypotheses.

Sixth, the extension pre-analysis plan stated that each extension treatment would include 180 participants. The implemented sample is slightly smaller, with 165 participants in MIXED, 165 in REPORTS, and 172 in STATISTICS. This difference reflects budget limitations.

# F Interventions

**Figure F1: Intervention in `EASY` and `MIXED`**

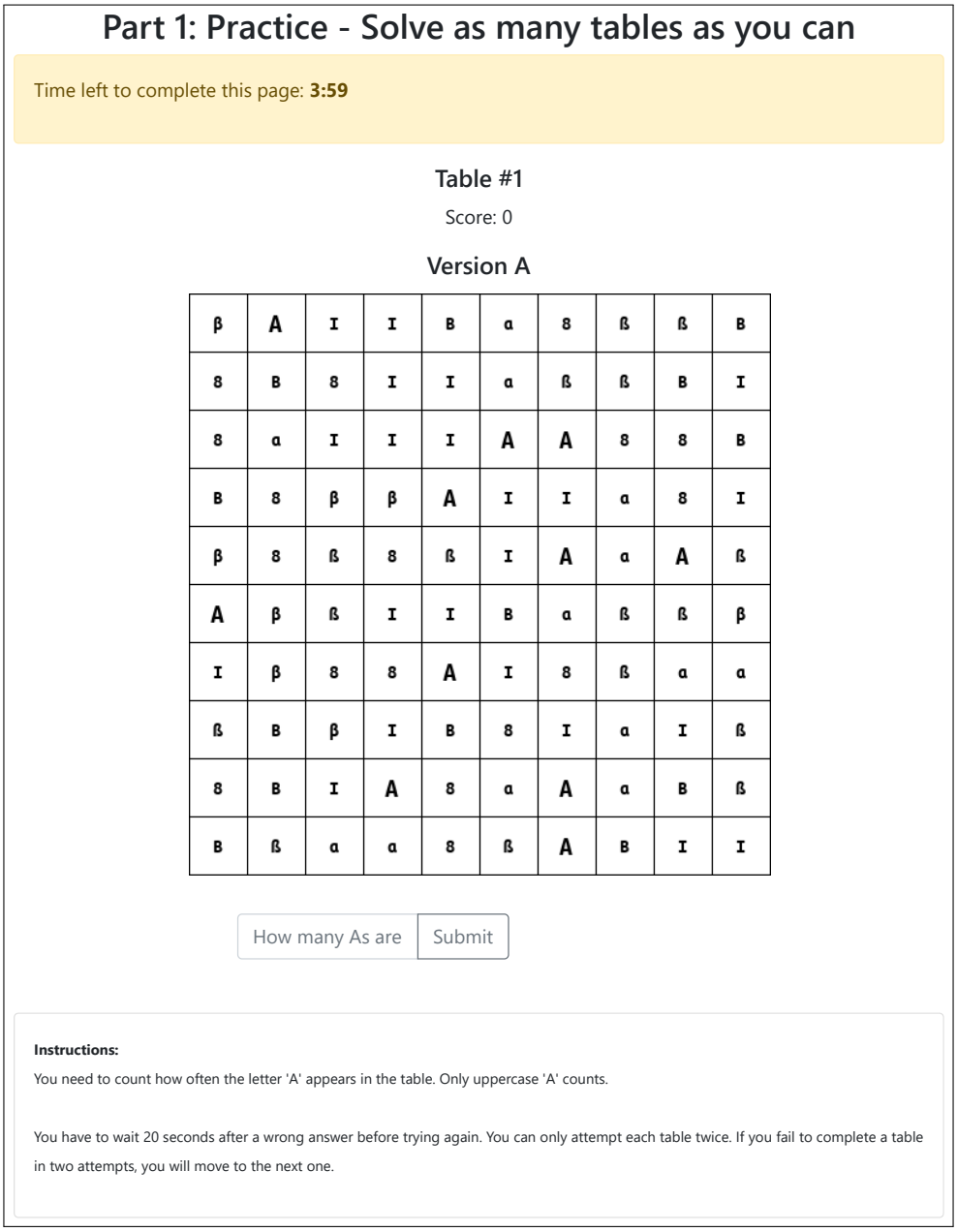


**Figure F2: Intervention in HARD and MIXED**

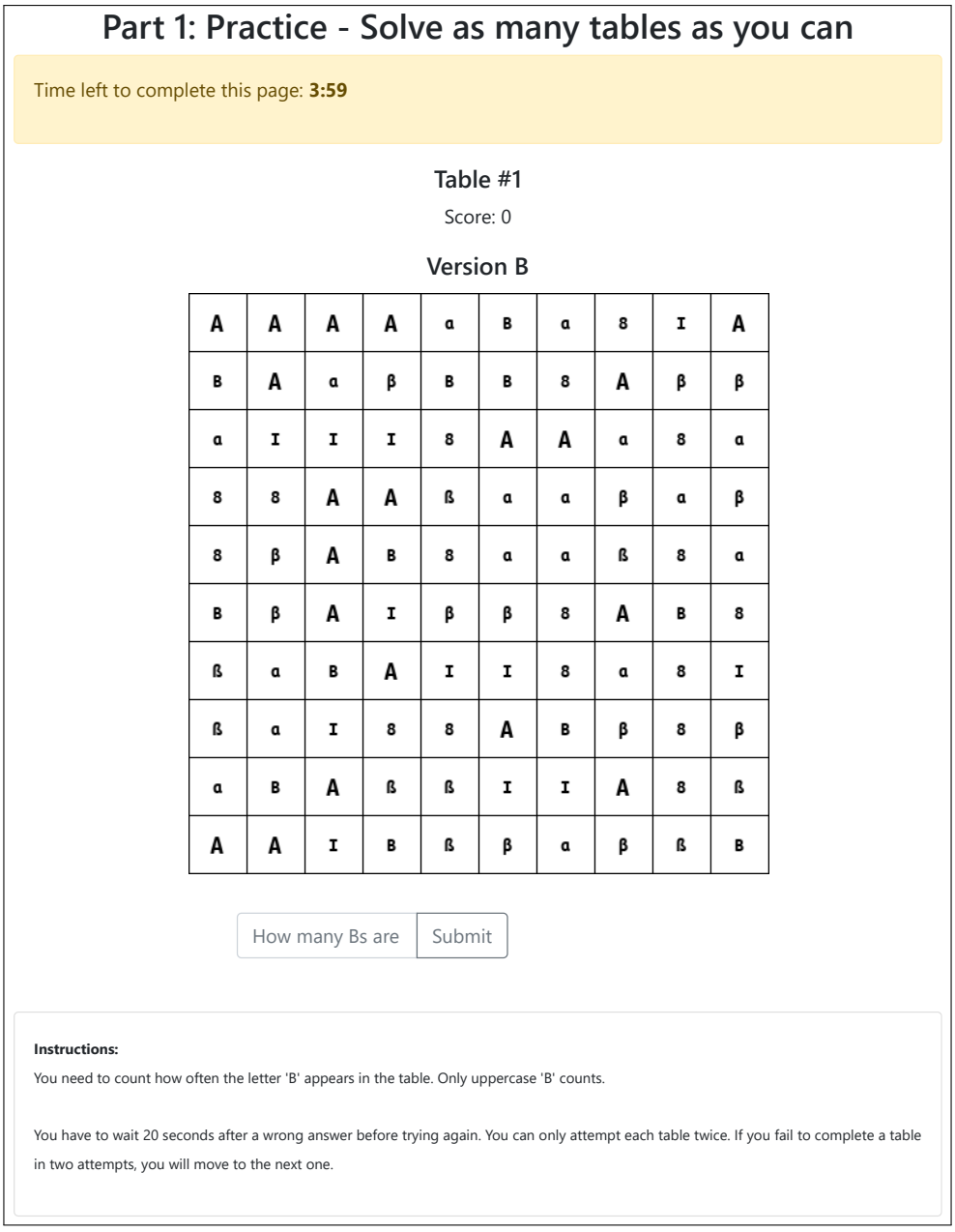


**Figure F3: Intervention in REPORTS**

**Agents A**

People assigned to version A generally described the task as easy but somewhat repetitive. Many reported scanning the grid carefully or following a systematic pattern to keep track of the letters. Some noted that the task required sustained attention over time, but overall they felt confident in their ability to count the letters correctly. Despite the repetitive nature, several participants described the task as enjoyable or puzzle-like.

**Agent A #22 wrote:** "*It was interesting. I found it pretty easy to do, because of pattern recognition. I was able to see the A pretty easily, and it made it easier to find them to count them.*"

**Agent A #60 wrote:** "*It was fairly straight forward and easy to understand. I also thought it was mentally engaging which gave me more motivation to do better.*"

**Agents B**

People assigned to version B generally described the task as challenging and mentally tiring. Many mentioned that the letters looked very similar or appeared in different formats, making them harder to count accurately. Some reported losing track of their count or needing to double-check frequently, while others noted that the task required a high level of concentration and could feel frustrating or time-consuming.

**Agent B #17 wrote:** "*I found it very difficult since all the letters looked very similar to each other so it was easy to miscount or get confused. It was also a bit stressful looking the timer.*"

**Agent B #69 wrote:** "*It was frustrating seeing how similar the characters were to each other and I blamed myself for choosing the wrong thing. I became frustrated.*"

**Figure F4: Intervention in STATISTICS**

**Agents A**

People assigned to version A of the task completed 11 tasks correctly in 4 minutes. They completed correctly 86% of the tasks they attempted. The best performing agent completed 23 tasks correctly, while the worst performing agent completed 0 tasks correctly. The probability that a person assigned to version A outperforms a person assigned to version B is 91%.

**Agents B**

People assigned to version B of the task completed 3 tasks correctly in 4 minutes. They completed correctly 49% of the tasks they attempted. The best performing agent completed 8 tasks correctly, while the worst performing agent completed 0 tasks correctly. The probability that a person assigned to version B outperforms a person assigned to version A is 6%.

# G Text Classification Procedure

This appendix documents the procedure used to classify spectators' open-text justifications into fairness types, in enough detail to be reproduced independently.

## G.1 Classification Task

Each spectator explained, in free text, why they allocated the $6 bonus between Agent A and Agent B the way they did. The classification task takes this text as the sole input and assigns it to exactly one of five categories. The amount actually redistributed is not used as an input to the classification in any way: two spectators reaching the same dollar split for different stated reasons receive different codes, and two spectators reaching different splits for the same stated reason receive the same code.

## G.2 Precise Prompt and Coding Categories

In this study, spectators were told that two people, Agent A and Agent B, each completed a counting task under randomly assigned, unequally difficult task versions (version A was generally easier than version B). Whoever solved more tables correctly was originally promised a $6 bonus in full. The spectator was then allowed to reallocate the $6 bonus between Agent A and Agent B however they wished, and explained their reasoning in free text. IMPORTANT: text only, never the split Classify using only the written explanation. Do not use, look up, or infer from how many dollars each agent actually received. You are coding reasoning, not outcomes. Two spectators can reach the exact same dollar split for entirely different reasons, and two spectators with entirely different splits can be reasoning identically. If you are given the dollar amounts alongside the text, ignore them completely — do not let "it was an equal split" or "it was a big/small adjustment" push you toward a code. The code should follow only from what the sentence says the reason is. (If a respondent's own explanation happens to mention a dollar figure — e.g., "I gave them $4 because..." — that's fine, it's part of their reasoning; the rule is about not using the allocation data as a separate signal, not about scrubbing participants' own words.) Your job is to read the free text and assign exactly one of five codes based on the reasoning given. The five codes are:

| Code | Name | Definition |
|---|---|---|
| LIB | Libertarian | The allocation should stay as originally promised/assigned, because changing it would violate the pre-set rule or agreement itself. |
| NCM | Non-compensating meritocrat | The allocation is based on observed performance (more tables, higher score, more effort), and the task-difficulty gap is either not mentioned or explicitly not used to adjust the allocation. |
| CM | Compensating meritocrat | The allocation is based on performance, but is explicitly adjusted because of the task-difficulty gap, randomness of assignment, or resulting unfairness. |
| EGAL | Egalitarian | The respondent justifies the allocation primarily by appealing to equality, equal entitlement, or equal treatment. The reasoning does not depend on who performed better or on one agent having faced the harder task. |
| OTHER | Other | No identifiable reason, incoherent/contradictory reasoning, or a reason unrelated to performance, effort, or fairness. |

**Table G1: Coding categories for spectators' bonus-allocation justifications.**

## G.3 Representative Quotations by Category

To illustrate how the coding categories map onto participants' own language, we reproduce below two representative explanations for each category, verbatim (including original spelling). All quotations below were assigned the same code by both GPT-5.5 Thinking and Claude.

**Libertarian.**

> *"The rule was whoever completed more tables got the bonus. The tasks were comparable so the rule should stand. Agent A completed more tables and should get to keep the full bonus. We were awarding completion, not effort."*

> *"The rules of the study were that whoever got the most points right earns the highest bonus, so I decided not to change that. If I thought the rules were unfair, I would have split it evenly."*

**Non-compensating meritocrat.**

> *"Agent A solved more tables correctly so they earned the bonus as far as I can tell. I wanted*

*to reward performance. I didn't consider the versions of the tasks, only on the person who performed better."*

*"I made that decision because A did an excellent job and deserve $4 and also B tried their best and they are supposed to be awarded $2 for the effort."*

**Compensating meritocrat.**

*"I chose to split the bonus equally because both agents dedicated the same amount of time to the study. I also factored in that agent B had a much more difficult and frustrating task, which explains why they solved fewer tables than agent A."*

*"The significant scoring gap between versions suggests Version B was much harder. Since assignment was random, I split the bonus equally to ensure fairness regardless of the task difficulty imbalance."*

**Egalitarian.**

*"I gave both agents an equal amount because they put in the same amount of effort and an equal time. I think the tasks are evenly split."*

*"Equal effort and time spent on the task between both groups should produce same pay. Task is relatively simple and time spent is most relevant factor to pay in this case."*

**Other.**

*"Based on my understanding I actually choose the allocation of amount I want to distributed."*

*"No just greedy. This is a very hard to calculate in my mind I should keep the version of my task and not give any to the other guy. As I said just greed."*

# H Questionnaires

## H.1 Spectators

This section includes the screenshots for spectator data collection. The screenshots are those of `EASY`. See Figures F1, F2, F3 and F4 for differences across treatment arms.

# Thanks for participating in this study!




This study is conducted by researchers from Tampere University and Erasmus University Rotterdam.

In this study you have the opportunity of earning money by answering questions about different versions of a task.

**Duration**
The study can be completed in around 30 minutes.

**Compensation**
You will receive a 3.00 GBP (around 4.00 USD) payment for completing the study.

**Informed consent form**
On the next page, you will find an informed consent form. Please read it carefully and confirm it if you want to proceed with this study.

Next

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# Informed consent

Before proceeding, please give your consent that you voluntarily participate in this study, that you have been sufficiently informed about the nature of the study, and that you explicitly give us your consent that:

- We can collect your anonymous, non-sensitive personal data (like age, income, etc).
- We can use this personal data for scientific purposes.
- We can store your anonymous, non-sensitive personal data on our safe-guarded university servers for up to 10 years.
- We can make anonymized data available to other researchers online.

We promise to protect your data according to the latest General Data Protection Regulation (GDPR) data regulation laws. Notice that you can also withdraw your consent anytime during the study by returning your submission before completing this study. In that case, your data will not be stored.

In case you have any questions at any point, please contact us at pogliano@ese.eur.nl.

I agree | I do not agree

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# General Instructions

This study has three parts.

**Part 1:**
You will learn about a task in which you count how many times a specific character appears in a table. There are two versions of the task, version A and version B.

**Part 2:**
You will guess the results of a previous study we completed. You can win an additional prize if your guesses are correct. You will also make a decision that can affect the earnings of two other real people.

**Part 3:**
You will answer some additional questions.

Next

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# Part 1

**Instructions**

You will now learn about a task. In this task, you need to count how often a specific character appears in a table like the one below.

Your goal is to solve as many tables as possible in 4 minutes. Every time you solve a table correctly, you increase your score by 1. Your final score is the total number of tables you solve correctly within 4 minutes.

Each table is filled with characters. You must count how many times the target character appears and submit the count. When you submit the correct count for a table, and a new one will appear with new characters in shuffled positions.

You have 2 attempts per table:

- If your first attempt is incorrect, you must wait 20 seconds before trying again.
- If your second attempt is incorrect, the table will disappear, and you will start over with a completely new one.

There are two versions of the task. The type of tables are the same in both versions, only the target character differs. In Version A, you count how many times uppercase “A” appears in the table (e.g., 17 in the example below). In Version B, you count how many times uppercase “B” appears in the table (e.g., 14 in the example below).

**Comprehension Questions:**

**What does your score depend on?**

- ○ The number of tables you solve correctly within 4 minutes
- ○ The number of As present in the tables
- ○ The number of attempts you make

**What do you have to do if you are assigned Version A?**

- ○ Count how many times character B appears in the table
- ○ Count how many times character A appears in the table
- ○ There is no Version A

**Are there any differences in the tables between versions?**

- ○ No
- ○ Yes

Next

*Example table:*

| B | B | β | α | α | A | β | I | 8 | I |
|---|---|---|---|---|---|---|---|---|---|
| I | I | I | 8 | α | A | A | α | 8 | I |
| B | I | ß | β | A | ß | β | β | A | α |
| I | β | 8 | α | 8 | I | β | ß | I | α |
| 8 | β | α | α | β | β | 8 | B | A | ß |
| β | 8 | β | 8 | β | 8 | β | I | β | α |
| 8 | B | β | ß | 8 | A | I | β | A | 8 |
| ß | B | A | A | A | β | ß | 8 | A | I |
| A | β | 8 | A | I | A | B | A | A | 8 |
| β | α | B | I | ß | B | I | A | A | 8 |

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You have answered all the comprehension questions correctly.

We will now ask you some questions related to the task.

Next

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**Imagine this situation:**
Two people are completing the task at the same time.

- One person is assigned Version A (counting 'A's)
- The other person is assigned Version B (counting 'B's)

The two are matched and their scores are compared.

**Answer the following question:**

**What matters most in determining who gets the higher score?**
Rank the following factors. Drag and drop to sort the factors from most relevant on top to least relevant on the bottom.

☰ **Ability** (how good a person is at counting quickly and accurately)

☰ **Effort** (how hard a person works during the task)

☰ **Version** (whether they receive Version A or Version B of the task)

Next

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**Answer the following question:**

**Do the two versions of the task feel similar to you?**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

Next

**Remember:**

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

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**Answer the following question:**

On a scale from 1 to 7, where 1 is 'Getting Version A is a major advantage' and 7 is 'Getting Version B is a major advantage',
**Do you think any version confers an advantage for achieving higher scores?**

- ○ 1: Getting Version A is a major advantage
- ○ 2
- ○ 3
- ○ 4: Both versions are approximately the same
- ○ 5
- ○ 6
- ○ 7: Getting Version B is a major advantage

Next

**Remember:**

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

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**Answer the following question:**

The following question is an attention check. Please select letter C.
**Select one of the following letters.**

- A
- B
- C
- D

Next

**Remember:**

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

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**You will now get more familiar with the task.**

You were randomly assigned to practice **Version A** of the task.

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

Solve as many tables as you can in 4 minutes.

Start

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# Part 1: Practice - Solve as many tables as you can

Time left to complete this page: **3:59**

## Table #1

Score: 0

### Version A

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| β | A | I | I | B | a | 8 | ß | ß | B |
| 8 | B | 8 | I | I | a | ß | ß | B | I |
| 8 | a | I | I | I | A | A | 8 | 8 | B |
| B | 8 | β | β | A | I | I | a | 8 | I |
| β | 8 | ß | 8 | ß | I | A | a | A | ß |
| A | β | ß | I | I | B | a | ß | ß | β |
| I | β | 8 | 8 | A | I | 8 | ß | a | a |
| ß | B | β | I | B | 8 | I | a | I | ß |
| 8 | B | I | A | 8 | a | A | a | B | ß |
| B | ß | a | a | 8 | ß | A | B | I | I |

How many As are | Submit

**Instructions:**

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

You have to wait 20 seconds after a wrong answer before trying again. You can only attempt each table twice. If you fail to complete a table in two attempts, you will move to the next one.

**You are midway through the practice.**

The following question is an attention check. Please select number 23.
**Select one of the following numbers.**

- ○ 12
- ○ 23
- ○ 30
- ○ 41

Next

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# Part 1: Practice - Solve as many tables as you can

Time left to complete this page: **1:58**

## Table #1

Score: 0

### Version A

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| β | A | I | I | B | a | 8 | ß | ß | B |
| 8 | B | 8 | I | I | a | ß | ß | B | I |
| 8 | a | I | I | I | A | A | 8 | 8 | B |
| B | 8 | β | β | A | I | I | a | 8 | I |
| β | 8 | ß | 8 | ß | I | A | a | A | ß |
| A | β | ß | I | I | B | a | ß | ß | β |
| I | β | 8 | 8 | A | I | 8 | ß | a | a |
| ß | B | β | I | B | 8 | I | a | I | ß |
| 8 | B | I | A | 8 | a | A | a | B | ß |
| B | ß | a | a | 8 | ß | A | B | I | I |

How many As are | Submit

**Instructions:**

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

You have to wait 20 seconds after a wrong answer before trying again. You can only attempt each table twice. If you fail to complete a table in two attempts, you will move to the next one.

**Answer the following question:**

Remember there are 2 versions of the task.
**Which version of the task did you just complete?**

- ○ Version A
- ○ Version B

Next

**Remember:**

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

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Congratulations, you completed **Part 1**!

To reward you for your effort so far, we have awarded you **an additional prize of 1 USD**.

Part 2 will start in the next screen.

Next

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# Part 2

Last week, we conducted another study here on Prolific. In that study, people had to complete the counting task you previously learned about. We had two groups of people, which we will refer to as Agents A and Agents B.

We now give you more details about our study.

Next

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# Part 2

**Context**

People taking part in our study were randomly assigned to two groups.

- People randomly assigned to be an Agent A completed Version A, in which they had to count the number of times the letter 'A' appeared in the tables.
- People randomly assigned to be an Agent B completed Version B, in which they had to count the number of times the letter 'B' appeared in the tables.

All agents completed the task for 4 minutes. They all received the same payment for their participation, regardless of their score.

We also informed the agents that they might get some additional bonus based on how many tables they solved and on others' decisions. More information about the bonus will follow later.

After they completed the task, we matched each Agent A with one Agent B to form pairs.

**Comprehension test:**

**1) The version of the task (Version A or Version B) is randomly assigned to each person.**

- ○ True
- ○ False

**2) Agents spend the same amount of time on the task and receive the same fixed payment to compensate them for their participation.**

- ○ True
- ○ False

**3) Which agent had to count the number of times the letter 'A' appears in the tables?**

- ○ Agent A
- ○ Agent B

Next

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You have answered all the comprehension questions correctly.

Next

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**Guess the Results**

In the next screens, we will ask you to make 3 guesses about the results of our study.

At the end of this survey, 1 out of these 3 guesses will be randomly selected. We will compare the selected guess with the actual results of our study. If your guess is correct, **you will receive a prize of 3 USD**.

Please read and think carefully about each guess.

Next

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**Guess #1:**

**On average, how many tables do you think Agents A solved correctly in 4 minutes?**

**How sure are you about your answer?**

- Not at all sure
- A little sure
- Somewhat sure
- Very sure
- Completely sure

Next

**Remember:**

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.

Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

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**Guess #2:**

**On average, how many tables do you think Agents B solved correctly in 4 minutes?**

**How sure are you about your answer?**

- Not at all sure
- A little sure
- Somewhat sure
- Very sure
- Completely sure

Next

**Remember:**

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.

Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

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**Guess #3:**

**Consider the following scenario:**
Agents A complete Version A of the task, in which they count the number of times the letter 'A' appeared in the tables. Agents B complete Version B of the task, in which they count the number of times the letter 'B' appeared in the tables. After all agents complete the task, we randomly match each Agent A with one Agent B to form pairs.

Suppose there are 100 such pairs of agents as described above.

**In how many pairs (out of the 100) do you think that the Agent B solves more correct tables than the Agent A?**
*Your guess is the number of pairs (out of 100) in which Agent B outperforms Agent A.*

**How sure are you about your answer?**

- ○ Not at all sure
- ○ A little sure
- ○ Somewhat sure
- ○ Very sure
- ○ Completely sure

Next

**Remember:**

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.

Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

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1 of the previous 3 guesses will be selected to determine whether you get the prize.

We will now continue asking you questions about the agents we studied.

Next

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**Read carefully:**
In the next screens, you will learn about two pairs of Agents that completed the study last week. On top of their participation payment, an additional bonus of 6 USD was initially assigned to the agent in each pair that solved more correct tables.

But you can now change that. We want to know how much bonus payment YOU would give each agent.

There are no correct or wrong answers. You can choose what you want, as long as the total bonus for each pair adds up to 6 USD.

**Important:**
At the end of this study, we will randomly select 1 out of every 10 participants. For the selected participants, we will randomly implement one of their two decisions for a pair of real people. This means that your decision could affect the actual earnings of real people, so please make it carefully.

- ○ I understand that my decision can affect the earnings of real people
- ○ Attention check: Do not select this option

Next

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**First pair**

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

Next

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**Information about the first pair.**
The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

**How would you split the 6 USD bonus?**

| | |
|---|---|
| Agent A | Starts with 6 |
| Agent B | Starts with 0 |
| **Total:** | **0** |

Decide

**Instructions**

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD.

**Time on screen**

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

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**Think about your decision.**

**Explain**, in at least 2 or 3 sentences, **the reasoning behind why you made this decision**.
Did you take into account the fact that the agents completed different versions of the task?
*Please try to spend 1 or 2 minutes answering this question.*

Next

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**Second pair**

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

**Read carefully:**

To change the initial bonus distribution in this pair, you need to pay.

You can keep the initial distribution with no cost, or you can change it by paying 0.20 USD from the 1 USD bonus you received earlier.

Next

Powered by oTree

**Information about the second pair.**
The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

You can keep the initial distribution with no cost, or you can change it by paying 0.20 USD from the 1 USD bonus you received earlier.

**How would you split the 6 USD bonus?**

| | |
|---|---|
| Agent A | Starts with 6 |
| Agent B | Starts with 0 |
| **Total:** | **0** |
| **Your Bonus:** | 1.00 USD |

Decide

**Instructions**

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD. Whenever your decision changes the initial distribution, you pay 0.20 USD from the 1 USD bonus you received earlier.

**Time on screen**

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

Powered by oTree

Now please think about the pair formed by an Agent A and an Agent B you just decided for. Remember, Agent A was initially assigned the bonus because they performed better.

You will be asked questions about this pair.

Next

Powered by oTree

**Answer the following question:**

**In this pair, how fair was that the agent that solved more correct tables was initially assigned all the bonus?**

- ○ Completely unfair
- ○ Somewhat unfair
- ○ Somewhat fair
- ○ Completely fair

Next

**Remember:**

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.

Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

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**How much do you agree with this statement?**

**In this pair, if Agent B had worked really hard they could have solved more correct tables than Agent A.**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

Next

**Remember:**

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.

Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

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**Answer the following question:**

Imagine that both agents in this pair completed Version B of the task.
**Who do you think would have solved more tables in that case?**

- ○ Agent A
- ○ Agent B
- ○ They would have completed the same amount approximately

Next

**Remember:**

In Version A of the task, you need to count the number of times the letter 'A' appeared in the tables.

In Version B of the task, you need to count the number of times the letter 'B' appeared in the tables.

Powered by oTree

Just for this question, think about all agents that completed the task.

Imagine all agents complete Version B of the task, but are still randomly assigned to be called either Agents A or Agents B.

**Answer the following question:**

**Which group of agents do you think would solve more correct tables?**

- ○ Agents A would solve many more correct tables
- ○ Agents A would solve more correct tables
- ○ Both Agents A and Agents B would solve about the same number of correct tables
- ○ Agents B would solve more correct tables
- ○ Agents B would solve many more correct tables

Next

**Remember:**

In Version A of the task, you need to count the number of times the letter 'A' appeared in the tables.

In Version B of the task, you need to count the number of times the letter 'B' appeared in the tables.

Powered by oTree

**Answer the following question:**

**In the pair you decided for, how much do these factors explain which of the agents solved more correct tables?**

Please give points to each factor so that the total adds up to 100 points.

| | |
|---|---|
| Effort exerted and agent's ability: | 0 |
| Version of the task assigned: | 0 |
| **Total:** | **0** |

Next

Powered by oTree

**Answer the following question:**

**Do you think the version of the task assigned to each agent provided any advantage?**

- ○ 1: Agent A had a major advantage
- ○ 2
- ○ 3
- ○ 4: Both agents were on equal footing
- ○ 5
- ○ 6
- ○ 7: Agent B had a major advantage

Next

Powered by oTree

**How much do you agree with these statements?**

**It is easy for me to imagine Agent A's experience with the task.**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

**It is easy for me to imagine Agent B's experience with the task.**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

Next

**Remember:**

Agents A completed Version A of the task, in which they had to count the number of times the letter 'A' appeared in the tables.

Agents B completed Version B of the task, in which they had to count the number of times the letter 'B' appeared in the tables.

Powered by oTree

**Now, we want to know how you felt when learning about this pair.**
Remember, the two agents completed the task for 4 minutes. Agent A completed Version A of the task, while Agent B completed Version B of the task. After they completed the task, we matched both agents to form a pair.

Agent A solved more correct tables than Agent B, and was initially assigned all of the 6 USD bonus.

**When learning about it, how much did you feel each of the following?**

**Anger**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

**Compassion**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

**Sadness**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

**Indifference**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

**Irritation**

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Completely

Next

Powered by oTree

We will now ask you to make decisions for three other pairs of Agents.

You can decide what you want with no cost to yourself.

Next

Powered by oTree

**Information about the pair.**
The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

**Read carefully: Agent B solved more correct tables and was initially assigned the full bonus**.

**How would you split the 6 USD bonus?**

| | |
|---|---|
| Agent A | Starts with 0 |
| Agent B | Starts with 6 |
| **Total:** | **0** |

Decide

**Instructions**

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD and there are no costs to yourself.

**Time on screen**

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

Powered by oTree

**Information about the pair.**
The pair was formed by Agent 1 and Agent 2.

**Read carefully: Both Agents completed Version B of the Task.**

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

Agent 1 solved more correct tables and was initially assigned the full bonus.

**How would you split the 6 USD bonus?**

| | |
|---|---|
| Agent 1 | Starts with 6 |
| Agent 2 | Starts with 0 |
| **Total:** | **0** |

Decide

**Instructions**

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD and there are no costs to yourself.

**Time on screen**

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

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**Information about the pair.**
The pair was formed by Agent 3 and Agent 4.

**Read carefully: Both Agents participated in a coin toss**.

Both Agents received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that won a coin toss.

Agent 3 won the coin toss and was initially assigned the full bonus.

**How would you split the 6 USD bonus?**

| | |
|---|---|
| Agent 3 | Starts with 6 |
| Agent 4 | Starts with 0 |
| **Total:** | **0** |

Decide

**Instructions**

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD and there are no costs to yourself.

**Time on screen**

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

Powered by oTree

We will now ask you to make a decision for a pair of Agents that completed a different task to the one you've seen so far.

Next

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**Consider the following situation:**
Two persons, Agent C and Agent D, are invited to complete a series of multiple choice questions.

Before completing the multiple choice questions, both Agent C and Agent D received learning materials on similar topic and length to prepare themselves.

- Agent C was randomly assigned materials that are of high relevance for answering correctly the questions.
- Agent D was randomly assigned materials that are of low relevance for answering correctly the questions.

Both Agents spent the same time on the questions and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent completing more multiple choice questions correctly.

You are informed that Agent C completed more multiple choice questions correctly and was initially assigned the full bonus.

**You can decide on the final bonus each agent gets.**

Next

Powered by oTree

**Information about the pair.**
Two persons, Agent C and Agent D, are invited to complete a series of multiple choice questions.

Before completing the multiple choice questions, both Agent C and Agent D received learning materials on similar topic and length to prepare themselves.

- Agent C was randomly assigned materials that are of high relevance for answering correctly the questions.
- Agent D was randomly assigned materials that are of low relevance for answering correctly the questions.

An additional bonus of 6 USD was initially assigned to the agent who completed more multiple choice questions correctly.

Agent C answered more questions correctly and was initially assigned the full bonus.

**How would you split the bonus?**

| | |
|---|---|
| Agent C | Starts with 6 |
| Agent D | Starts with 0 |
| **Total:** | **0** |

Decide

**Instructions**

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD and there are no costs to yourself.

**Time on screen**

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

Powered by oTree

We now want you to think back about the first pair of Agents you saw.

Remember, the two agents completed a task for 4 minutes. Agent A completed Version A of the task, while Agent B completed Version B of the task. After they completed the task, we matched both agents to form a pair.

Next

Powered by oTree

Here are 5 types of views about how earnings should be distributed within the pair.

**Type 1:**
People in Type 1 always want agents to earn roughly the same.

**Type 2:**
People in Type 2 always want agents who are initially assigned more earnings to earn more, no matter why they were assigned more.

**Type 3:**
People in Type 3 don't want earnings to be decided by a coin toss. They want agents who solve more correct tables to earn more, no matter the version of the task they had to solve.

**Type 4:**
People in Type 4 don't want earnings to be decided by a coin toss. They want earnings to reflect the number of tables correctly solved, but take into account the version of the task each agent had to solve.

**Type 5:**
People in Type 5 don't fall into the other 4 types.

**Which type do you feel is a good description of yourself?**

- ○ Type 1
- ○ Type 2
- ○ Type 3
- ○ Type 4
- ○ Type 5

Next

Powered by oTree

Congratulations, you completed **Part 2**!

Part 3 will start in the next screen.

Next

Powered by oTree

# Part 3

**Instructions**

In this part you will answer some additional questions.

Next

Powered by oTree

**Answer the following questions:**

**What was your sex assigned at birth?**

- ○ Female
- ○ Male
- ○ Other
- ○ Prefer not to disclose

**What is your age?**

**What is your ethnicity?**
Select all that apply. You can leave blank if you prefer not to disclose.

- ☐ White
- ☐ Black or African American
- ☐ Asian
- ☐ American Indian or Alaska Native
- ☐ Native Hawaiian or Other Pacific Islander
- ☐ Other

**Are you of Hispanic, Latino, or Spanish origin?**

- ○ Yes
- ○ No
- ○ Prefer not to disclose

**In which state do you currently live?**

--------

Next

Powered by oTree

**Answer the following questions:**

**What is the highest level of education you have completed?**

--------

**Do you have a diagnosed condition of dyslexia?**

- No
- Yes
- Prefer not to disclose

Next

Powered by oTree

**Answer the following questions:**

**When it comes to social issues, how liberal or conservative would you say that you are?**

- ○ 1: Very Liberal
- ○ 2
- ○ 3
- ○ 4
- ○ 5
- ○ 6
- ○ 7
- ○ 8
- ○ 9
- ○ 10: Very Conservative

**When it comes to economic issues, how liberal or conservative would you say that you are?**

- ○ 1: Very Liberal
- ○ 2
- ○ 3
- ○ 4
- ○ 5
- ○ 6
- ○ 7
- ○ 8
- ○ 9
- ○ 10: Very Conservative

**Which political party do you more strongly identify with?**

- ○ Democrat
- ○ Republican
- ○ Independent

Next

Powered by oTree

**To what extent do you agree with the following statements?**

**Government should redistribute income from the better off to those who are less well off.**

- ○ Strongly Disagree
- ○ Disagree
- ○ Neither Agree nor Disagree
- ○ Agree
- ○ Strongly Agree

**The context from which a person comes from is important for getting ahead in society.**

- ○ Strongly Disagree
- ○ Disagree
- ○ Neither Agree nor Disagree
- ○ Agree
- ○ Strongly Agree

**Exceptional talent can overcome bad luck.**

- ○ Strongly Disagree
- ○ Disagree
- ○ Neither Agree nor Disagree
- ○ Agree
- ○ Strongly Agree

**I find it easy to put myself in someone else's shoes.**

- ○ Strongly Disagree
- ○ Disagree
- ○ Neither Agree nor Disagree
- ○ Agree
- ○ Strongly Agree

Next

Powered by oTree

## H.2 Agents

# Thanks for participating in this study!




This study is conducted by researchers from Tampere University and Erasmus University Rotterdam.

In this study you have the opportunity of earning money by completing a task.

**Duration**
The study can be completed in around 10 minutes.

**Compensation**
You will receive a 1.00 GBP (around 1.30 USD) payment for completing the study.

**Informed consent form**
On the next page, you will find an informed consent form. Please read it carefully and confirm it if you want to proceed with this study.

Next

## Informed consent

Before proceeding, please give your consent that you voluntarily participate in this study, that you have been sufficiently informed about the nature of the study, and that you explicitly give us your consent that:

- We can collect your anonymous, non-sensitive personal data (like age, income, etc).
- We can use this personal data for scientific purposes.
- We can store your anonymous, non-sensitive personal data on our safe-guarded university servers for up to 10 years.
- We can make anonymized data available to other researchers online.

We promise to protect your data according to the latest General Data Protection Regulation (GDPR) data regulation laws. Notice that you can also withdraw your consent anytime during the study by returning your submission before completing this study. In that case, your data will not be stored.

In case you have any questions at any point, please contact us at pogliano@ese.eur.nl.

I agree  I do not agree

**Instructions**
In this task, you need to count how often a specific character appears in a table like the one below.

Your goal is to solve as many tables as possible in 4 minutes. Every time you solve a table correctly, you increase your score by 1. Your final score is the total number of tables you solve correctly within 4 minutes.

Each table is filled with characters. You must count how many times the target character appears and submit the count. When you submit the correct count for a table, and a new one will appear with new characters in shuffled positions.

You have 2 attempts per table:

- If your first attempt is incorrect, you must wait 20 seconds before trying again.
- If your second attempt is incorrect, the table will disappear, and you will start over with a completely new one.

There are two versions of the task. The type of tables are the same in both versions, only the target character differs. In Version A, you count how many times uppercase “A” appears in the table (e.g., 17 in the example below). In Version B, you count how many times uppercase “B” appears in the table (e.g., 14 in the example below).

You will be randomly assigned one version of the task.

**Bonus Payment**
You can earn bonus payment depending on your score.
Your chances of earning a bonus payment increase the higher your score is.

*Notice: You need to have completed at least one table to be eligible to the bonus payment.*

**Comprehension Questions:**

**What does your score depend on?**

- ○ The number of tables you solve correctly within 4 minutes
- ○ The number of As present in the tables
- ○ The number of attempts you make

**What do you have to do if you are assigned Version A?**

- ○ Count how many times character B appears in the table

- ○ Count how many times character A appears in the table
- ○ There is no Version A

**Are there any differences in the tables between versions?**

- ○ No
- ○ Yes

Next

*Example table:*

| B | B | β | α | α | A | β | I | 8 | I |
|---|---|---|---|---|---|---|---|---|---|
| I | I | I | 8 | α | A | A | α | 8 | I |
| B | I | ß | β | A | ß | β | β | A | α |
| I | β | 8 | α | 8 | I | β | ß | I | α |
| 8 | β | α | α | β | β | 8 | B | A | ß |
| β | 8 | β | 8 | β | 8 | β | I | β | α |
| 8 | B | β | ß | 8 | A | I | β | A | 8 |
| ß | B | A | A | A | β | ß | 8 | A | I |
| A | β | 8 | A | I | A | B | A | A | 8 |
| β | α | B | I | ß | B | I | A | A | 8 |

You have answered all the comprehension questions correctly.

Next

**Answer the following question:**

The following question is an attention check. Please select letter C.
**Select one of the following letters.**

- A
- B
- C
- D

Next

**Remember:**

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

You were randomly assigned to **Version A** of the task.

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

Solve as many tables as you can in 4 minutes.

Start

# Solve as many tables as you can

Time left to complete this page: **3:58**

## Table #1

Score: 0

### Version A

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| B | 8 | B | ß | β | a | A | 8 | ß | A |
| β | I | B | 8 | a | A | B | A | β | B |
| B | 8 | β | I | 8 | 8 | A | B | B | ß |
| B | β | 8 | I | a | a | B | a | 8 | 8 |
| β | β | A | β | A | A | β | B | β | a |
| I | ß | I | β | 8 | I | a | ß | B | A |
| a | ß | ß | I | I | A | A | A | a | a |
| I | a | I | ß | ß | 8 | 8 | ß | β | I |
| β | I | I | a | ß | I | A | 8 | B | 8 |
| I | 8 | 8 | β | 8 | a | A | I | ß | a |

How many As are in the table? Submit

**Instructions:**

You need to count how often the letter 'A' appears in the table. Only uppercase 'A' counts.

You have to wait 20 seconds after a wrong answer before trying again. You can only attempt each table twice. If you fail to complete a table in two attempts, you will move to the next one.

**Answer the following question:**

Remember there are 2 versions of the task.
**Which version of the task did you just complete?**

- ○ Version A
- ○ Version B

Next

**Remember:**

In Version A, you count the number of times the letter 'A' appears in the table.

In Version B, you count the number of times the letter 'B' appears in the table.

Congratulations, you completed the task!

We still have a few more questions for you to answer.

Next

**Context**

People taking part in this study are paired. Each pair consists of one person who completed Version A (referred to as Agent A) and one person who completed Version B (referred to as Agent B).

In each pair, Agent A and Agent B compete to solve as many correct tables as possible. On top of the participation payment, an additional bonus of 6 USD is initially assigned to the agent that wins.

**Read carefully:**

You completed Version A, so you are Agent A in your pair.

We want you to imagine that you won the competition against Agent B in your pair.

Next

In the next screen, you will learn about one pair of Agents that already completed the study.
*Notice: You are not part of the pair, so you won't be affected.*

We want to know how much bonus payment YOU would give each agent.

There are no correct or wrong answers. You can choose what you want, as long as the total bonus adds up to 6 USD.

Next

The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

Next

**Information about the pair.**
The pair was formed by one Agent A and by one Agent B.

Both Agents spent 4 minutes on the task and received a payment for their participation.

An additional bonus of 6 USD was initially assigned to the agent that solved more correct tables.

The Agent A solved more correct tables and was initially assigned the bonus of 6 USD.

**How would you split the 6 USD bonus?**

| | |
|---|---|
| Agent A | Starts with 6 |
| Agent B | Starts with 0 |
| **Total:** | **0** |

Decide

**Instructions**

Write how much bonus payments you want each agent to receive. Remember the sum of bonus payments must equal 6 USD.

**Time on screen**

Your answers need to sum 6 USD. If not, you will not be able to submit them and proceed to the next page.

In the next screen, we will ask you some final questions.

Next

**Answer the following questions:**

**What was your sex assigned at birth?**

- ○ Female
- ○ Male
- ○ Other
- ○ Prefer not to disclose

**What is your age?**

**What is your ethnicity?**
Select all that apply. You can leave blank if you prefer not to disclose.

- ☐ White
- ☐ Black or African American
- ☐ Asian
- ☐ American Indian or Alaska Native
- ☐ Native Hawaiian or Other Pacific Islander
- ☐ Other

**Are you of Hispanic, Latino, or Spanish origin?**

- ○ Yes
- ○ No
- ○ Prefer not to disclose

**Which political party do you more strongly identify with?**

- Democrat
- Republican
- Independent

Next